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MASTER'S THESIS

Design and evaluation of a system for determining electric power quality parameters

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Tytuł pracy

Projekt i badania układu do wyznaczania parametrów jakości energii elektrycznej

Streszczenie

W pracy przedstawiono projekt, wykonanie oraz badania, niskokosztowego miernika parametrów jakości energii dla obwodów jednofazowych. Urządzenie realizuje galwanicznie izolowany pomiar napięcia i prądu z wykorzystaniem przetworników Sigma-Delta współpracujących z mikrokontrolerem STM32H750. Z pomocą algorytmu synchronizacji z siecią SOGI-PLL wyznaczane są wartości skuteczne napięcia i prądu, moce czynna, bierna i pozorna, oraz współczynniki zawartości harmonicznych i współczynniki mocy. Wyniki pomiarów przesyłane są do aplikacji nadrzędnej przez interfejs Ethernet (stos TCP/IP). W ramach pracy przeprowadzono kalibrację oraz weryfikację urządzenia w zestawieniu z analizatorem klasy A Fluke 435-II. Przeprowadzono także badania poboru energii przez wybrane urządzenia domowe oraz analizę dobowego profilu obciążenia instalacji mieszkaniowej.

Słowa kluczowe

jakość energii elektrycznej, pomiary mocy, przetworniki Sigma-Delta, SOGI-PLL

Thesis title

Design and evaluation of a system for determining electric power quality parameters

Abstract

This thesis presents the design, construction and testing of a low-cost power quality meter for single-phase circuits. The device performs galvanically isolated voltage and current measurement using Sigma-Delta converters cooperating with an STM32H750 microcontroller. With the aid of the SOGI-PLL grid synchronization algorithm, the RMS values of voltage and current, active, reactive and apparent power, as well as total harmonic distortion and power factors, are determined. Measurement results are transmitted to a host application via the Ethernet interface (TCP/IP stack). Within the framework of the thesis, calibration and verification of the device against a class A Fluke 435-II analyzer were carried out. The power consumption of selected household appliances was also investigated and the daily load profile of a residential installation was analysed.

Key words

power quality, power measurements, Sigma-Delta converters, SOGI-PLL

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Chapter 1

Introduction

1.1 Introduction to power quality issues

Power quality (PQ) is currently one of the key aspects of the reliable operation of both modern industry and households. Along with the dynamic development of technology and the ongoing digitalisation, the requirements imposed on power grids have become considerably more stringent. The basic parameters defining the quality of the energy supplied to consumers are specified in the PN-EN 50160 standard [1], which defines the permissible voltage deviations, asymmetry, and harmonic content. Another critical parameter is the mains voltage frequency, which in the interconnected European ENTSO-E system (*European Network of Transmission System Operators for Electricity*) must be maintained at a rigorous level of 50 Hz, since deviations from this value indicate a disturbance of the active power balance in the system.

Nowadays, a growing problem is observed with maintaining high power quality, which results from the changing nature of the loads used. In the past, linear devices predominated (e.g. traditional incandescent lamps, simple motors). Today, non-linear loads are commonly used in both industrial and domestic environments. The introduction of a vast number of switched-mode power supplies, mobile device chargers, inverters, and LED-based lighting causes the current drawn to deviate from an ideal sinusoidal waveform. Although individual household appliances are characterised by relatively low power, their mass scale means that, in aggregate, they introduce significant distortion into the grid (expressed by the THD coefficient) and generate problems related to reactive power consumption. This results in reduced transformer efficiency, overloading of neutral conductors, and a negative impact on the service life of other devices connected to the grid.

1.2 Formulation of the research problem

The growing awareness of the impact of grid distortion on electrical devices necessitates the use of appropriate measurement systems. When analysing the market for devices measuring power quality parameters, two main classes of equipment can be distinguished. The first consists of professional power quality analyzers (manufactured, among others, by Fluke or Sonel), certified in class A or S in accordance with the IEC 61000-4-30 standard [2]. These are advanced instruments used for specialised audits, the selection of reactive power compensation systems, and the recording of phenomena such as voltage dips, swells, and flicker. The main barrier to their widespread use is their very high cost, ranging from several thousand to several tens of thousands of zlotys. [3]

The second, opposite class consists of cheap consumer meters – both simple plug-in devices (wattmeters) and "Smart Home" modules mounted on a DIN rail. Despite their attractive price (often below a few hundred zlotys), their functionality is usually limited to measuring RMS values and active energy consumption (kWh), without the possibility of in-depth distortion analysis or advanced power parameters.

Between these two classes, a trend of *open-source* devices is developing (e.g. the IoTaWatt projects or solutions based on ESP32 or Raspberry Pi microcontrollers) [4, 5, 6]. Nevertheless, these devices are usually based on non-invasive current measurement using clamp-on transformers (*CT clamp*). Although convenient to install, this solution requires relatively expensive, good-quality measurement components, which significantly increases the total cost of building the system. Moreover, the signal from the transformers is most often fed to simple analog-to-digital converters built into popular microcontrollers. The limited resolution and relatively low sampling frequency of such circuits make it impossible to accurately determine parameters such as THD or to perform precise phase analysis using advanced algorithms (e.g. SOGI-PLL).

The above analysis reveals a clear need to design and build an inexpensive, integrated power quality meter based on an *open-source* architecture. In order to address the described limitations, this thesis opted for precise series measurement using shunt resistors and isolated Sigma-Delta modulators. The use of such converters ensures a very high sampling frequency. As a result, it is possible to flawlessly record and take into account in the calculations the higher harmonics of the distorted current and voltage waveforms. This hardware approach, combined with advanced digital signal processing (DSP), enables very accurate measurement of grid parameters and flexible communication, e.g. via an Ethernet interface.

1.3 Aim and scope of the thesis

The main aim of this thesis is to design, build and program an electronic device capable of precise measurement and analysis of power quality parameters for a single-phase circuit (including, among others, voltage, current, active power P, reactive power Q, apparent power S, total harmonic distortion THD, frequency f, and power factors $\cos(\phi)$ and PF).

Achieving this aim required the completion of a number of design and research tasks. The scope of the thesis included:

- selection of electronic components, with particular emphasis on isolated measurement converters and a high-performance microcontroller,
- design of the schematic diagram and the printed circuit board (PCB) of the device,
- assembly of the components and commissioning of the hardware,
- implementation of the software in C using the FreeRTOS real-time operating system, including, among others, support for the hardware digital filters (DFSDM) and the grid synchronisation algorithm (SOGI-PLL),
- commissioning of the network stack and communication protocols using the Ethernet interface,
- calibration of the measurement channels,
- verification of the correctness of the measurements by comparing the results with a professional certified Fluke analyzer,
- conducting practical tests consisting in the analysis of the power consumption of selected household electrical appliances.

1.4 Structure of the thesis

This thesis is divided into eight main chapters. After the Introduction, the second chapter discusses the theoretical foundations concerning power quality and the algorithms and protocols used. The third chapter presents the general concept and assumptions of the entire system. The fourth chapter contains the hardware specification, describing the selection of components and the PCB design. The fifth chapter is devoted to the description of the implemented software (firmware). The sixth chapter presents the calibration process and the metrological verification of the built meter. The seventh chapter contains the results of the tests and the analysis of the power consumption of selected loads. The thesis ends with the eighth chapter, which contains a summary of the achieved results and proposals for the further development of the project.

Chapter 2

Analysis of the topic

2.1 Power quality parameters and power theory

2.1.1 Definition and phenomena degrading power quality

Power quality (PQ) is a set of parameters describing the properties of voltage and current in the power grid, defining the degree of their deviation from an ideal sinusoidal waveform of constant amplitude and frequency. In an ideal AC power supply system, the voltage should have the shape of a pure sine wave; however, under real operating conditions it is subject to numerous distortions.

The most common phenomena degrading power quality, defined, among others, by the standards PN-EN 50160 [1] and IEEE 1459 [7], include:

- **Frequency variations** – deviations from the nominal value of 50 Hz, resulting from an imbalance of active power between generation and demand in the power system.
- **Voltage dips and swells** – short-duration (lasting from fractions of a second to several minutes) decreases or increases in the RMS value of the voltage, often caused by switching on large loads or short-circuits in the grid.
- **Flicker (light flicker)** – the subjective impression of instability of illumination caused by rapid fluctuations of the supply voltage.
- **Higher harmonics** – components whose frequencies are multiples of the fundamental frequency (for Europe $n \cdot 50$ Hz), distorting the time-domain waveform of the signal.

A particular challenge in the era of modern electronics is the presence of higher harmonics, which are a direct consequence of the widespread use of non-linear loads such as switched-mode power supplies, LED lighting, photovoltaic inverters or converter-fed drives.

2.1.2 Powers in linear and non-linear circuits

The classical AC power theory for linear circuits operates on three fundamental quantities, which can be illustrated by means of the so-called flat power triangle. These are: active power (P), defining the actual work performed by the load; reactive power (Q), associated with the exchange of energy in the magnetic and electric fields of reactive elements; and apparent power (S), being the geometric sum of the previous two ($S^2 = P^2 + Q^2$).

The above relation is, however, true only for ideal, sinusoidal voltage and current waveforms. In the case of non-linear loads drawing a distorted current, the classical power triangle ceases to apply. According to the theory of Constantin Budeanu from 1927, an additional component, called distortion power (or deformation power) and denoted as D, appears in distorted circuits. It results from the presence of higher harmonics [8].

The relationship between these powers can be illustrated by means of the power cuboid (Fig. 2.1). In this approach, the total apparent power S constitutes the main diagonal of the cuboid, and the power equation takes a spatial form:

$$S^2 = P^2 + Q^2 + D^2 \tag{2.1}$$

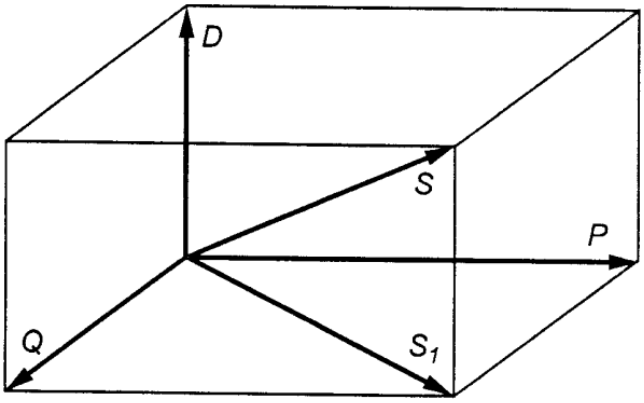


Figure 2.1: Power components in an electrical circuit with distorted waveforms (power cuboid). Source: [8]

According to the theory formulated by S. Fryze, the components that do not participate in the useful conversion of energy (i.e. reactive power and distortion power) can be grouped into a single quantity called non-active power N (often denoted in Western literature also as Q_{tot}). This relationship is defined by the equation:

$$N = \sqrt{S^2 - P^2} \tag{2.2}$$

In metrological practice, many commercial power quality analysers indicate the value of N as the resultant reactive power of a non-linear circuit [7, 9].

2.1.3 Power factors and harmonic content (THD)

The presence of distortion power means that the traditionally understood power factor $\cos(\varphi)$ becomes insufficient for assessing the energy efficiency of a device. It is therefore necessary to distinguish between two concepts:

- **Displacement power factor (DPF** – Displacement Power Factor) – relates solely to the fundamental component (50 Hz) and corresponds to the classical $\cos(\varphi_1)$. It describes the phase-shift angle between the first harmonic of the voltage and the first harmonic of the current.
- **True power factor (TPF** – True Power Factor) – defined as the ratio of the total active power to the total apparent power (P/S). It accounts for both the phase shift and the signal distortion [10].

The measure of waveform distortion is the total harmonic distortion THD (Total Harmonic Distortion). It is defined as the ratio of the RMS value of the higher harmonics to the RMS value of the fundamental component:

$$THD = \frac{\sqrt{\sum_{h=2}^{\infty} X_h^2}}{X_1} \tag{2.3}$$

The relationship between the individual factors is defined by the parameter DF (Distortion Factor), linking DPF and TPF by the relation: $TPF = DPF \cdot DF$ [10].

2.1.4 Methodology for calculating parameters using orthogonal signals

For the correct determination of the above-mentioned parameters, especially the fundamental components, the use of advanced signal processing is indispensable. The SOGI algorithm used in the project (whose detailed theory is discussed in subsection 2.3) makes it possible to generate, on an ongoing basis, two orthogonal signals for the current and voltage: an in-phase signal (α axis) and a signal shifted by 90° (β axis). This makes it possible to avoid the need to use the highly computationally expensive FFT algorithm.

In the developed measurement device, the parameters are calculated cyclically, and their averaging is performed on the basis of a time window T equal to 10 periods of the fundamental signal, which is consistent with the recommendations of metrological standards [9].

The total RMS values of the voltage (U_{rms}) and current (I_{rms}) are calculated from the definition on the basis of the collected discrete samples u_x and i_x:

$$U_{rms} = \sqrt{\frac{1}{T}\int_0^T u^2(t)dt}, \quad I_{rms} = \sqrt{\frac{1}{T}\int_0^T i^2(t)dt} \tag{2.4}$$

The RMS values solely for the fundamental component (U_{1rms}, I_{1rms}) are determined on the basis of the orthogonal output signals from the SOGI blocks (u_α, u_β and i_α, i_β) [11]:

$$U_{1rms} = \frac{1}{\sqrt{2}}\sqrt{u_\alpha^2 + u_\beta^2}, \quad I_{1rms} = \frac{1}{\sqrt{2}}\sqrt{i_\alpha^2 + i_\beta^2} \tag{2.5}$$

The total active power (P), responsible for supplying useful energy and accounting for all harmonics:

$$P = \frac{1}{T}\int_0^T u(t) \cdot i(t)dt \tag{2.6}$$

The active (P_1) and reactive (Q_1) powers for the fundamental component are calculated directly from the signals in the $\alpha\beta$ frame [12]:

$$P_1 = \frac{1}{2}(u_\alpha i_\alpha + u_\beta i_\beta) \tag{2.7}$$

$$Q_1 = \frac{1}{2}(u_\beta i_\alpha - u_\alpha i_\beta) \tag{2.8}$$

According to recognised methodologies, only the fundamental reactive power Q_1 is used to assess the nature of the load and its compensation by means of classical capacitor banks, since the higher harmonics do not participate in the classical exchange of reactive energy [9]. Bearing in mind, however, the total burden of the grid with non-productive power, the system also computes the total non-active power (N), often treated as an indicator of the resultant reactive burden:

$$N = \sqrt{S^2 - P^2} \tag{2.9}$$

This enables precise diagnostics – if N is significantly higher than Q_1, the load on the grid is dominated by distorted currents. The total apparent power S and the total power of the fundamental component amount to:

$$S = U_{rms}I_{rms}, \quad S_1 = U_{1rms}I_{1rms} \tag{2.10}$$

The total harmonic distortion factors for the voltage and current are calculated from a transformed formula that uses the previously determined RMS values [10]:

$$THD_V = \frac{\sqrt{U_{rms}^2 - U_{1rms}^2}}{U_{1rms}}, \quad THD_I = \frac{\sqrt{I_{rms}^2 - I_{1rms}^2}}{I_{1rms}} \tag{2.11}$$

The true displacement power factor (PF_1, corresponding to DPF) and the distortion factor (DF) amount to [10]:

$$PF_1 = DPF = \frac{P_1}{S_1} \tag{2.12}$$

$$DF = \frac{1}{\sqrt{1+THD_V^2}} \times \frac{1}{\sqrt{1+THD_I^2}} \tag{2.13}$$

Finally, the total, true power factor of the device (TPF) is calculated as [10]:

$$TPF = PF_1 \cdot DF \tag{2.14}$$

2.2 Isolation and signal processing using Sigma-Delta converters

In the field of measurement of electrical quantities, traditional analogue channels dominated for many years. Today, however, the aim is to digitise the signal as early as possible, which makes it possible to minimise the influence of electromagnetic interference and phenomena such as the temperature drift of discrete components. One of the newer solutions used for this purpose is the analog-to-digital converter (A/D) based on the Sigma-Delta ($\Sigma\Delta$) architecture. In contrast to classical converters (e.g. those using successive approximation – SAR), $\Sigma\Delta$ circuits transfer a significant part of the processing burden from the analogue domain to the digital domain [13].

The complete processing channel in the $\Sigma\Delta$ architecture consists of two main blocks: an analogue modulator and a digital decimation filter. In many modern industrial applications, these blocks are physically separated (Fig. 2.2). The modulator is located in a dedicated integrated circuit close to the measurement point, while the data stream is transmitted to a digital circuit (e.g. a microcontroller with a built-in hardware filter, a DSP or an FPGA), where the filtering and decimation process takes place [14].

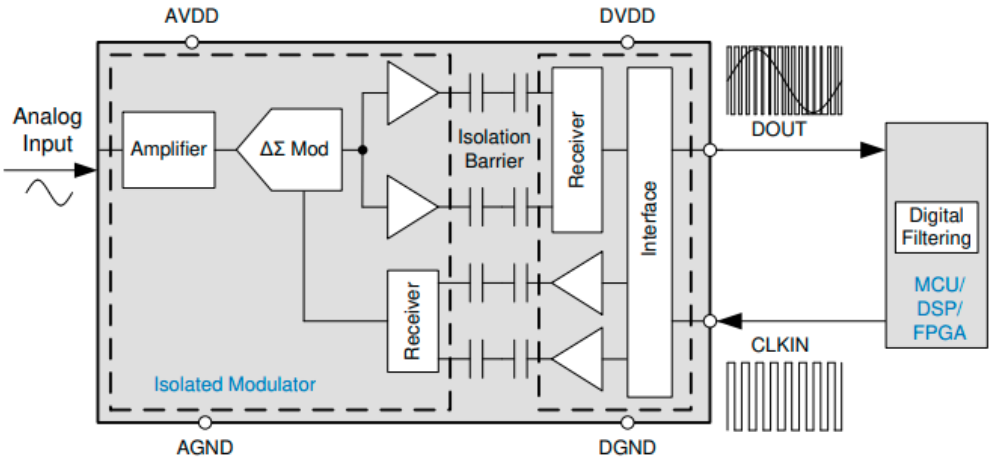


Figure 2.2: Block diagram of an isolated Sigma-Delta converter connected to a microcontroller. Source: [14]

2.2.1 The modulator and the noise-shaping phenomenon

The heart of the analogue part of the converter is the ΣΔ modulator. Its task is to sample the input analogue signal at a frequency (f_s) considerably higher than the final output frequency of the converter (f_d). This technique is called oversampling.

The structure of a classical first-order modulator (Fig. 2.3) consists of a difference amplifier (summing node), an integrating block (integrator) and a single-bit quantiser (comparator) [13]. The input signal reaches the summing node, where the value originating from the single-bit digital-to-analog converter (D/A) located in the negative-feedback loop is subtracted from it. The resulting error signal undergoes integration. The voltage at the integrator output rises or falls at a rate dependent on the magnitude of the error. This signal is then compared with a reference value by the comparator, which generates at its output a logic state of high (1) or low (0) voltage.

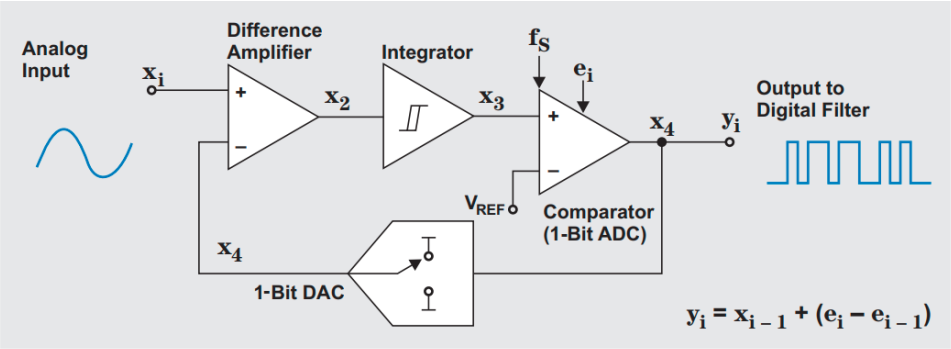


Figure 2.3: Structure of a classical first-order ΣΔ modulator in the time domain. Source: [13]

The output of the modulator is a very fast, single-bit data stream whose duty cycle is proportional to the amplitude of the input signal (Fig. 2.4). For example, if the value of the input signal is 0 V (relative to the reference voltage), the output stream will consist of alternating zeros and ones, giving an average value at the level of 50% [15].

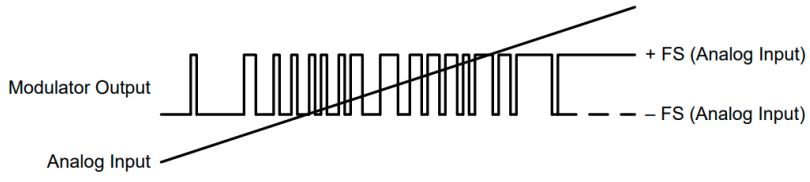


Figure 2.4: Principle of generation of the bit stream (bitstream) corresponding to the input analogue signal. Source: [15]

A key property of the ΣΔ architecture is the noise-shaping mechanism. The use of an integrator makes the feedback loop act as a low-pass filter for the useful signal, but at the same time as a high-pass filter for the quantisation error. As a result, the quantisation

noise is shifted in the frequency domain from the useful band (low frequencies) to the high-frequency band, close to the sampling frequency f_s [13].

To remove the quantisation noise from the useful band even more effectively, higher-order modulators are used (containing two or more integrators connected in cascade). As illustrated in Fig. 2.5, they push the quantisation error into the high-frequency region much more strongly. The price for this improvement is, however, an increase in the noise level in the higher parts of the spectrum and a greater degree of hardware complexity of the circuit.

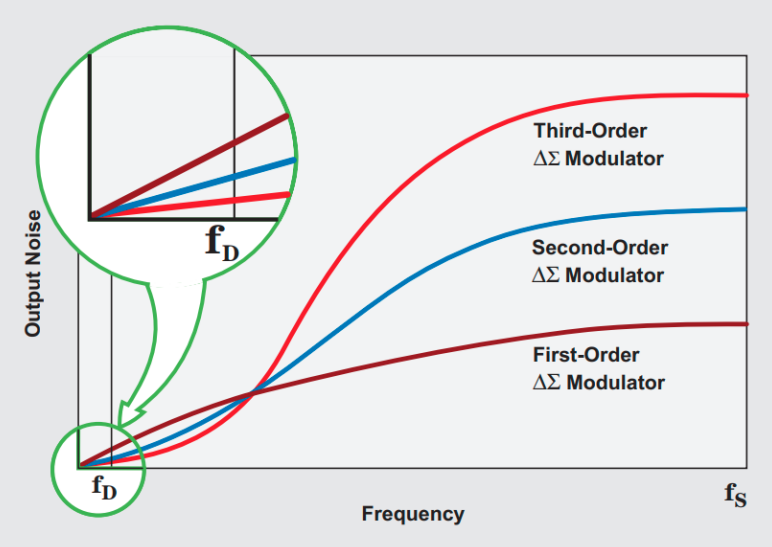


Figure 2.5: The noise-shaping phenomenon as a function of the order of the ΣΔ modulator. Source: [13]

2.2.2 Digital filtering and decimation

The bit stream generated by the modulator must be processed in order to obtain a high-resolution, multi-bit result. This is accomplished by means of a digital low-pass filter and a decimation process [16].

One of the most widely used solutions is the *Sinc* filter (often implemented as a comb filter, Cascaded Integrator-Comb). In its simplest form, this algorithm acts as a moving-average filter, counting and averaging the logic states of the ones in the bit sequence over a defined time window. Decimation is inextricably linked with the filtering process. It reduces the very high sampling frequency of the modulator (f_s) to the useful output data frequency (f_d). The ratio of these two frequencies is referred to as the oversampling ratio OSR (Oversampling Ratio):

$$OSR = \frac{f_s}{f_d} \tag{2.15}$$

Discarding samples in the decimation process does not lead to a loss of information, because the signal has previously been suitably low-pass filtered [16].

In order to obtain better attenuation for frequencies above half the output frequency ($f_d/2$), filters are connected in cascade, yielding higher-order filters, e.g. $Sinc^3$ or $Sinc^5$ (Fig. 2.6).

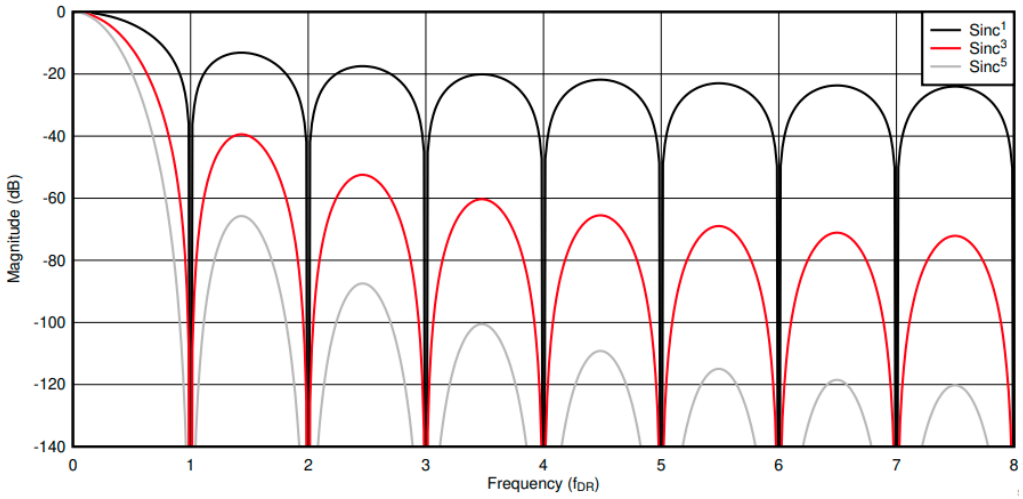


Figure 2.6: Frequency response of digital filters of the $Sinc^1$, $Sinc^3$ and $Sinc^5$ types. Source: [17]

Higher-order filters provide considerably stronger attenuation of the high-frequency quantisation noise. The price for a higher filter order is, however, an extended settling time, which introduces a greater delay in the measurement channel in response to a sudden change in the signal (Fig. 2.7) [17].

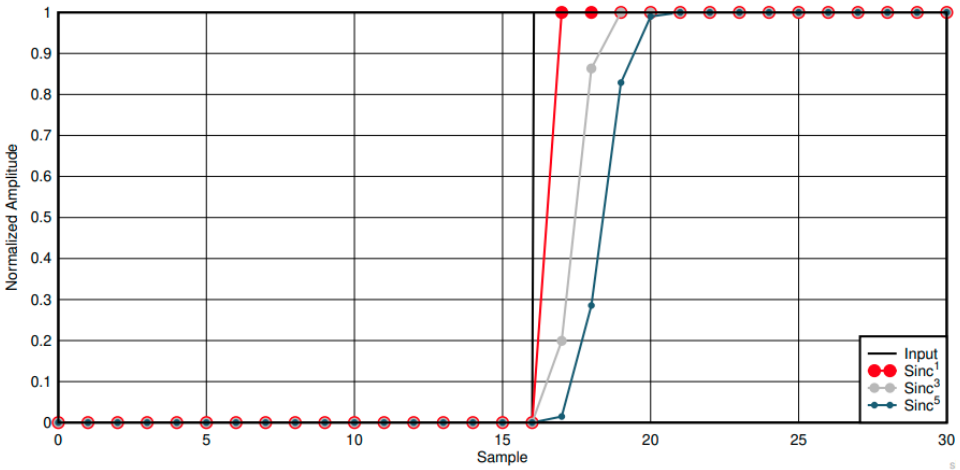


Figure 2.7: Step response in the time domain for filters of the $Sinc^1$, $Sinc^3$ and $Sinc^5$ types. Source: [17]

The selection of the OSR parameter and the filter order constitutes an engineering trade-off. Increasing the OSR makes it possible to obtain a very high measurement resolution (an increase in the ENOB parameter – Effective Number of Bits), which, however, directly results in a narrowing of the measurement bandwidth and a slowdown of the

circuit's operation. An example compilation of the relationships between OSR, accuracy and delay is presented in Table 2.1.

Table 2.1: Relationship between the OSR ratio, the ENOB resolution, the settling time, the delay and the bandwidth for the $Sinc^3$ filter (for a 20 MHz clock). Source: [14]

OSR	**ENOB [bits]**	**Settling time [μs]**	**Delay [μs]**	**Bandwidth [kHz]**
8	4.65	1.2	0.6	1250.00
16	7.57	2.4	1.2	625.00
32	10.02	4.8	2.4	312.50
64	12.30	9.6	4.8	156.25
128	13.51	19.2	9.6	78.13
256	14.11	38.4	19.2	39.06
512	14.39	76.8	38.4	19.53

Thanks to the physical separation of the analogue and digital parts, it is possible on the microcontroller side to process the same data stream in parallel. The bitstream from a single modulator can be processed by several filter modules configured in different ways. In practice, this makes it possible to create a precision channel (high OSR) for accurate power quality measurements and an independent, parallel low-latency channel (low OSR), used, for example, for the immediate detection of short-circuit currents.

2.2.3 Galvanic isolation of the measurement barrier

In measurement systems connected to the power grid, it is necessary to provide galvanic isolation. An architecture in which the transmitted signal is already a modulated bit stream greatly facilitates this task. In contrast to analogue circuits, digital transmission protects the information from degradation when crossing the isolation barrier [14].

This isolation is implemented, among others, on the basis of a capacitive barrier integrated inside the structure of the integrated circuit (Fig. 2.2). The digital signal is modulated onto a high-frequency carrier (e.g. by means of OOK – On-Off Keying), transmitted through the built-in isolation capacitors and demodulated on the low-voltage side [15]; the signal modulation is presented in Fig. 2.8. This guarantees high measurement precision while at the same time providing excellent immunity to rapid changes in the common-mode voltage (Common-Mode Transient Immunity – CMTI).

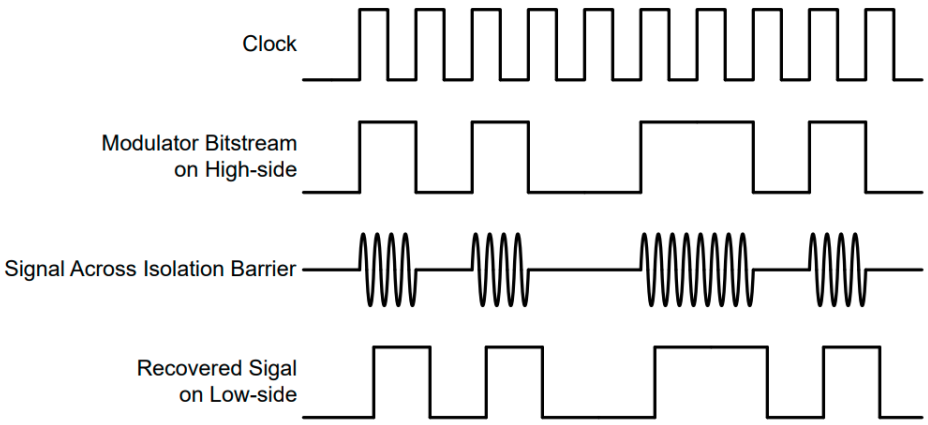


Figure 2.8: OOK modulation scheme. Source: [15]

2.3 Algorithms for synchronisation with the power grid

2.3.1 The necessity of synchronisation and the classification of methods

In measurement systems connected to the power grid, a key issue is the precise determination of the current frequency and the phase angle of the grid voltage. According to the standard PN-EN 61000-4-7 (specifying methods of measuring harmonics and power quality parameters), the determination of parameters such as active power or the THD factor should be carried out by averaging the results over a time interval corresponding exactly to 10 periods of the fundamental grid frequency (which for 50 Hz amounts to 200 ms). Since the frequency in the power system is subject to continuous fluctuations, a rigid setting of the measurement window would lead to the so-called spectral leakage phenomenon and errors in numerical calculations [18]. For this reason, it is necessary to dynamically synchronise the measurement algorithms with the current phase angle of the voltage.

Synchronisation methods can be divided into two main categories: open-loop systems and closed-loop feedback systems. The simplest open method is the detection of the zero crossing of the signal (Zero-Crossing Detection – ZCD). Although it is easy to implement, it is characterised by very poor dynamics (the information is updated only twice per period) and is highly susceptible to interference, noise and voltage dips, which can generate false zero crossings [18].

Considerably better parameters are offered by closed-loop methods, which include phase-locked loops PLL (Phase-Locked Loop), frequency-locked loops FLL (Frequency-Locked Loop) and adaptive filters. The ability to continuously track the signal and adapt to dynamically changing conditions means that, in converter and measurement applica-

tions, PLL algorithms have become an industrial standard [19].

2.3.2 Operating principle of the SOGI-PLL algorithm

The implementation of the PLL algorithm in three-phase systems is relatively simple, because the availability of three phase-shifted signals makes it possible to directly apply the Clarke and Park transformations in order to move to the rotating dq reference frame. In single-phase systems, there is only one reference signal. For this reason, the single-phase algorithm must first generate a virtual orthogonal signal (shifted by 90° with respect to the original signal) in order to create a two-phase $\alpha\beta$ frame. An advanced and highly effective way of accomplishing this task is the use of a second-order generalised integrator, hence the name of the entire algorithm: SOGI-PLL (Second Order Generalized Integrator - Phase Locked Loop) [19].

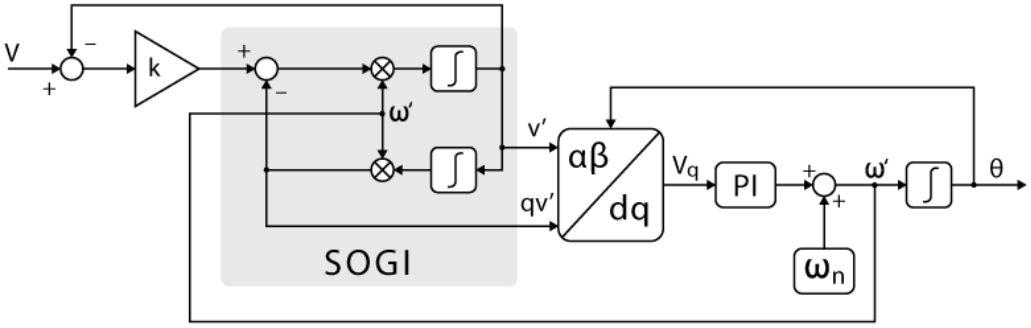


Figure 2.9: General block diagram of a single-phase SOGI-PLL loop. Source: [19]

As shown in the block diagram (Fig. 2.9), the SOGI-PLL structure consists of two main blocks: the orthogonal signal generator (SOGI) and the phase-locked loop proper. The SOGI block, through a system of feedback loops and two integrators, processes the input voltage v and generates two output signals:

- v' (α axis) – a signal in phase with the input signal, band-pass filtered,
- qv' (β axis) – a signal delayed in phase by 90°, low-pass filtered.

The obtained orthogonal signals v' and qv' are subjected to the Park transformation, which transfers them from the stationary $\alpha\beta$ reference frame to the rotating dq frame. The phase angle θ necessary to perform this transformation comes from the output of the feedback loop.

In a correctly synchronised dq frame, the component V_d corresponds to the amplitude of the grid voltage, while the component V_q represents the phase error. Accordingly, the component V_q is fed to the input of a proportional-integral (PI) controller. The task of the controller is to bring the value of V_q to zero. The control signal from the PI controller represents the angular-frequency deviation $\Delta\omega$, which is added to the nominal grid angular

frequency ω_n (for 50 Hz, $\omega_n \approx 314.15$ rad/s). Finally, the obtained angular frequency ω' is integrated, which makes it possible to obtain the estimated phase angle θ, thus closing the error loop [18].

2.3.3 Filtering properties and transfer functions of the SOGI block

An advantage of the SOGI block, apart from the generation of the orthogonal signal, is its natural ability to filter interference and higher harmonics. By analysing the structure of the SOGI block, two operator transfer functions can be determined that describe the ratio of the generated signals to the input signal:

$$D(s) = \frac{v'}{v}(s) = \frac{k\omega' s}{s^2 + k\omega' s + \omega'^2} \tag{2.16}$$

$$Q(s) = \frac{qv'}{v}(s) = \frac{k\omega'^2}{s^2 + k\omega' s + \omega'^2} \tag{2.17}$$

where ω' denotes the currently estimated fundamental angular frequency of the grid, and k is the damping factor (the gain of the SOGI loop).

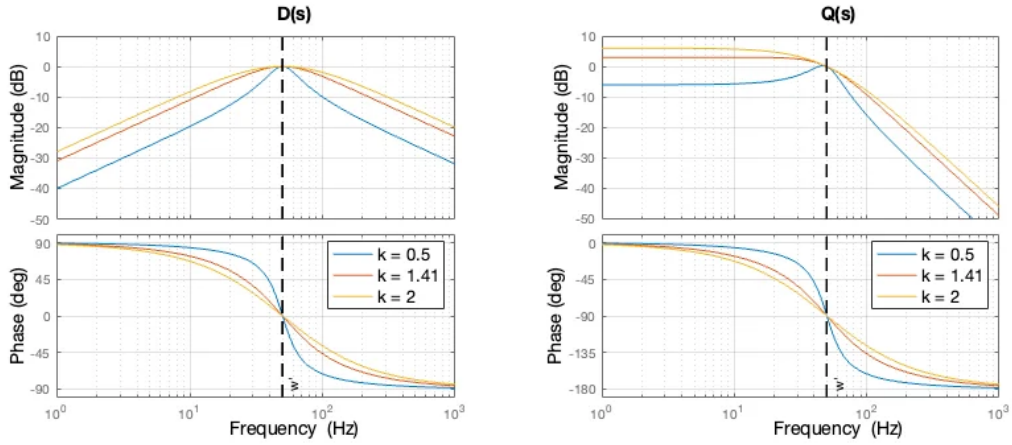


Figure 2.10: Frequency characteristics (Bode plots) of the transfer functions $D(s)$ and $Q(s)$ for different values of the parameter k. Source: [19]

The magnitude-phase characteristics (Bode plots) for both transfer functions are presented in Fig. 2.10. The transfer function $D(s)$ has the character of a band-pass filter (BPF) with unity gain (0 dB) and zero phase shift exactly at the point of the resonant angular frequency ω'. In turn, the transfer function $Q(s)$ acts as a second-order low-pass filter (LPF), introducing unity gain and the required phase shift equal to -90° for the fundamental frequency.

The coefficient k determines the width of the pass band and the dynamics of the system's response. Reducing the value of k improves frequency selectivity (stronger attenuation of interference) but considerably slows down the system's response to step

changes in the grid voltage. In practical applications, the value $k = \sqrt{2}$ has come to be commonly used, which corresponds to a damping factor of $\zeta = 0.707$, guaranteeing an optimal compromise between the settling time and the overshoot (a response similar to that of a Butterworth filter) [19].

2.3.4 Practical implementation of the orthogonal channel for current and voltage

In a typical measurement application analysing power flows, it is necessary to generate orthogonal signals for both the voltage and the current. Theoretically, this would require the use of two independent, complete synchronisation loops (e.g. SOGI-PLL) for each of these signals. In practice, however, this is a suboptimal and error-prone solution.

The voltage signal in a stiff power grid usually retains a shape close to a sine wave, which guarantees stable frequency estimation. In turn, the current signal, especially in the case of non-linear loads (e.g. rectifier circuits), is often heavily distorted. For this reason, an asymmetric architecture was used in the project, consistent with the methodology described, among others, in the literature on smart energy meters [10].

A full SOGI-PLL loop was implemented in the voltage channel. Its main task is to reconstruct a stable phase angle θ and to estimate the current grid angular frequency ω. The angular frequency determined in this way is then fed directly to the second SOGI block, responsible solely for generating the orthogonal signals i_α and i_β for the measured current. Such an approach, in which the frequency determined from the voltage channel is shared by the remaining SOGI blocks in the system, prevents errors in calculations during transient states, eliminates the risk of losing synchronisation with heavily distorted load currents, and also reduces the computational load of the microcontroller [10].

2.3.5 Limitations and advanced modifications of SOGI

The classical SOGI-PLL structure has certain limitations when the grid voltage is heavily distorted. The main problem is the influence of the DC component (DC offset). By analysing the transfer function $Q(s)$ (equation 2.17 and Fig. 2.10), it can be noticed that it acts as a low-pass filter and does not attenuate a zero-frequency signal. The appearance of a DC component at the input of the system translates into an error in the phase and frequency estimation, generating oscillations at twice the grid frequency (2ω). The solution to this problem is a modification of the SOGI loop that transforms the filter for the β axis into a band-pass system (the so-called Modified SOGI), which, however, requires additional mathematical operations in the microcontroller [19]. The modified circuit and its magnitude-phase characteristics are presented in Fig. 2.11 and 2.12.

Another problem is the occurrence of strong interference from specific, odd harmonics. In such a case, extensive networks of cross-coupled feedback loops with several SOGI

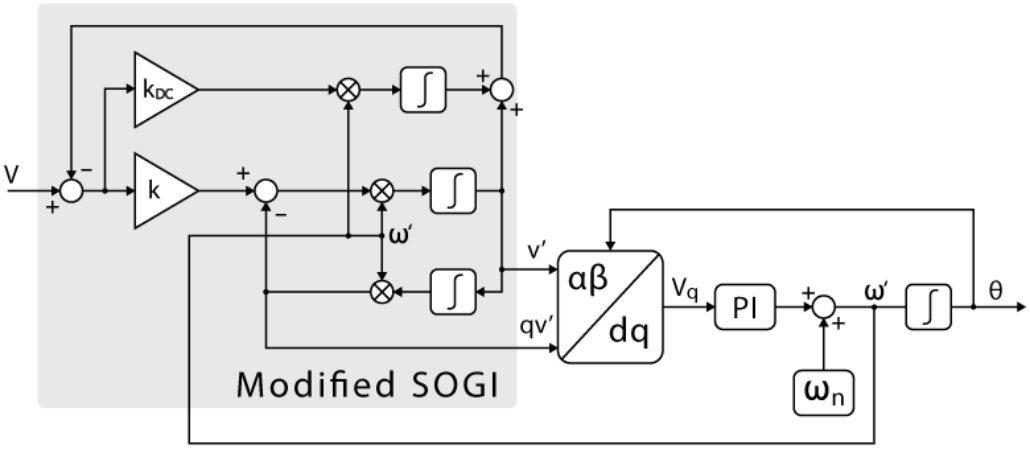


Figure 2.11: General block diagram of a single-phase SOGI-PLL loop modified for DC-component rejection. Source: [19]

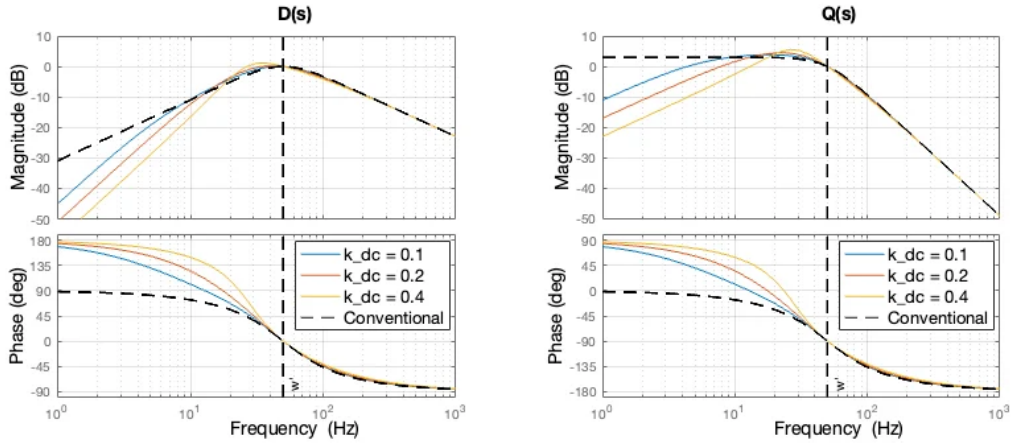


Figure 2.12: Frequency characteristics (Bode plots) of the transfer functions $D(s)$ and $Q(s)$ for the modified circuit with DC-component rejection. Source: [19]

blocks tuned to different frequencies are used (the so-called MSOGI-PLL – Multiple SOGI-PLL). This circuit effectively eliminates the influence of selected harmonics, which, however, takes place at the cost of a drastic increase in computational complexity. The general block diagram is presented in Fig. 2.13.

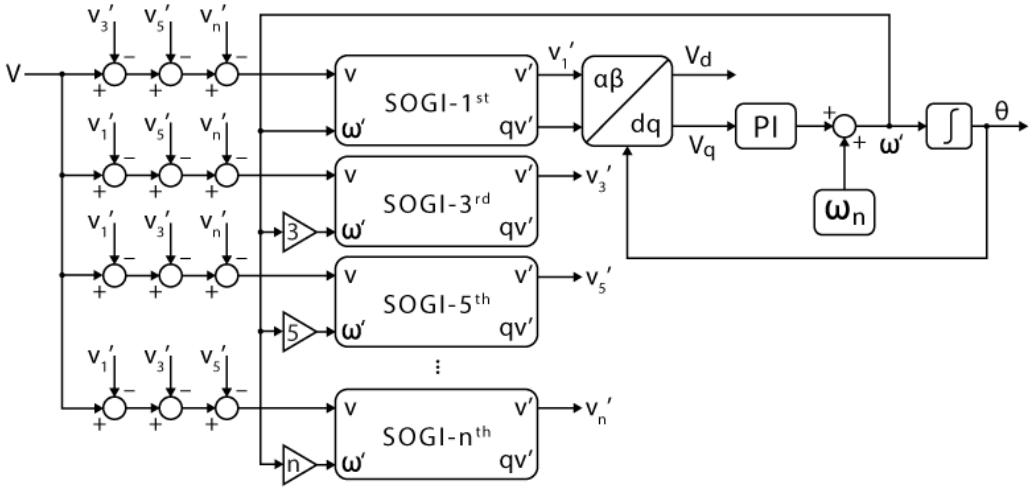


Figure 2.13: Frequency characteristics (Bode plots) of the transfer functions $D(s)$ and $Q(s)$ for the modified circuit with selective harmonic rejection. Source: [19]

Bearing in mind the computational power of the STM32-family microcontroller used in the project, as well as the simplicity of the basic SOGI-PLL structure, the classical structure was implemented in the device. This was deemed sufficient in the context of the assumed design objectives.

2.4 Communication standards and protocols

Modern measurement devices rarely function as isolated units. They are required to provide remote configuration, real-time data reading and integration within larger distributed systems. For this reason, the project uses widely recognised communication standards: the TCP/IP model based on the Ethernet physical layer and the SCPI protocol used for transmitting control commands.

2.4.1 The TCP/IP model and the Ethernet standard

Network communication in the device is based on the TCP/IP model (Transmission Control Protocol / Internet Protocol), which is the foundation of modern computer networks. This model divides the communication process into four cooperating layers: the application layer, the transport layer, the internet (network) layer and the network access (data link) layer [20].

At the lowest level (the network access layer), the Ethernet standard is used. In embedded systems based on microcontrollers, the Ethernet interface is architecturally divided

into two separate blocks: the media access control MAC (Media Access Control) and the physical layer PHY (Physical Layer) circuit [21].

- **MAC** – performs logical functions such as the formation of Ethernet frames, the handling of physical addresses (MAC addressing) and the verification of data integrity by means of checksums (FCS). In modern high-performance microcontrollers, such as the STM32H7 used in the project, the MAC controller is built directly into the structure of the processor.

- **PHY** – is most often an external integrated circuit (transceiver) that is responsible for converting the digital signals from the microcontroller into the appropriate electrical signals transmitted over a copper cable (twisted pair). It performs, among others, signal coding and speed auto-negotiation (usually 10 Mbps or 100 Mbps in the Fast Ethernet standard).

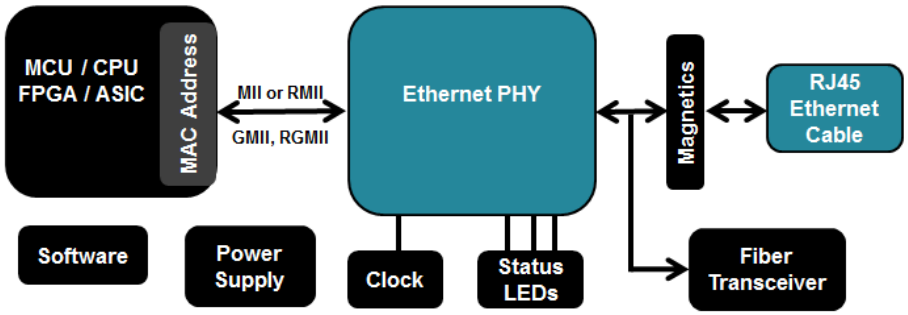


Figure 2.14: Architecture of the connection between the built-in MAC controller and the external PHY circuit by means of the RMII bus. Source: [22]

The exchange of data between the built-in MAC controller and the external PHY circuit takes place by means of standardised digital interfaces. Historically, the MII bus (Media Independent Interface) was used; however, owing to the large number of required signal lines (16 to 18 pins), its reduced version – RMII (Reduced Media Independent Interface), requiring only 7 to 9 signal lines, with a clock signal at a frequency of 50 MHz, is now the standard [21].

The higher layers of the model are implemented in software (e.g. by the LwIP – Lightweight IP stack). The internet layer (the IP protocol) is responsible for the logical addressing of devices in the network and for packet routing, using IP addresses for this purpose. In turn, the transport layer (the TCP protocol) guarantees error-free, ordered delivery of data between devices. The TCP protocol operates in a connection-oriented manner, which means that a session is established before data is exchanged. It is based on the concept of ports, which allows the microcontroller to function as a TCP server, listening for incoming connections on a defined logical port and smoothly transmitting the collected measurement data to the connected client (e.g. a PC) [20].

2.4.2 The SCPI protocol

Remote configuration and changing of the device's operating modes is accomplished by means of the SCPI standard (Standard Commands for Programmable Instruments). This is a unified communication standard introduced in 1990 as an extension of the IEEE 488.2 standard. Today it constitutes a universal control language used by leading manufacturers of measurement instrumentation, such as Keysight, Rohde & Schwarz or Fluke, to operate devices such as oscilloscopes, generators and advanced multimeters [23, 24].

The main advantage of SCPI is its complete independence from the hardware layer. This means that the instructions – which are, in essence, a set of human-readable characters in ASCII code – can be transmitted via any transmission medium, such as a UART serial interface, USB or an Ethernet network (in the TCP layer).

The syntax of the SCPI protocol is based on a hierarchical, tree-like structure of commands, which considerably facilitates their categorisation. The individual nodes of the tree (subsystems) are separated from one another by a colon (`:`). An example configuration command looks as follows:

```
MEASure:VOLTage:DC
```

The standard defines flexible rules regarding letter case. In the official documentation and application notes, a convention has been adopted of writing commands using upper- and lower-case letters (e.g. `MEASure`). The upper-case letters denote the mandatory, abbreviated version of the command, while the part written in lower-case letters may be omitted in order to optimise the size of the transmitted messages [25]. Accordingly, the command `MEASure:VOLTage:DC` will be interpreted by the parser in exactly the same way as its abbreviated form `MEAS:VOLT:DC`. The device on the receiving side treats the commands in a case-insensitive manner.

In SCPI, two basic types of commands are distinguished: setting commands and queries. Queries are characterised by a question mark (`?`) placed at the end of the character string and indicate that the controlling device expects a return transmission of measurement data (e.g. `MEAS:VOLT:DC?`). Commands can also be supplied with arguments, which are separated from the main command by a single space character (e.g. `MEAS:VOLT:DC CHannel1`) [24].

An advanced feature of the SCPI standard, appreciated in metrological applications, is the built-in support for units of measure and exponential notation. This makes it possible to send values with engineering prefixes to the device (e.g. `200MHz` instead of `200000000`) or to precisely read fractions with exponential notation (e.g. `-7.89E-001`) [24].

Owing to the complexity of the hierarchical command tree and the need to parse units, implementing SCPI support from scratch can be time-consuming. For this reason, ready-made *open-source* software is widely used in embedded applications. In the present project, the decoding of character strings was entrusted to the popular *SCPI Parser v2* library

(the so-called *libscpi*), which offers full compatibility with the IEEE 488.2 standard and considerably simplifies the mapping of textual commands onto specific functions written in the C language [26].

Chapter 3

System concept and component selection

3.1 Main design assumptions

The device design process began with the formulation of functional and technical assumptions. The module under development was intended to be a comprehensive solution combining precise analog data acquisition with advanced digital processing and flexible communication. The most important design requirements included:

- implementation of a precise, galvanically isolated measurement of voltage and current for a single phase while maintaining a high sampling frequency,
- powering the entire circuit from a single 5V supply voltage,
- implementation of an Ethernet interface with TCP/IP stack support for fast and reliable transmission of measurement data,
- performing the analysis of power quality parameters (mathematical calculations) directly on the microcontroller, in real time,
- providing optional wired communication via a UART serial port or USB,
- bringing out additional input/output lines (GPIO) for possible control of external devices,
- design and manufacture of a dedicated printed circuit board.

The initial outline of the concept also assumed equipping the circuit with physical control buttons and a local display. The general structure of the system, taking into account all the planned functional blocks, is presented in the block diagram (Fig. 3.1). It should be noted, however, that in the course of the project's evolution, the physical

user interfaces (buttons and display) and the USB interface connector were abandoned. A detailed justification of these modifications is discussed in the chapter summarising the thesis 7.2.

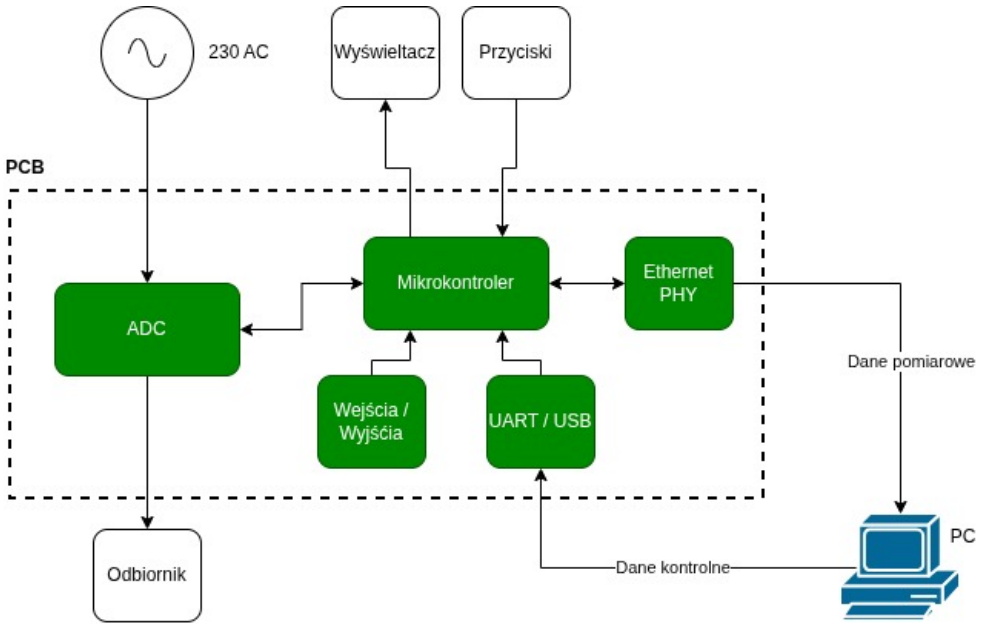


Figure 3.1: General block diagram of the concept of the proposed measurement device. Source: own elaboration.

3.2 Selection of the main hardware components

3.2.1 Isolated Sigma-Delta modulators

For the implementation of the measurement channel, modern, isolated $\Sigma\Delta$ modulators from the AMC family by Texas Instruments were selected (the AMC3336 for the voltage channel and the AMC3306 for the current channel, respectively). These are converters with a double-integrating feed-forward architecture, optimised specifically for precise measurements in power grids.

The greatest advantage of the selected devices is the built-in double galvanic isolation. It covers not only the transmission of the modulated digital signal, but also the power supply (the devices feature an integrated, isolated DC/DC converter). As a result, the high-voltage (measurement) part of the converter does not require a separate power supply – the device independently generates the required voltage (approx. 3.2 V) on the basis of the supply provided from the safe side. This significantly simplifies the design of the power supply on the PCB. The structure of the device (AMC3306M25) is presented in the technical documentation (Fig. 3.2)

These modulators can be clocked with an external clock signal at a frequency of up to 20 MHz. With the use of a $Sinc^3$ digital filter in the microcontroller with a typical

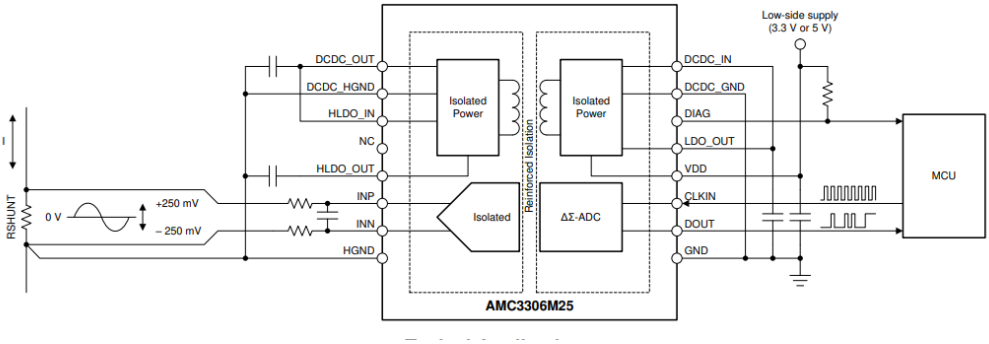


Figure 3.2: Internal structure of the AMC3306M25 device. Source: [27]

oversampling ratio of $OSR = 256$, an output data frequency (the sampling frequency of the useful signal) of approx. 78.125 kHz is obtained. According to the Nyquist-Shannon theorem, this makes it possible to analyse signals with frequencies of up to approx. 39 kHz, which is amply sufficient for measuring the higher harmonics of the grid signal (e.g. the 50th harmonic, equal to 2.5 kHz) [15, 27].

3.2.2 Main microcontroller

A 32-bit microcontroller from the *High-Performance* series – the STM32H750VBT6 by STMicroelectronics, based on the powerful ARM Cortex-M7 core – was selected as the central computing unit. Its choice was determined by a set of unique hardware features that fit the project assumptions perfectly:

- **The DFSDM module** (Digital Filter for Sigma-Delta Modulators) – the microcontroller has a dedicated hardware block for the direct reception and filtering of single-bit data streams from external ΣΔ modulators. This module allows the hardware implementation of *Sinc* filters, supporting up to 8 channels and 4 independent filters.

- **DMA support** (Direct Memory Access) – the filtered measurement data from the DFSDM module is transferred directly to RAM without any involvement of the processor core. The core only receives an interrupt when the data buffer is ready for analysis.

- **Hardware Ethernet MAC controller** – the processor has a built-in MAC module supporting the MII and RMII interfaces, which allows direct and easy connection of an external physical-layer (PHY) device.

- **Computational performance** – the device can be clocked with a signal at a frequency of up to 480 MHz. Moreover, it is equipped with a hardware double-precision floating-point unit FPU (Floating-Point Unit), which is crucial when implementing algorithms such as SOGI filters or real-time RMS value calculation.

An additional advantage was the large capacity of the built-in RAM and the wide availability of a rich development environment for the STM32 family [28].

3.2.3 Ethernet physical-layer device (PHY)

The LAN8740A device by Microchip was selected to handle the physical layer of network communication. It is a popular, industrially proven 10/100 Mbps Ethernet transceiver. This device can communicate with the STM32 microcontroller via the reduced RMII interface, which allows a significant saving of the processor pins compared with the standard MII bus, while maintaining full data-transfer stability in the TCP/IP protocol [29].

3.3 Tool software used

The realisation of the complex project required the use of a number of specialised development environments, supporting both hardware design and embedded software as well as data analysis.

- **KiCad EDA** – free, cross-platform *open-source* software for designing schematic diagrams and printed circuit boards (PCB). This software, developed, among others, with the support of CERN, is characterised by a huge library base and powerful track-routing capabilities, which were used when designing the multi-layer PCB of the meter. The software can be downloaded from [30]

- **STM32CubeMX** – the official graphical tool by STMicroelectronics used for initialising microcontrollers. The program allows easy visual assignment of functions to the physical pins of the processor, configuration of the clock tree, and setting up the peripherals (including Ethernet, DFSDM, DMA). The result of the program's work is a generated C code skeleton, ready for further development. A view of an example configuration in CubeMX is presented in Fig. 3.3.

- **Visual Studio Code (VS Code)** – a lightweight, universal code editor whose functionality was extended with dedicated plugins. The project used the official extension by STMicroelectronics [32], which integrates seamlessly with projects generated by CubeMX, providing compilation, firmware uploading, and live hardware debugging capabilities.

- **Serial Studio** – versatile, flexible *open-source* software for visualising and recording telemetry data. This tool supports various communication protocols, including TCP/IP, UDP, serial ports, and Bluetooth. Its greatest advantage is the ability to create custom visualisation panels based on JSON configuration files. This allows

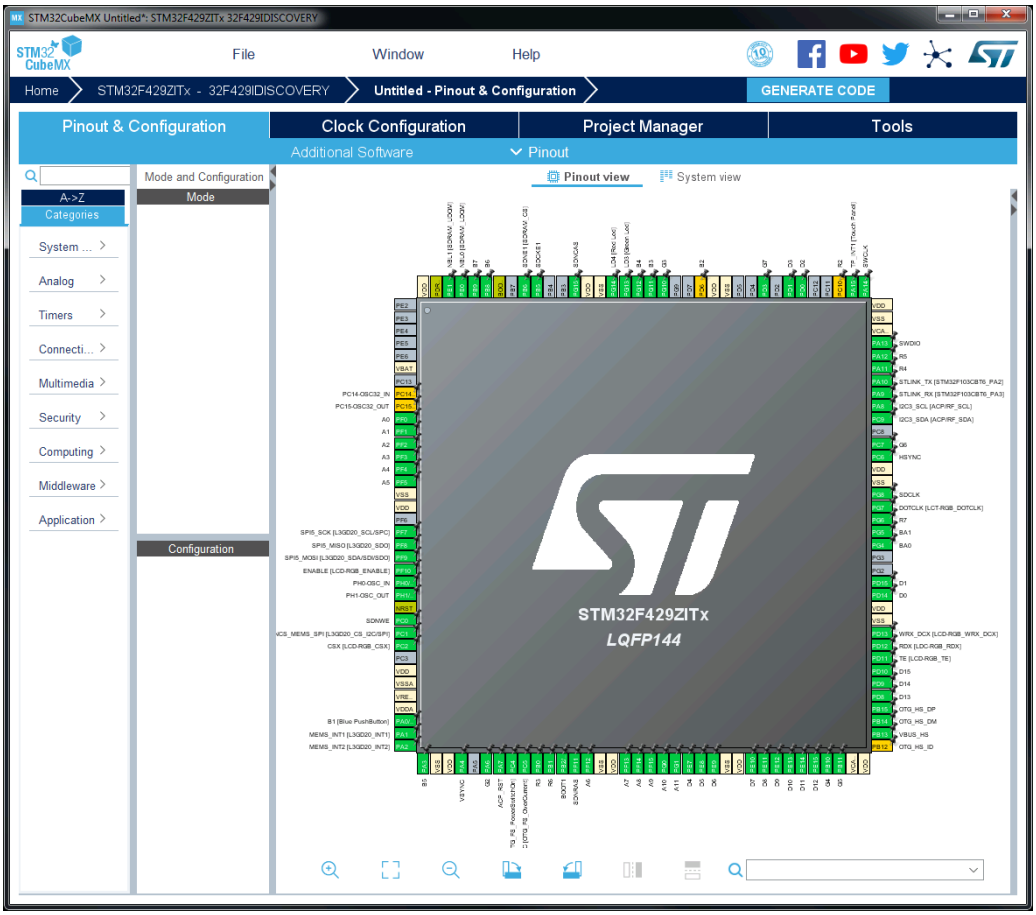


Figure 3.3: Graphical configuration of the peripherals and pins of the STM32H7 microcontroller in STM32CubeMX. Source: [31]

mapping of the non-standard binary or text data frames sent from the microcontroller and displaying them in the form of readable indicators and charts in real time. The program also allows saving the transmitted data in `.csv` files. The free version of the software can be downloaded and compiled from the official GitHub page [33].

- **GNU Octave** – a free computing environment with a syntax largely compatible with MATLAB. Within the thesis it was used to write scripts automating the post-processing process – loading and mathematically analysing historical measurement data saved by the host application in the `.csv` format.

Chapter 4

Hardware design of the device

4.1 Structure of the schematic diagram

The design of the schematic diagram and the printed circuit board (PCB) was carried out using the KiCad environment in version 9.0. A more detailed description of the program can be found in 3.3.

In order to ensure high clarity of the design and to facilitate the analysis of the relationships between the individual subassemblies, the schematic diagram was divided into a hierarchical structure comprising the following functional blocks:

- **Power supply** – the section responsible for filtering, protection, and stabilisation of the supply voltages (5 V and 3.3 V),
- **GRID-ADC (measurement channel)** – the block containing the high-voltage input circuits and the isolated Sigma-Delta modulators from the AMC family,
- **STM32** – the main microcontroller block together with its peripherals,
- **Buttons** – the circuits conditioning the user input signals,
- **Outputs (EXT-OUT)** – the circuit of the controllable digital outputs,
- **Ethernet** – the physical layer of the network interface (PHY device) together with the RJ45 connector,
- **USB-FS** – the USB connector and protection

The general outline of the relationships between the listed functional blocks on the PCB is presented in Fig. 4.1. The complete list of the electronic components used (BOM) is included in Appendix A (Technical documentation).

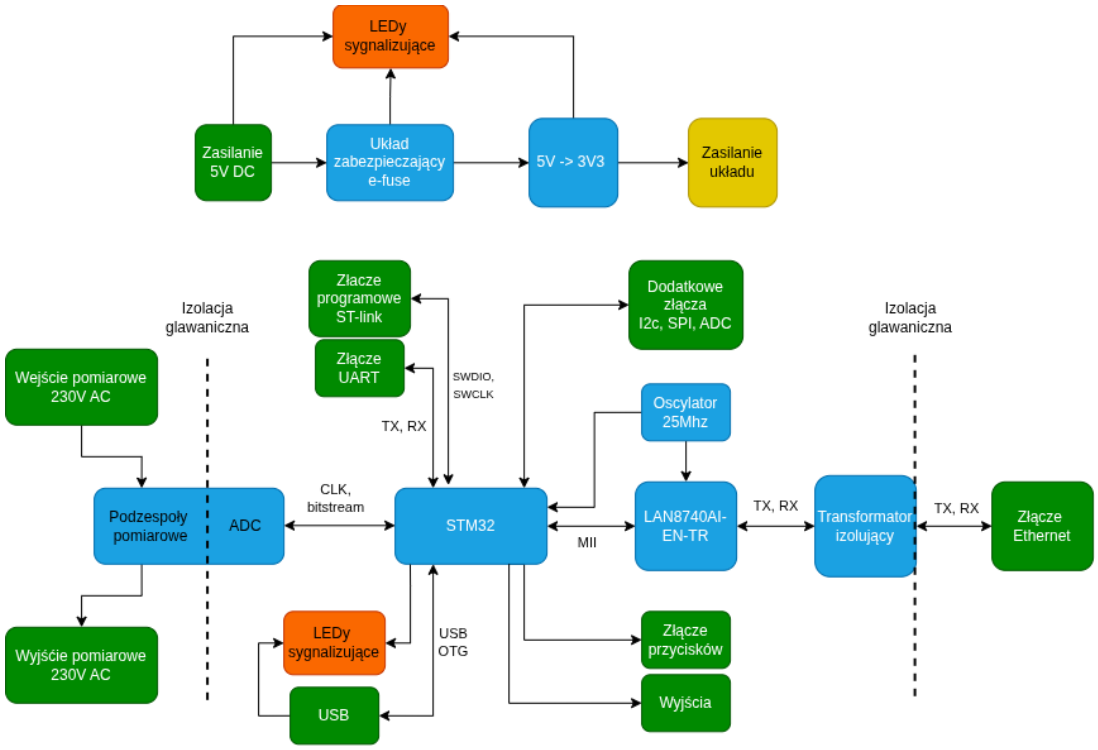


Figure 4.1: Block diagram of the relationships between the modules on the printed circuit board. Source: own elaboration.

4.2 Power supply and protection module

The task of the power supply block is to accept an external voltage with a nominal value of 5 V, to provide protection of the components against fault conditions, and to stabilise the 3.3 V voltage required to power the microcontroller and the digital logic. The schematic of the power supply module is presented in Fig. 4.2.

4.2.1 Input protection – the eFuse circuit

An integrated electronic fuse, the TPS2596 by Texas Instruments, was used at the power supply input. It is a highly integrated device protecting the circuit against overvoltages, undervoltage, overcurrents, and limiting the inrush current. Additionally, the device was equipped with bidirectional transient-voltage-suppression (TVS) diodes at the input and output, protecting against electrostatic discharge (ESD) and fast transients [34].

The operating parameters of the TPS2596 were tightly matched to the system requirements by means of external components. The voltage-protection trip threshold was set by means of a resistive divider connected to the OVLO and EN/UVLO pins. In accordance with the datasheet, the resistor $R_3 = 47$ kΩ was adopted. In order to ensure a safe operating window for the device (shutdown when approx. 6 V is exceeded), the sum of the

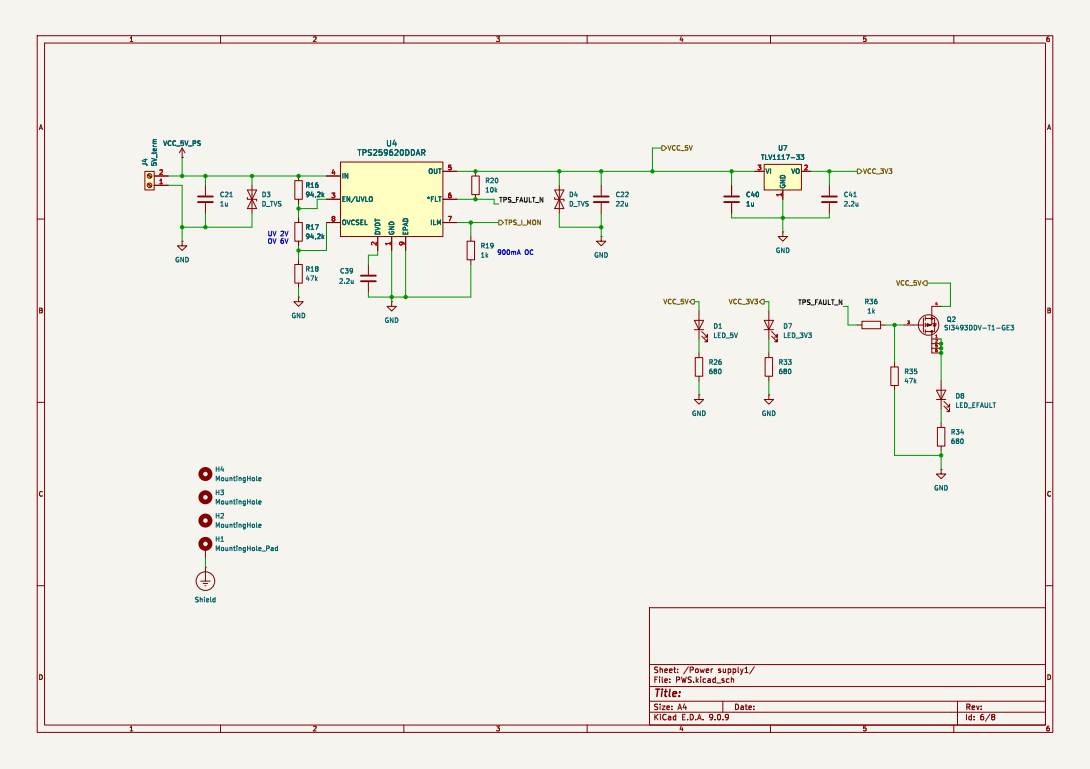


Figure 4.2: Schematic of the device power supply. Source: own elaboration.

resistances was calculated:

$$R_1 + R_2 = 188 \text{ k}\Omega \tag{4.1}$$

Finally, standard resistors from the series were selected:

$$R_1 = R_2 = 94.2 \text{ k}\Omega \tag{4.2}$$

which provided threshold parameters close to the ideal theoretical assumptions.

The overcurrent protection (I_{lim}) was originally planned at the level of 1 A. Using the manufacturer's formula, the required programming resistance was 913.23 Ω. In order to optimise and facilitate component availability, the following resistor was used:

$$R_{lim} = 1 \text{ k}\Omega \tag{4.3}$$

which results in an actual current limit at a safe level:

$$I_{lim} \approx 0.914 \text{ A} \tag{4.4}$$

The device also enables precise control of the output-voltage rise time. On the basis of the capacitance $C_{dvdt} = 2.2$ μF, the following SR coefficient value was obtained:

$$SR = 0.019 \text{ V/ms} \tag{4.5}$$

Assuming no current consumption by the circuit, the total output capacitance on the 5 V line amounts to $C_{out} = 22$ μF; under this assumption, the inrush current (I_{inrush}) is limited to merely:

$$I_{inrush} = 0.42 \text{ mA} \tag{4.6}$$

and the power dissipated in the fuse structure during start-up amounted to approx. 1 mW. The total voltage rise time (t_{rise}) amounts to approximately:

$$t_{rise} = 261 \text{ ms} \tag{4.7}$$

Although this is a relatively long start-up time, an immediate start is not required in the designed measurement device. Such a soft start significantly minimises thermal and electrical stresses in the circuit, which has a positive effect on its reliability.

4.2.2 3.3 V voltage regulator

The main power line of the microcontroller, the PHY device, and the digital part of the converters is supplied with 3.3 V, obtained by means of a fixed-output linear regulator (LDO) – the TLV1117-33 (Texas Instruments). This device is characterised by a maximum output current of 800 mA, which is a value fully covering the current demand of the device [35].

To ensure stable operation of the regulator and to suppress interference, a 1 μF capacitor with a low equivalent series resistance (low ESR, X7R dielectric) was used at its input, while a 2.2 μF capacitor was placed at the output. The selected LDO model has a number of built-in protections, such as internal current limiting, thermal protection, under-voltage lockout (UVLO), and an active output-discharge circuit. Although some of these functions, such as inrush-current reduction, overlap with the tasks already performed by the input eFuse, their cascaded use significantly increases the overall reliability of the entire power supply channel.

4.2.3 Power supply status indication

For diagnostic purposes, optical indicators were implemented in the power supply block. LEDs (together with appropriate limiting resistors) were placed on the 5 V and 3.3 V lines, indicating the presence of the correct operating voltages.

An additional diode associated with the fault output (`FLT`) of the TPS2596 was also provided, intended to give immediate visual indication of the eFuse protection tripping (e.g. in the event of a short circuit). However, at the schematic design stage a logic error was made, which prevented it from functioning correctly. A detailed analysis of this hardware defect and the methods of resolving it are described in the chapter summarising the thesis 7.2.

4.2.4 Mounting holes and earthing

In order to ensure stable mechanical mounting of the module in the target enclosure, the printed circuit board was equipped with four mounting holes, suitable for standard M4 screws.

An important aspect from the point of view of electromagnetic compatibility (EMC) is the way the enclosure and the shields are earthed. One of the mounting holes was galvanically connected to the shield signal (*Shield*) coming from the Ethernet connector. The use of such an arrangement forces the connection of the earthing to the enclosure only at a single, predetermined point. This is a key principle preventing the formation of the so-called ground loop. Multi-point earthing could lead to the flow of unwanted equalising currents and, consequently, to the induction of interference.

4.3 Design of the measurement channel

The main measurement circuit of the device was designed to be connected in series between the power source (the power grid) and the load under test. Screw connectors for the phase (L), neutral (N), and protective earth (PE) conductors were brought out at the input and output of the circuit. The central elements of the channel are two independent, isolated Sigma-Delta modulators: the AMC3336 device dedicated to voltage measurement and the AMC3306M25 device intended for current measurement.

The complete schematic diagram of the entire measurement block is presented in Fig. 4.3. The application of both devices was implemented on the basis of the manufacturer's (Texas Instruments) recommendations. The devices are supplied from the safe (digital) side with a single 3.3 V voltage. The output pins of the built-in converters (DCDC_OUT, HLDO_OUT) and the reference inputs were suitably decoupled with a set of low-ESR capacitors, which ensures the stability of the internal voltages supplying the isolation barrier. The diagnostic signals (`DIAG`), informing about the correct operation of the built-in converters, were pulled up to the supply voltage via resistors.

Both the voltage and the current converter share the same clock signal (CLKIN) generated by the STM32 microcontroller, which guarantees strict time synchronisation of both measurements. The output data streams (DOUT) were routed to independent channels of the hardware DFSDM module inside the microcontroller.

4.3.1 Voltage measurement channel

The AMC3336 converter was used to measure the grid voltage. This device is characterised by a very high input resistance (at the level of 1 GΩ) and a linear differential voltage range of ±1 V. Moreover, it exhibits excellent metrological parameters: a maximum zero-offset error of ±0.3 mV and high immunity to common-mode interference

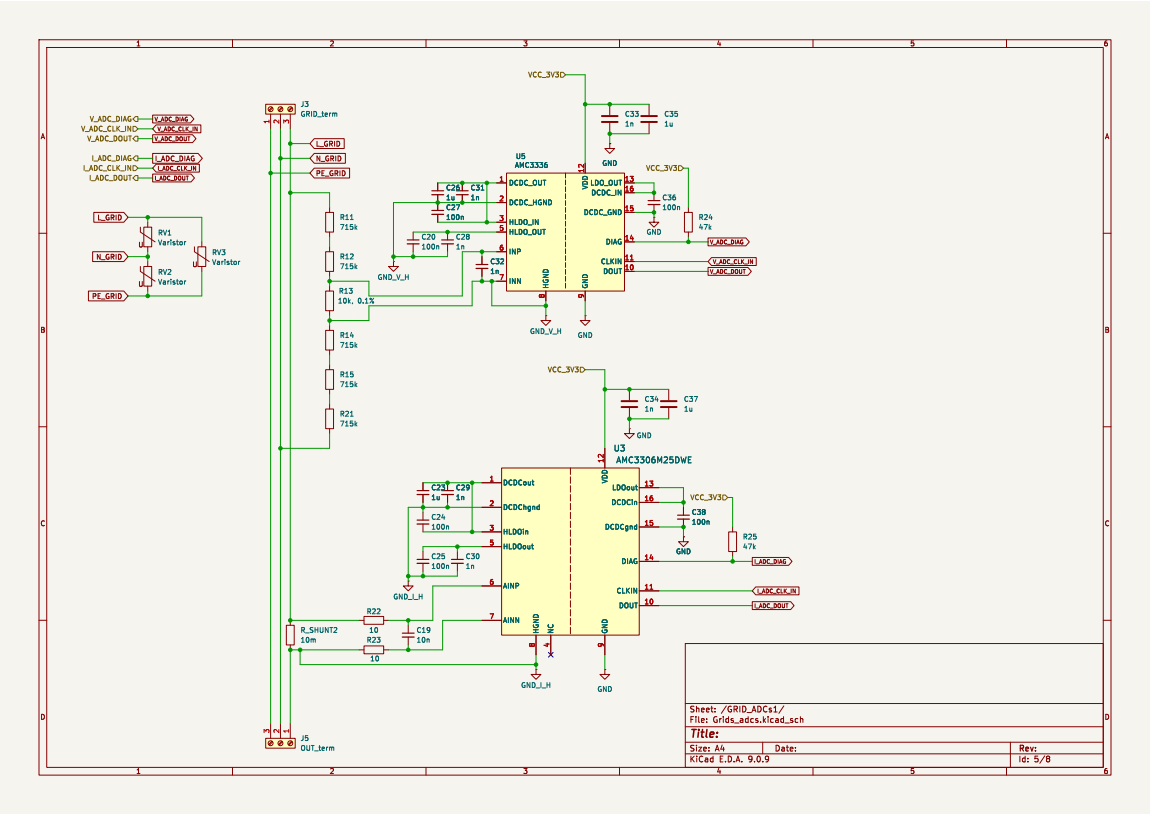


Figure 4.3: Schematic diagram of the voltage and current measurement channel based on converters from the AMC family. Source: own elaboration.

(CMTI of min. 90 kV/μs).

The adaptation of the 230 V AC grid voltage to the converter's input range was implemented by means of a resistive divider. The resistor selection methodology was based on the calculations presented in the manufacturer's datasheet [15]. In accordance with the assumptions, it was adopted that the maximum peak voltage at the device input would be $U_{max} = 360$ V. In order to obtain a maximum safe voltage of $U_{ADC} = 1$ V at the differential input of the converter, the required component values were determined from the following relationships:

$$I_{divider} = \frac{U_{max}}{R_{tot}} \quad \text{and} \quad U_{ADC} = I_{divider} \cdot R_{shunt_v} \tag{4.8}$$

where: $I_{divider}$ – the current flowing through the divider branch, R_{tot} – the total resistance of the divider, R_{shunt_v} – the resistance parallel to the converter input.

Assuming the lower resistance of the divider $R_{shunt_v} = 10$ kΩ, the required total resistance R_{tot} was calculated:

$$R_{tot} = \frac{U_{max} \cdot R_{shunt_v}}{U_{ADC}} = \frac{360 \text{ V} \cdot 10 \text{ k}\Omega}{1 \text{ V}} = 3.6 \text{ M}\Omega \tag{4.9}$$

In order to meet the requirements concerning the maximum operating voltage of individual

SMD components (typically 75 V for 0603 or 0805 packages), the series resistance was implemented as a chain of five resistors connected in series, each with a value of 715 kΩ.

In accordance with the recommendations for a typical application, a filtering capacitor $C_f = 1$ nF [15] was placed in parallel with the measurement resistor (R_{shunt_v}). Together with the divider, it forms a passive (anti-aliasing) low-pass filter, whose cut-off frequency amounts to:

$$f_c = \frac{1}{2\pi \cdot R_{shunt_v} \cdot C_f} = \frac{1}{2\pi \cdot 10\ \text{k}\Omega \cdot 1\ \text{nF}} \approx 15.92\ \text{kHz} \tag{4.10}$$

The cut-off frequency safely attenuates high-frequency interference and eliminates the aliasing phenomenon relative to the modulator clock frequency (of the order of MHz), while not affecting the amplitude and phase of the useful band of the grid signal (0–2.5 kHz for the 50th harmonic).

4.3.2 Current measurement channel

The AMC3306M25 device, whose linear measurement range is ±250 mV, was selected for the acquisition of the load current. The measurement was implemented by the direct method, using a series shunt resistor.

The design assumption was to adapt the current channel to the measurement of loads with a nominal current consumption of $I_{RMS} = 16$ A. The maximum peak current value therefore amounts to:

$$I_{peak} = I_{RMS} \cdot \sqrt{2} = 16\ \text{A} \cdot \sqrt{2} \approx 22.62\ \text{A} \tag{4.11}$$

In order to ensure that the voltage dropped across the shunt does not exceed the linear range of the converter, the maximum permissible value of the resistance R_{shunt} was calculated as:

$$R_{shunt_max} = \frac{U_{AMC_max}}{I_{peak}} = \frac{250\ \text{mV}}{22.62\ \text{A}} \approx 11.05\ \text{m}\Omega \tag{4.12}$$

In the project, it was decided to use the safe value $R_{shunt} = 10$ mΩ. With such a resistance, the theoretical maximum measurable peak current amounts to 25 A (approx. 17.6 A RMS).

A key problem in shunt circuits is the power dissipated in the form of heat. The rated power loss for a current of 16 A amounts to:

$$P_{loss} = I_{RMS}^2 \cdot R_{shunt} = 16^2 \cdot 0.01\ \Omega = 2.56\ \text{W} \tag{4.13}$$

In order to minimise the heating of the printed circuit board, a parallel connection of two specialised wire-wound resistors, each with a value of 20 mΩ, in OAR (Open Air Resistor) technology was used. This type of construction is characterised by the elevation of the resistive element above the laminate, which facilitates heat dissipation by convection. The selected components feature a very low temperature coefficient of resistance (TCR =

20 ppm/°C), which minimises the measurement error associated with the heating of the circuit during continuous operation. It should be noted, however, that a mistake was made at the design stage, the consequences and mitigation methods of which are described in more detail in the concluding chapter 7.2.

Similarly to the voltage channel, an input differential RC filter was implemented in front of the AMC3306 converter input. Its structure and component values were taken directly from the typical application recommended in the manufacturer's datasheet [27]. This filter consists of two series resistors $R_f = 10\ \Omega$ and a capacitor $C_f = 10$ nF.

4.3.3 Protective components (varistors)

At the schematic design stage, the mounting of components protecting the circuit against high-energy overvoltages was planned at the main grid power input. For this purpose, three metal-oxide varistors (MOVs) operating in the L-N, L-PE, and N-PE configurations were provided. Their task was to absorb the voltage spikes generated, among others, when inductive loads are switched off.

Ultimately, however, at the physical assembly stage of the printed circuit board, the decision was made not to mount these components. Metal-oxide varistors are characterised by a relatively large parasitic self-capacitance (of the order of several hundred to even several thousand picofarads). In measurement circuits, the parasitic capacitance connected in parallel with the input could form additional resonant circuits with the line impedance or introduce phase shifts of high-frequency signals that are difficult to compensate. Given that the overriding goal of building the prototype was the verification of the accuracy of the harmonic-analysis (THD) algorithms and the precise estimation of phase angles, this protection stage was deliberately abandoned in favour of maximising the quality of the measured signal.

4.4 Microcontroller module

The central element of the block under discussion is the microcontroller from the STM32 family, whose schematic diagram together with the necessary surroundings is presented in Fig. 4.4. The argumentation behind the choice of this particular device and a detailed description of its parameters were discussed in subsection 3.2.2. Conditioning and filtering circuits as well as the leads of the peripheral interfaces were designed around the microcontroller.

In order to ensure stable operation of the digital circuit, a 3.3 V voltage was routed to all the supply pins (VDD), decoupled by means of local 100 nF filtering capacitors and one central 4.7 μF capacitor. The supply of the analog part of the microcontroller (the VDDA pin) was implemented via a series filter using a ferrite bead. This treatment is

aimed at separating the high-frequency interference originating from the digital part and improving the precision of the built-in A/D converters.

A *tact switch* was connected to the `NRST` pin, enabling a forced hardware reset of the device. The `BOOT0` pin was brought out to a pin header (jumper), which allows it to be biased to the ground potential (GND) or the supply voltage (VCC). This makes it possible to start the microcontroller in the built-in bootloader mode, which is crucial for service purposes – for example, in the case of an incorrect configuration of the clock signals that prevents standard programming of the device.

The clock source is an external oscillator. A 25 MHz device appears in the original schematic; however, in the course of commissioning the prototype, this value was changed to 50 MHz. This modification resulted from the need to provide synchronous clocking of both the microcontroller and the PHY device for the RMII interface. The problem with the original Ethernet-interface clocking assumption was analysed in more detail in the chapter summarising the thesis. 7.2

The schematic includes a number of connectors bringing out the communication and measurement interfaces, including: I2C, SPI, additional ADC inputs, and UART, used mainly for transmitting diagnostic logs. A standard SWD connector (the `SWDIO` and `SWCLK` signals) was provided for programming and hardware debugging of the microcontroller. The remaining GPIO pins, unused in the main application, were brought out to an additional expansion connector. Additionally, for diagnostic purposes, two LEDs were connected to selected digital outputs, serving as visual indicators of the program's operating state.

It should be noted that, in order to maintain the readability of the design documentation, the circuits responsible for handling the buttons, the outputs, the USB connector, and the Ethernet-interface PHY device were separated into separate hierarchical sheets, with which the microcontroller is connected by means of global signal labels.

4.4.1 Input signal conditioning circuits

The project provides support for external user interfaces consisting of three momentary push buttons and an encoder with a built-in button. In total, six digital signals were introduced into the circuit. The schematic diagram of the block under discussion is presented in Fig. 4.5.

An undesirable phenomenon inseparably associated with mechanical contact elements is the so-called contact bounce. In order to eliminate the need for software-level mitigation of this phenomenon, a hardware-based debouncing was chosen. For this purpose, a passive RC low-pass filter was used on each of the six inputs. The circuit consists of 10 kΩ pull-up resistors to the supply voltage and 100 nF capacitors connected to ground. This filter effectively attenuates the fast, high-frequency voltage pulses arising during the switching

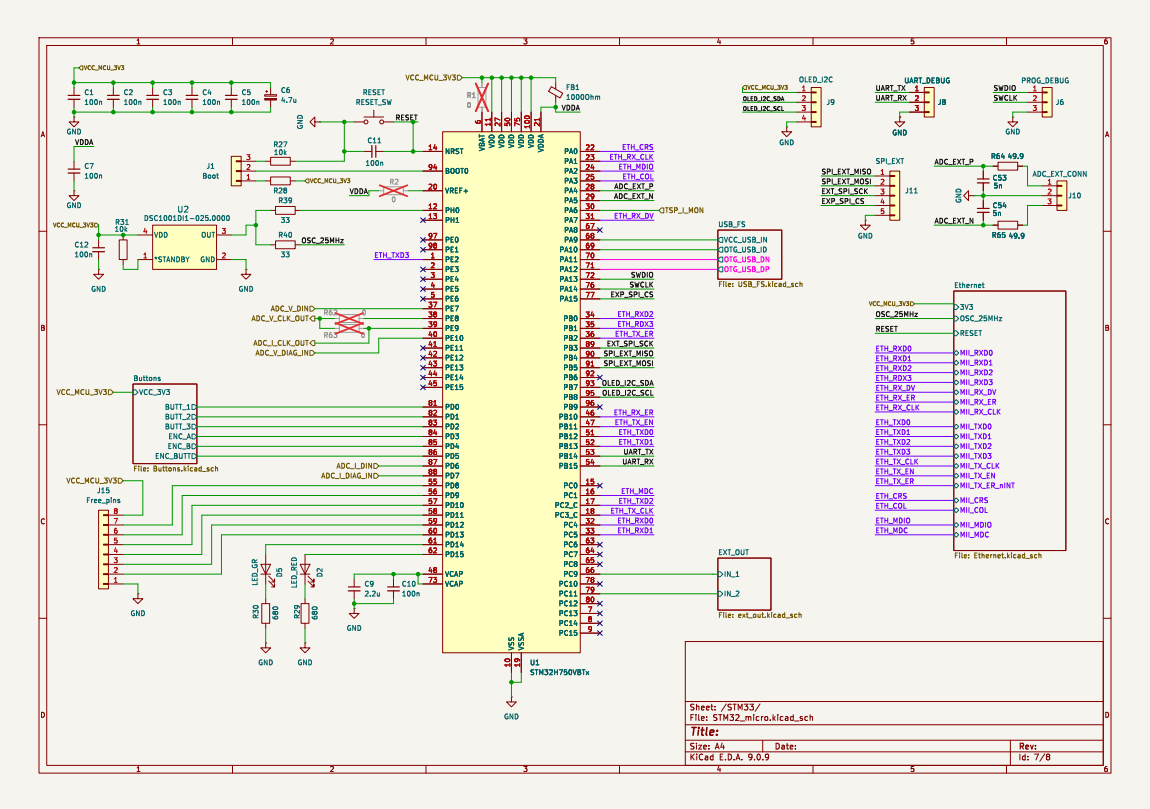


Figure 4.4: Schematic diagram of the STM32 microcontroller block and its pins. Source: own elaboration.

of the contacts.

The use of the RC filter causes a significant lengthening of the signal rise and fall times. Feeding such a slowly changing, non-linear signal (in the form of an exponential curve) directly to the digital input pins of the microcontroller could lead to the signal remaining in the so-called forbidden state and, consequently, to an erroneous reading of the logic states. To prevent this, the signals from the filters were routed to the inputs of a sixfold inverting Schmitt-trigger buffer – the SN74LVC14A device [36].

The voltage hysteresis built into the device guarantees that the slowly changing signal edges from the RC circuit are converted into steep, clean, and unambiguous rectangular edges, suitable for digital logic circuits. Such a hardware approach significantly simplifies the implementation of the embedded software, relieving the processor of the need to continuously poll the state of the inputs, implement timers, and count unstable states.

4.4.2 Isolated digital outputs

In order to enable the control of external actuating devices, the design of the measurement device provides two universal digital outputs. The schematic diagram of this block is presented in Fig. 4.6.

For safety reasons and in order to protect the digital circuits of the microcontroller

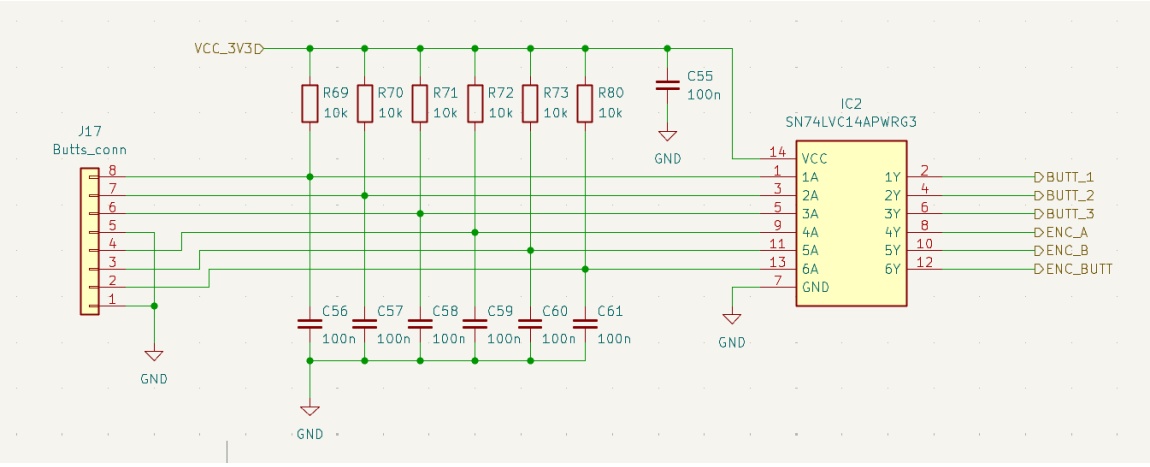


Figure 4.5: Schematic diagram of the input circuits (buttons and encoder) with RC filters and Schmitt triggers. Source: own elaboration.

against interference, these outputs were galvanically separated from the rest of the system. The PS2802-1 optocouplers [37] were used to accomplish this task. These devices are equipped with a Darlington-transistor output, which guarantees high current-carrying capacity and a high current transfer ratio (CTR), enabling safe driving of external circuits.

The emitter diode of each optocoupler is supplied directly from the microcontroller's GPIO output (3.3 V logic signals). In order to limit the diode forward current, a series connection of two resistors, each with a value of 150 Ω, was used.

The physical connection of the external signals was implemented by means of standard screw connectors, ensuring reliable mechanical and electrical mounting of the wires. In order to protect the output stage, additional overvoltage protections were implemented. Matrices of bidirectional ESDA5V3L (TVS) protection diodes [38] were connected in parallel with the output terminals of each channel (between the collector and the emitter of the phototransistor).

4.4.3 USB communication interface

The device provides the possibility of alternative wired communication by means of the Universal Serial Bus (USB). The schematic of the input circuit for this interface is presented in Fig. 4.7.

A Micro USB type B socket (designated as J2) was used as the physical connector. It has leads for four main signals: power (`VBUS`), hardware device identification (`ID`), and a data pair (`D+` and `D-`). Moreover, the connector body (shield) was connected directly to the circuit ground (GND) in order to shield against electromagnetic interference. The signal lines from the socket were routed to the dedicated pins of the hardware USB OTG (On-The-Go) controller built into the microcontroller. The `VBUS` (5 V) supply line was additionally connected to an indication circuit consisting of an LED and a current-limiting

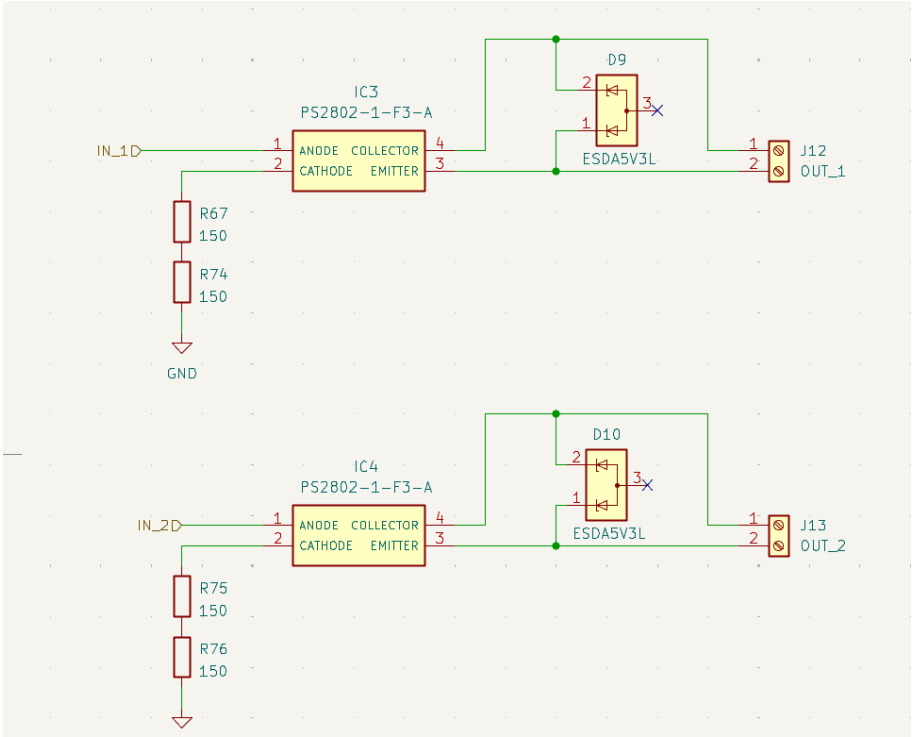


Figure 4.6: Schematic diagram of the isolated digital outputs controlled by the microcontroller. Source: own elaboration.

resistor. This solution enables immediate, visual verification of the correct connection of the USB cable.

The USB connector, as an external interface led out beyond the device enclosure, is particularly exposed to electrostatic discharges (ESD) originating from the user. In order to protect the delicate ports of the microcontroller against damage, an integrated matrix of USBLC6-4SC6 [39] protection diodes was used on the signal and supply lines. The internal structure of this component is based on diodes reverse-biased between the protected data lines and the `VBUS` and `GND` rails. In the event of an overvoltage, the fast diodes begin to conduct, clipping the voltage spike and draining the charge to the supply or the ground of the circuit. The low parasitic capacitance of this protection circuit does not disturb data transmission.

In the schematic diagram, the data lines `OTG_USB_DP` (`D+`) and `OTG_USB_DN` (`D-`) were marked in a different (pink) colour. This was intended to highlight the signals transmitted in the differential standard. These traces require an appropriate characteristic impedance (typically 90 Ω for the USB standard). This is critical for maintaining the integrity of high-speed digital signals and avoiding reflections of the electromagnetic wave at the line ends.

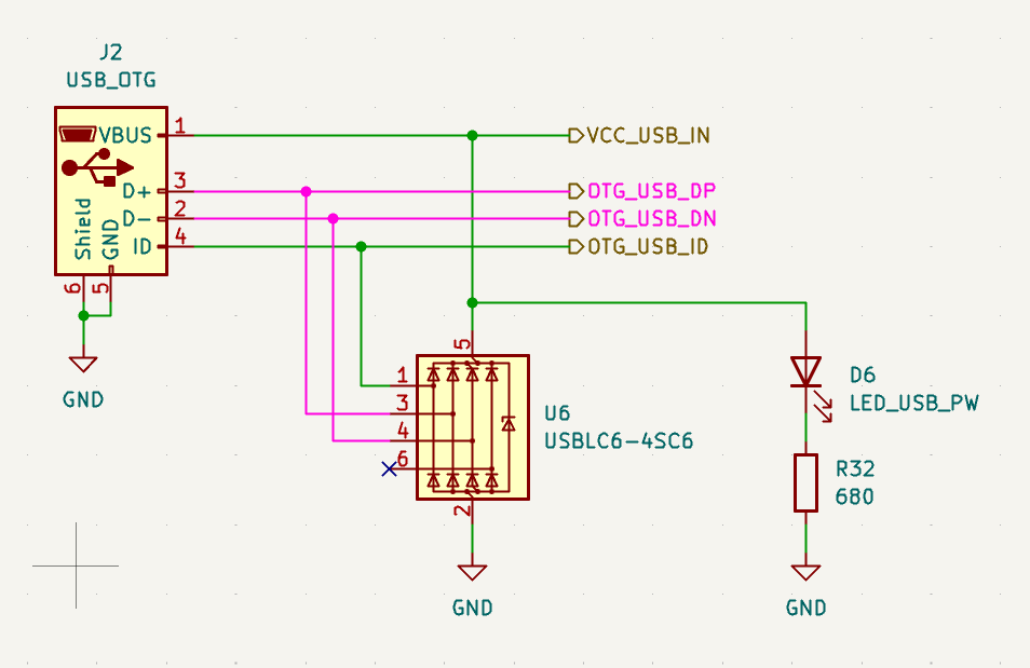


Figure 4.7: Schematic diagram of the USB interface together with the electrostatic protection circuit. Source: own elaboration.

4.5 Ethernet network interface

The wired communication module in the Ethernet standard was implemented on the basis of the physical-layer (PHY) device LAN8740A by Microchip [40]. It is responsible for translating the signals from the microcontroller's MAC interface into the physical differential signals of the 10/100Base-TX standard. From the design point of view, the block under discussion can be divided into three main sections: the PHY controller with matching circuits, the isolation transformer, and the RJ45 connector. The schematic of this circuit is presented in Fig. 4.8.

In order to ensure error-free data transmission, in accordance with the manufacturer's guidelines [41], the supply of the analog part of the PHY controller (`VDDA`) was separated from the main supply by means of a low-pass filter. This filter consists of a ferrite bead with an impedance of 1000 Ω (@ 100 MHz) and two 2.2 μF capacitors. Additionally, ceramic decoupling capacitors, minimising high-frequency noise, were placed right at the supply pins of the integrated circuit.

The communication between the STM32 microcontroller and the PHY device is based on a standardised interface. In the original schematic, the lines were routed for the MII standard; however, at the prototype commissioning stage, it was necessary to modify the traces on the printed circuit board in order to force operation in the RMII mode. A detailed analysis of this design error and its repair process is provided in the concluding chapter 7.2. Series resistors with a value of 56 Ω were used on the signal lines of the digital interface. Their task is to match the impedance of the traces and to suppress

the oscillation phenomenon during high-frequency switching of logic states. The common clocking for the interface is provided by an external crystal oscillator.

The analog section of the device connects to the ALANL100X1 isolation transformer via two differential pairs: the transmitting (TX) and receiving (RX) pair. The line drivers in the LAN8740A operate in current mode, which requires the use of pull-up resistors to the 3.3 V supply with a precise value of 49.9 Ω (1% tolerance). They simultaneously constitute a DC bias circuit and an impedance match to the characteristic line impedance of 100 Ω. Capacitors with a value of 10 pF were added in parallel with the signal lines, acting as a low-pass filter that suppresses the emission of electromagnetic interference (EMC). The centre taps of the transformer on the PHY-device side were connected to the 3.3 V voltage and decoupled to ground with a 22 nF capacitor.

The galvanic barrier in the form of a transformer separates the device electronics from the external cable network. On the RJ45-connector side, a common-mode termination circuit, known as the Bob Smith termination, was implemented. The centre taps of the secondary windings and the unused twisted-pair conductors (pins 4-5 and 7-8) were connected through 75 Ω matching resistors to high-voltage (2 kV) capacitors, draining the charge to the connector shield. Such an arrangement effectively absorbs the harmful RF interference induced in the long CAT5 cables.

The connector shield (`Shield`), being a local earth, was deliberately separated from the digital ground of the device (`GND`). A parallel circuit consisting of a 4.7 nF (2 kV) capacitor and a 1 MΩ bleed resistor was used to couple them. This constitutes a safe path for draining electrostatic charges (ESD), while at the same time preventing the formation of unwanted ground loops.

4.6 Printed circuit board (PCB) design

4.6.1 Layer stack-up and design rules

The printed circuit board design process began with the selection of an appropriate layer stack-up and the definition of the design rules. Owing to the degree of complexity of the circuit, the need to route impedance-controlled signals, and the desire to miniaturise the device, it was decided to use a four-layer laminate.

The configuration was based on the standard JLC04161H-3313 profile by JLCPCB. The total thickness of the laminate amounts to approximately 1.56 mm. The following functions were assigned to the individual layers:

- **L1 (Top)** – the signal and mounting layer for SMD components. The pouring of ground planes on it was abandoned in order to avoid disturbing the impedance of the signal traces.

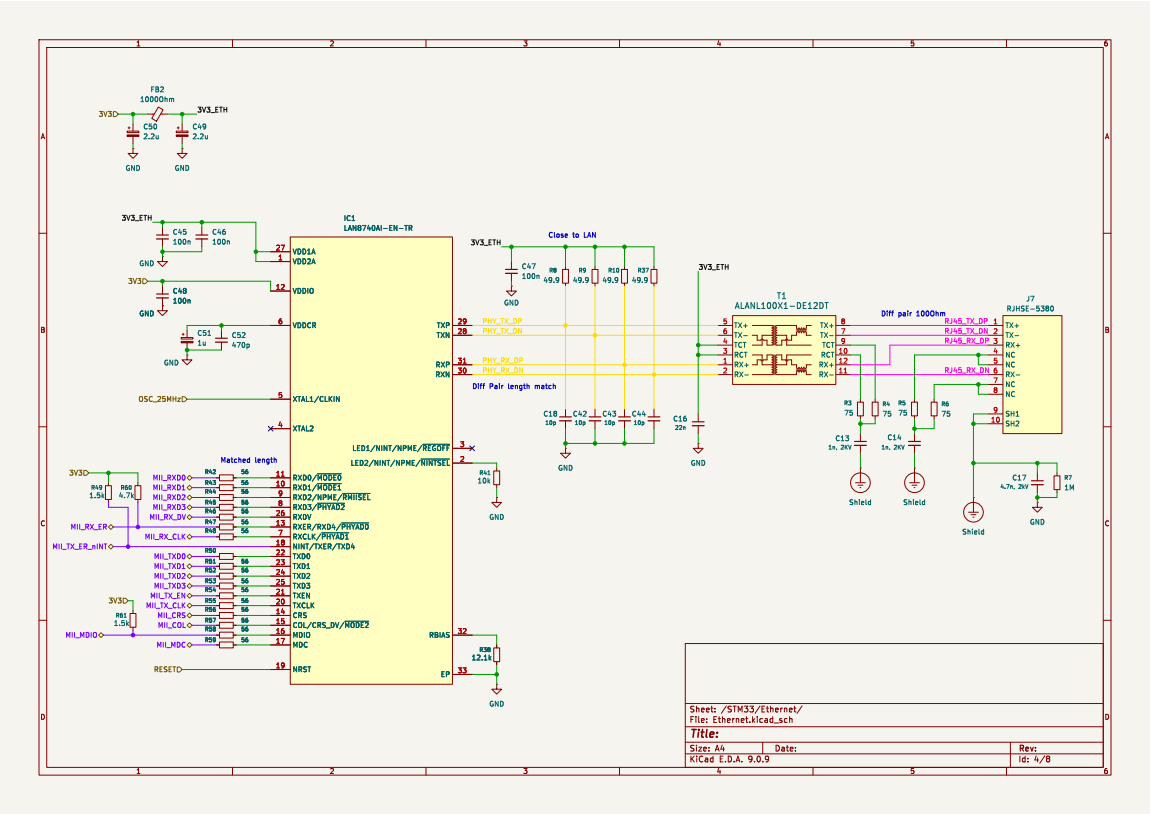


Figure 4.8: Schematic diagram of the Ethernet interface with the PHY device, impedance-matching circuits, and the isolation transformer. Source: own elaboration.

- **L2 (In1)** – an internal, continuous ground plane (GND), constituting an ideal reference for the fast digital signals routed on the top layer (L1).
- **L3 (In2)** – an internal power layer, dedicated to the distribution of the main operating voltages of the system (3.3 V and 5 V).
- **L4 (Bottom)** – the bottom signal and mounting layer.

Detailed parameters of the thicknesses of the individual dielectric materials (prepreg, core) and copper are presented in Table 4.1.

The use of a ground plane on the second layer made it possible to route the critical traces on the top layer in a *microstrip* topology for single-ended signals, and a *differential microstrip* topology for differential pairs.

For the USB and Ethernet communication buses, in order to minimise reflections of the electromagnetic wave, it was necessary to maintain impedance control. The trace widths and the spacings between them were not determined analytically by hand, but were selected using a dedicated impedance calculator provided by the printed circuit board manufacturer – JLCPCB [42]. This tool automatically accounts for the actual dielectric constant of the 3313 prepreg used and the physical constraints imposed by the manufacturing process. The dimensions calculated in this way for the individual signal classes are compiled in Table 4.2.

Table 4.1: Layer stack-up of the printed circuit board based on the JLC04161H-3313 profile. Source: own elaboration based on the manufacturer's data.

Layer	**Material**	**Copper thickness [mm]**	**Dielectric thickness [mm]**
L1 (Top)	Copper (1 oz)	0.0350	—
Dielectric 1	Prepreg 3313 RC57%	—	0.0994
L2 (In1)	Copper (0.5 oz)	0.0152	—
Dielectric 2	Core FR4	—	1.2650
L3 (In2)	Copper (0.5 oz)	0.0152	—
Dielectric 3	Prepreg 3313 RC57%	—	0.0994
L4 (Bottom)	Copper (1 oz)	0.0350	—
Total laminate thickness:		**≈ 1.56 mm**	

Table 4.2: Compilation of the design rules for impedance-controlled signals. Source: own elaboration based on [42].

Signal class	**Impedance type**	**Target**	**Width**	**Spacing**
`USB_90_Ohm`	Differential (L1-L2)	90 Ω	0.158 mm	0.203 mm
`ETH_100_Ohm`	Differential (L1-L2)	100 Ω	0.109 mm	0.152 mm
`MII_58_Ohm`	Single-ended (L1-L2)	≈ 50 Ω	0.101 mm	—
`Default`	—	—	0.200 mm	—

A special case from the design perspective was the section of the Ethernet differential signals routed from the isolation transformer to the RJ45 connector. In accordance with good electromagnetic-compatibility practices, all the internal copper planes were deliberately removed in this area in order to prevent galvanic breakdown and to maintain high isolation of the equipment. This means that the differential lines were deprived of a direct reference plane. Under such conditions, the classical impedance-calculation models do not apply. In order to minimise the impact of this phenomenon, the distance from the transformer to the connector was routed along the shortest possible symmetrical path (traces with a width of approx. 0.368 mm and a separation of 0.127 mm), which, at the low frequencies of the 100Base-TX interface (maximum 50 MHz), makes the impact of the impedance mismatch negligible.

4.6.2 Component placement on the printed circuit board

When designing the topography of the printed circuit board, the principle of functional division was followed. The board was logically divided into three main zones:

- **The left zone (high-voltage/measurement):** It includes the screw connectors, the current shunt resistors, the voltage divider, and the isolated AMC converters. This area is characterised by large isolation clearances, ensuring safety when working with the 230 V grid voltage.

- **The middle zone (digital):** The heart of the device, where the STM32H7 microcontroller, the power supply (eFuse, LDO), and the connectors of the user interfaces (buttons, LEDs, SWD programmer connectors, and additional I/O) are located.
- **The right zone (communication):** It includes the LAN8740A PHY device, the isolation transformer, and the RJ45 connector, separated by an isolation barrier from the digital layer.

A complete set of views of the printed circuit board, showing the detailed routing of the traces on all layers (Top, In1, In2, Bottom), is included in the technical documentation in Appendix A (Fig. D1. D2).

4.6.3 Design of the measurement-channel circuit

The measurement block, located in the left part of the printed circuit board, is a key element of the entire system. It consists of the screw connectors, the high-voltage resistive divider, the shunt resistor, and the isolated $\Sigma\Delta$ modulators from the AMC family. A view of the trace topography for this segment is presented in Fig. 4.9.

The AMC3306M25 device (responsible for the current channel) was placed on the top layer (Top), while the AMC3336 device (responsible for the voltage channel) was mounted on the bottom layer (Bottom). The input and decoupling circuits cooperating with the converters were designed strictly in accordance with the manufacturer's recommendations contained in the datasheets [15, 27].

Particular attention was paid to the design of the current traces for the phase (L1) and neutral (N) lines. Owing to the assumed maximum load currents of the order of 16–20 A, the current pours were routed symmetrically on all four layers of the laminate. These layers were densely interconnected with a grid of vias. This was intended to multiply the effective copper cross-section, reduce the thermal resistance, and minimise the heating of the circuit. Despite the measures applied, during extreme load tests at continuous currents of the order of 20 A, an excessive temperature rise was observed on the neutral line. In order to increase the current-carrying capacity, in the assembled prototype this trace was additionally thickened by applying a thick layer of solder to the exposed part of the pour on the bottom layer.

The shunt resistor was placed in series in the phase line. The routing of the measurement signals from the shunt was implemented using a technique similar to a Kelvin connection. The traces of the differential pair were routed fully symmetrically, being led out from the inner, central parts of the resistor pads. This minimises the measurement error resulting from the voltage drops across the solder fields themselves, which are loaded with a high operating current.

A critical aspect for ensuring the safety of the entire device was the correct implementation of the galvanic barrier. As can be observed in the circuit design, the digital

ground (GND) and power pours on the internal layers (In1, In2) were cut off before the central axis of the AMC devices. This ensures the maintenance of appropriate isolation clearances under the integrated circuits.

Owing to the availability of free space in the far-left part of the module, two of the four main mounting holes, providing mechanically stable fixing of the circuit in the target enclosure, were also placed there.

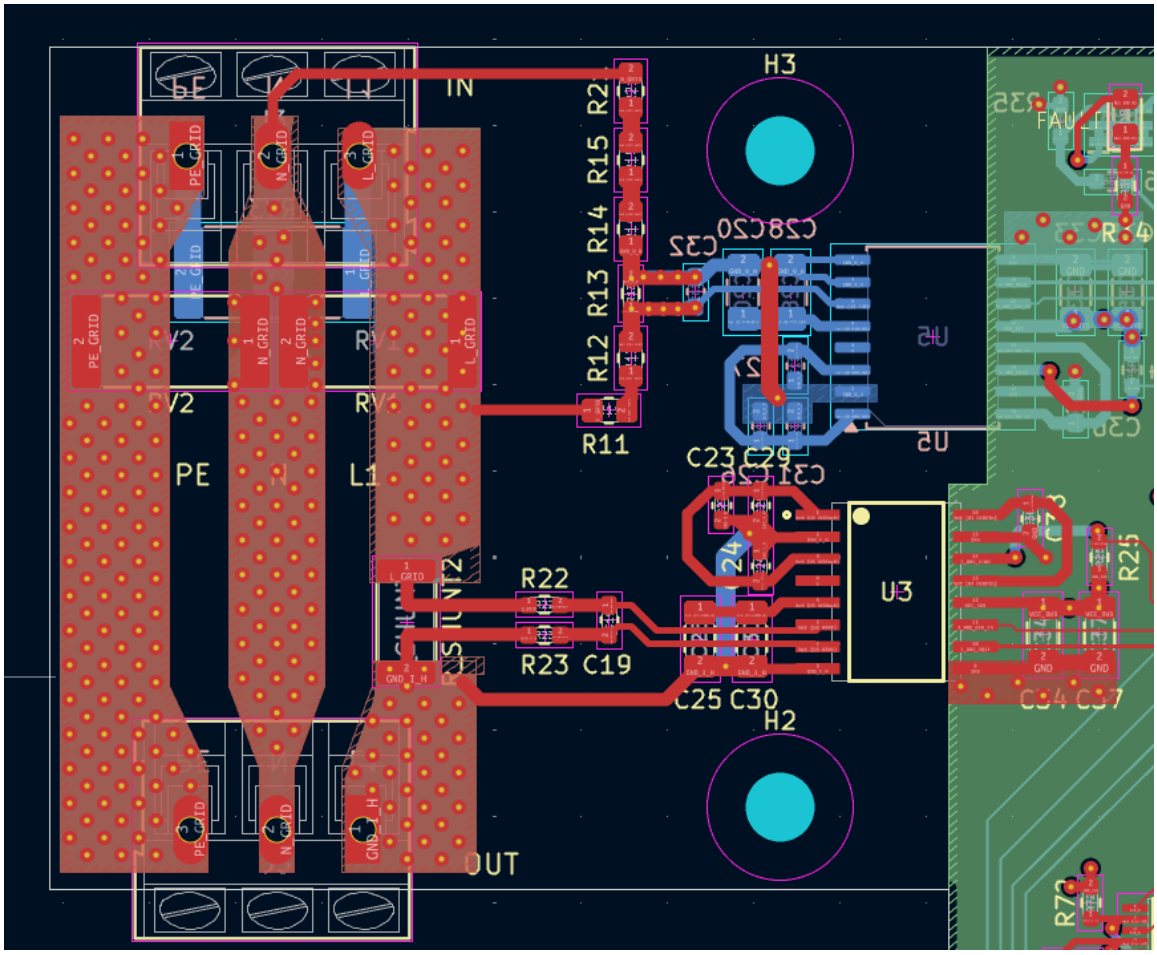


Figure 4.9: View of the traces and pours of the measurement part of the printed circuit board. Source: own elaboration.

4.6.4 Design of the Ethernet interface circuit

The Ethernet interface communication block was located in the far right part of the printed circuit board. A view of the trace topography for this segment – with the internal layers hidden in order to improve readability – is presented in Fig. 4.10.

Owing to the placement of the main reference plane (GND) on the second layer (L2), the vast majority of the components and signal traces were routed on the top layer (Top). This minimised the need to use vias on the fast MII interface lines, which has a positive effect on the integrity of the transmitted digital signals.

Around the PHY controller itself (LAN8740A), the necessary passive components were densely placed. In accordance with good design practices for circuits with a high clock frequency, the ceramic decoupling capacitors were placed at the smallest possible distance from the supply pins of the integrated circuit. This reduces the parasitic inductance of the traces and ensures a fast supply transient response.

Particular rigour was maintained when routing the differential pairs (transmitting and receiving) connecting the PHY device, the isolation transformer, and the RJ45 connector. In accordance with the design guidelines for maintaining impedance matching, these lines were routed in close proximity along almost their entire length, being separated from each other only in the immediate vicinity of the target pads. The construction of the selected transformer (T1), which has centre taps led out in the central part of the package, significantly facilitated the optimal and symmetrical routing of the supply and bias traces without collisions with the differential signals.

The galvanic separation of the circuits was ensured by introducing a *keep-out* zone (an area devoid of any copper pours on all four layers of the laminate) in the space between the isolation transformer and the RJ45 connector. Additionally, along the right edge of the board, a separate pour acting as a shield was poured around the connector.

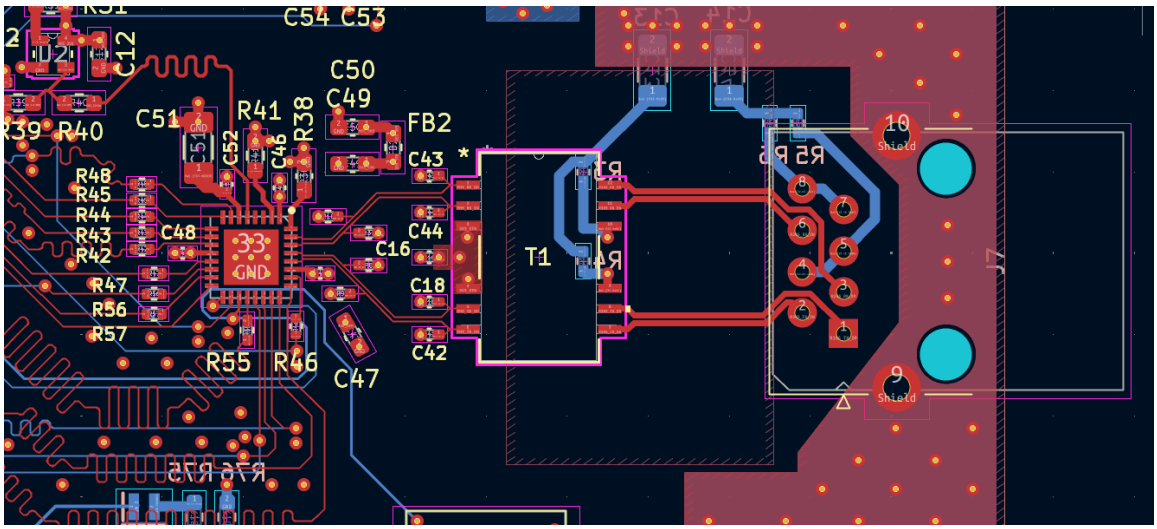


Figure 4.10: View of the topography of the top (Top) and bottom (Bottom) layers for the Ethernet interface block. The internal layers were hidden to improve readability. Source: own elaboration.

4.6.5 Design of the digital zone and the microcontroller

The central section of the printed circuit board is the heart of the digital part of the device. A view of the trace topography for this area, divided into the top (Top) and bottom (Bottom) layers, is presented in Figs. 4.11 and 4.12, respectively. As with the previous sections, the internal layers were hidden in order to improve the readability of

the analysed circuit.

The main STM32 microcontroller was located in the centre of the board, which optimised the distribution of signals to the individual zones of the device. The connectors of the peripheral interfaces were grouped along the top and bottom edges of the laminate. Such a layout significantly facilitates service access and minimises the tangle of wires inside the target enclosure. The diodes indicating the presence of the supply voltages were placed in the upper-left section in the vicinity of the 5 V input, while the USB-interface indicator diode and the diagnostic diodes were placed near the bottom edge. The distribution of the main supply voltages (on the outer layers) was implemented by means of wide copper pours, which minimises the resistance and the parasitic inductance of the supply traces.

The USB interface signal lines were routed in a differential-pair topology with a constant spacing. The matrix of electrostatic-discharge (ESD) protection diodes was placed in the immediate vicinity of the connector. This is a critical design measure ensuring that any electrostatic charge is drained to ground before it reaches the sensitive structures of the microcontroller, thus minimising the parasitic inductance of the discharge loop.

Exceptional attention was paid to the routing of the communication bus (the MAC-PHY interface) between the microcontroller and the LAN8740A. In accordance with the manufacturer's recommendations, equalisation of the signal propagation times was performed. For this purpose, a trace-meandering technique was used, visible in the design in the form of characteristic bends. This guarantees appropriate phase alignment of the clock and data signals. Care was also taken to ensure that all the fast signal lines were routed over a continuous reference plane (GND) on the internal L2 layer, avoiding the crossing of any gaps. This ensures an uninterrupted return path for high-frequency currents and drastically reduces the emission of electromagnetic interference (EMI).

The common crystal oscillator was mounted at an optimal, averaged distance between the microcontroller and the Ethernet physical-layer device. Moreover, series oscillation-suppressing resistors were used on the fast digital lines. They were placed on the basis of the source-termination principles – for the transmitting lines (`TX`) the resistors are located right at the transmitter pins of the STM32, while for the receiving lines (`RX`) they were placed in the immediate vicinity of the LAN8740A. Such a topology ensures impedance matching and signal integrity.

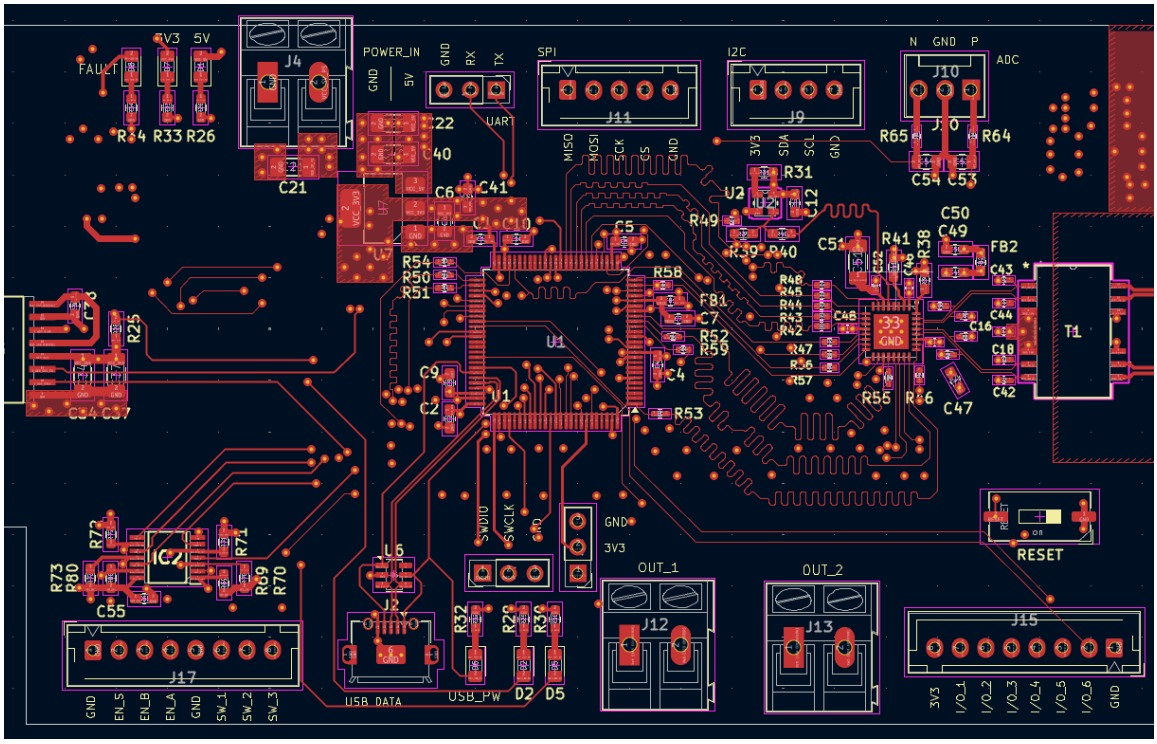


Figure 4.11: View of the topography of the top layer (Top) for the digital section with the microcontroller. Source: own elaboration.

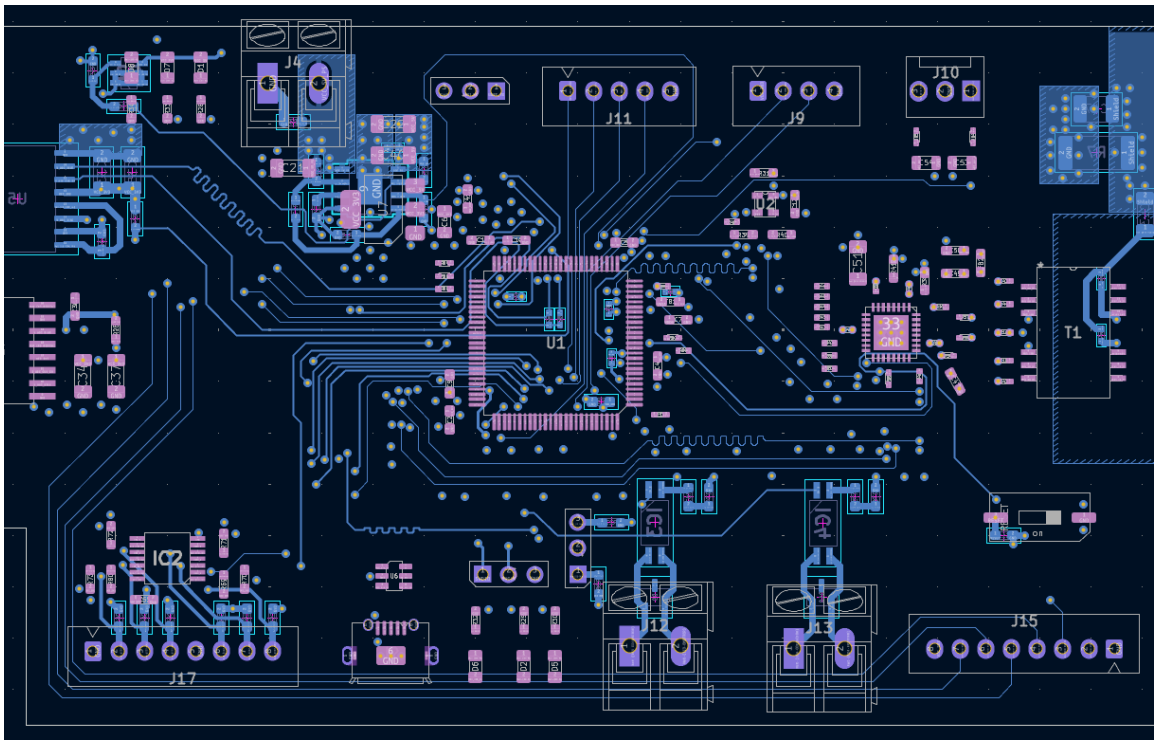


Figure 4.12: View of the topography of the bottom layer (Bottom) for the digital section. Source: own elaboration.

Chapter 5

Embedded software (firmware)

5.1 Hardware configuration and system initialisation

The process of creating the embedded software began with the generation of the project skeleton using the STM32CubeMX graphical tool. It makes it possible to automate the process of initialising the key microcontroller peripherals and to configure the clock tree. A preview of the pin map defined in the program is presented in Fig. 5.1.

A frequency of 480 MHz was selected as the main clock source for the entire circuit, which guarantees the highest available computing power of the Cortex-M7 core. Using appropriate prescalers, a 120 MHz clock signal dedicated to supplying the DFSDM module was generated. It was then divided internally, resulting in a clock signal driving the external $\Sigma\Delta$ modulators (the `CLKOUT` signal) at a frequency of 20 MHz. By configuring the hardware digital Sinc filter and adopting an oversampling ratio of $OSR = 256$, a final sampling frequency of the useful signal at the level of approximately 78.125 kHz was obtained. Both measurement channels in the DFSDM module were tightly synchronised with each other, operating on the basis of a common clock source, which prevents the formation of phase-shift errors between the current and the voltage. For the DFSDM module, support for the direct memory access (DMA) controller and the associated interrupts was also activated.

A significant part of the configuration was the network interface. The built-in MAC controller was activated, to which a fixed physical address was assigned; the necessary RAM areas for the receive and transmit buffers (the so-called descriptors) were defined, and the system interrupts from the Ethernet interface were enabled. Moreover, the remaining peripherals used in the project, such as the I2C, SPI, and UART interfaces as well as the control pins (SWDIO/SWCLK) used for programming and debugging, were initialised in the program.

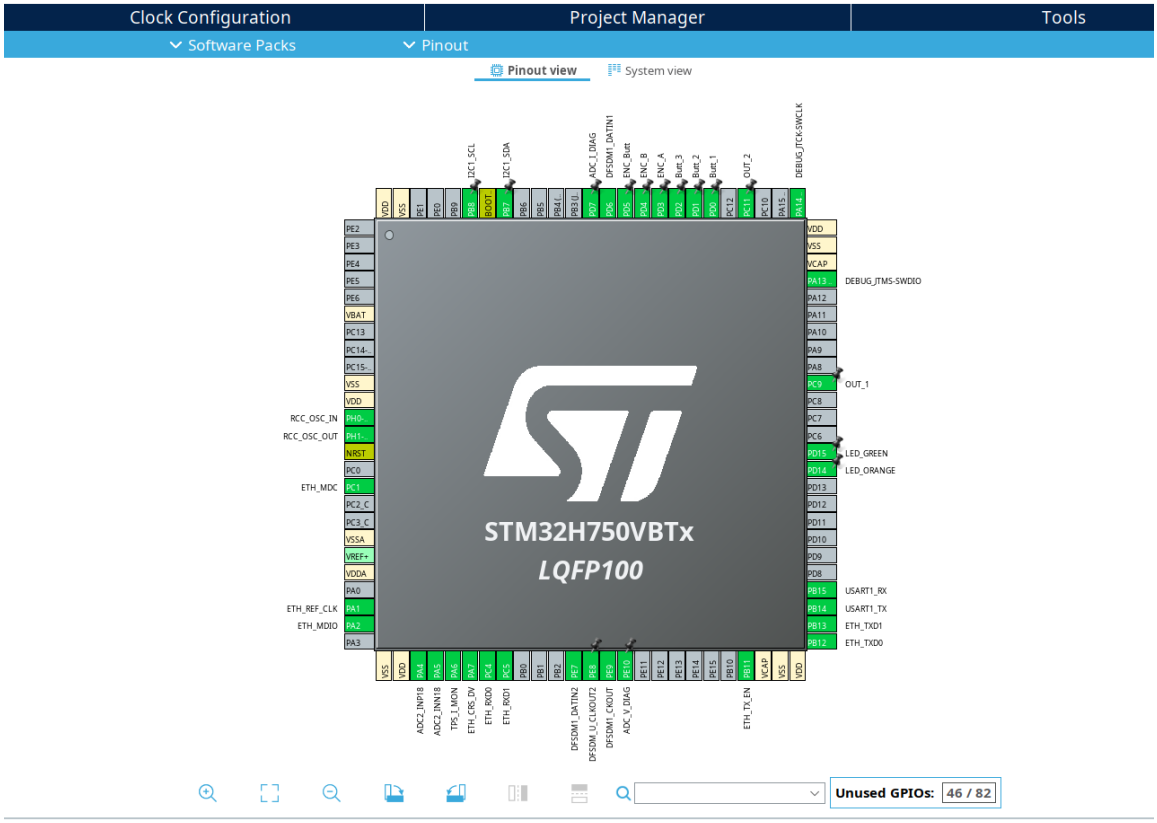


Figure 5.1: Map of the assignment of functions to the pins of the STM32H7 microcontroller created in STM32CubeMX. Source: own elaboration.

5.2 Middleware and libraries

In addition to the hardware abstraction layer (HAL), the STM32CubeMX tool was used to enable the middleware packages. The FreeRTOS real-time operating system, which constitutes the backbone of the entire program's architecture, and the LwIP (Lightweight IP) network protocol stack were integrated.

The configuration of the FreeRTOS system included the allocation of the heap memory, the determination of the operating-system tick frequency, and the definition of priorities. In turn, the configuration of the LwIP stack required the adaptation of the network parameters to the module's operating environment. Support for the TCP and UDP protocols was activated, and the mode of assigning the IP address (static IP) was selected.

The source code generated in this way was then imported into the Visual Studio Code environment. At this stage, two key external libraries were added to it:

- **libscpi** – an *open-source* library implementing a parser of standardised SCPI commands, used to receive control and configuration commands.
- **CMSIS-DSP** – the official library developed by ARM, providing highly optimised mathematical functions. It was used to implement fast trigonometric calculations, square-root extraction, and the determination of RMS values using the hardware

FPU floating-point unit.

5.3 Software architecture and file structure

The application was written in C, maintaining a modular character of the code. In order to separate the configuration part (generated automatically) from the business logic, a dedicated `Mine` folder was created. It contains the author's own source files, responsible for the actual functionality of the meter. A compilation and description of these files, most important from the point of view of the device's operation, is presented in Table 5.1.

Table 5.1: Structure of the author's own source files and description of their functionality. Source: own elaboration.

File name (.c / .h)	**Functional description**
`main`	Initialisation of the main peripherals, reading of the state stored in the non-volatile memory, and invocation of the system start-up procedures.
`freertos`	Initialisation of the FreeRTOS threads (tasks).
`outputs`	Abstraction layer for controlling the actuating elements (among others, the LEDs and the isolated digital outputs).
`EEPROM_I2c`	Implementation of the hardware driver for the I2C protocol for communication with the external EEPROM memory. Writing and reading of the calibration coefficients and the configuration.
`scpi_def`	Definitions, callbacks, and the tree structure of the custom text commands in the SCPI standard.
`sigma_delta_task`	The main loop of the measurement process. Management of the communication with the AMC converters, handling of the DMA interrupts, control of the device operating modes, and transmission of the calculated parameters.
`signal_processing`	The mathematical core and digital signal processing. Calculation of the power quality parameters on the basis of the discrete current and voltage samples. Implementation of the SOGI-PLL filter.
`eth_comms`	Configuration of the TCP server (LwIP), definitions of the structures of the data frames sent to the host software.

The use of the real-time operating system imposed a multi-threaded architecture of operation. The software logic is based on two main, explicitly defined tasks:

1. **Default Task** – a system task with standard priority. It is responsible for cyclically toggling the LED (indicating the stable operation of the microcontroller) and for

the global initialisation of the higher-order libraries, including the LwIP stack and the listening TCP server. It should be mentioned that the LwIP library internally creates its own asynchronous subtasks responsible for receiving Ethernet packets.

2. **Sigma Delta Task** – a dedicated high-priority measurement thread. This task remains in a sleep state for most of the time, waiting for the event flags reported by the DMA controller. Once half or all of the buffer is filled with data from the converters, the task wakes up, fetches the raw signals, and delegates them to the `signal_processing` module in order to drive the SOGI-PLL algorithm and perform the metrological calculations.

5.4 Configuration interface and SCPI commands

The external *libscpi* [26] library was used to implement a flexible and readable configuration interface. This tool not only implements the set of mandatory instructions required by the standardised IEEE 488.2 [23] standard, but also provides mechanisms allowing the creation of custom, nested command structures. The library was integrated with the microcontroller's UART interface, which acts as a character buffer passing the received text strings directly to the SCPI parser.

In order to fully cover the functionality of the designed meter, a command tree was developed. The implemented commands can be divided into the following functional groups:

- **Output management:** `OUTputs:LED`, `OUTputs:1`, `OUTputs:2`
  These commands (together with their query counterparts with the `?` suffix) are responsible for reading the current state and directly driving the indicator diodes and the opto-isolated digital outputs of the device.

- **Zero-offset calibration:** `CALIBration:[Voltage/Current]:ZEROoffset`
  The module makes it possible to remove the constant error of the measurement channel. The *offset* value can be entered manually or the auto-calibration process can be started (the `:START` command). In the automatic mode, the microcontroller fetches and averages the samples over a period of one second (with the measurement inputs short-circuited). The calculated raw value is written directly to the hardware compensation registers of the DFSDM module. As a result, the peripheral automatically subtracts the zero offset from each subsequent sample, completely relieving the processor core.

- **Gain calibration:** `CALIBration:[Voltage/Current]:GAINfactor`
  Analogously to the zero correction, the gain-factor calibration can be carried out manually or automatically. The auto-calibration requires providing in the command the actual value of the reference parameter (voltage or current). The meter records

the signal for one second, determines the averaged value from the ADC converters, and computes the correction factor in the form of a floating-point number. In the normal operating mode, all the raw data from the DFSDM module are multiplied by this factor.

- **SOGI-PLL filter parameterisation:** `CALIBration:SOGI:KP`, `CALIBration:SOGI:KI`
  A set of commands enabling dynamic changes of the settings (the proportional gain K_p and the integral time K_i) of the virtual PI controller implemented in the core of the SOGI-PLL algorithm.

- **Network settings:** `CONNection:DEVice:IP`
  Commands allowing remote reconfiguration of the device's static IP address in the LwIP stack structures.

5.4.1 Measurement mode management

Three main operating modes, corresponding to the type of data sent to the host device, were also implemented in the software. The mode is changed by means of the SCPI command `CONNection:MEASuring`. The device can operate in one of three main modes:

1. **RAW (RAW_DATA)** – the mode of transmitting raw, direct samples of the modulated voltage and current signal after preliminary calibration. Owing to the TCP protocol overhead and the performance limitations of the network server, it was impossible to stream data at the full DFSDM filter frequency ($\approx$ 78 kHz per channel). Therefore, in the RAW mode, the program applies additional hardware decimation, sending every second sample (approx. 39 kHz). The initial tests assumed the use of the UDP protocol for the transmission of the full stream; however, this concept was abandoned in favour of the packet-delivery guarantee offered by TCP.

2. **CONTROL (CONTROL_DATA)** – a diagnostic mode dedicated to verifying the correct operation of the phase-synchronisation algorithm. The data frame – sent with a tenfold decimation relative to the fundamental frequency – contains the raw samples of the input signals supplemented with the internal states of the SOGI-PLL loop: the estimated angle θ, the angular frequency ω, and the generated orthogonal signals $U_{\alpha/\beta}$ and $I_{\alpha/\beta}$.

3. **GRID (GRID_PARAM_DATA)** – the main, target operating mode of the measurement device (active by default after the power supply is switched on). In this mode, the continuous streaming of the time waveforms was abandoned in favour of the transmission of aggregated power quality parameters. In accordance with the metrological standards, the transmitted frame contains the values averaged over a

time window corresponding to 10 periods of the fundamental signal. It includes, among others, the RMS values, the powers (P, Q, S), the frequency, the peak values (PEAK), and the THD factors.

Apart from the listed modes, the system also has an internal **CALIBRATION** state, which is triggered only for the duration of the one-second converter auto-calibration procedures, blocking the transmission of other telemetry data within this time window.

5.5 Storing parameters in non-volatile memory

The correct operation of the measurement device requires the retention of the determined zero auto-calibration parameters, the gain factors, the SOGI-PLL controller settings, and the network configuration after the power supply is disconnected.

Owing to the specifics of the STM32H750VBT6 microcontroller used, writing data to the internal Flash memory directly from the running software proved technically impossible. This limitation results directly from the architecture of the device.

In order to solve this problem at the hardware level and to ensure permanent storage of the parameters, an external EEPROM memory communicating with the processor via the I2C bus was used in the system. At the software layer, this task was entrusted to a dedicated `EEPROM_I2c.c` module, which performs transparent writing and reading of the defined data structures. On every system restart, the microcontroller first communicates with the external memory, loading the previously stored values and thus restoring the meter to a ready, calibrated operating state.

5.6 Data acquisition and memory optimisation

In order to relieve the main core of the microcontroller, the process of continuous acquisition of the rapidly changing measurement signals was implemented using the direct memory access (DMA) controller. Owing to the complex, matrix architecture of the hardware buses of the STM32H7 device, it was necessary to precisely locate the buffers in the appropriate RAM domains (in this case `RAM_D2`), to which the DMA has collision-free access. This required manual modification of the linker script (the `.ld` file) and the use of special compiler directives, as presented in Listing 5.1.

Listing 5.1: Configuration of the memory section in the linker script and the declaration of the DMA buffers in C code.

```
1    /* Dedicated section for DMA buffers in RAM (the .ld file)
     */
2    .dma_buffers (NOLOAD) :
3    {
```

```
4         . = ALIGN(4);
5         *(.dma_buffers)
6         . = ALIGN(4);
7     } >RAM
8
9     /* Declaration of constants and assignment of arrays to the
      section */
10    #define DFSDM_BUF_SIZE 6250
11    #define HALF_BUF_SIZE (DFSDM_BUF_SIZE / 2)
12
13    __attribute__((section(".dma_buffers"))) int32_t
      dfsdm_V_buffer[DFSDM_BUF_SIZE];
14    __attribute__((section(".dma_buffers"))) int32_t
      dfsdm_I_buffer[DFSDM_BUF_SIZE];
```

The capacity of a single buffer (`DFSDM_BUF_SIZE`) was set to 6250 samples. At the assumed sampling frequency, this size makes it possible to store data corresponding to merely about four full periods of the grid voltage. The deliberate resignation from immediately buffering the full 10 periods (required for calculating the power quality parameters) resulted from the need to optimise the microcontroller's RAM resources.

5.6.1 Pipelined processing and period synchronisation

Data reception is carried out in the hardware circular mode. The software uses two system interrupts from the DMA controller: the half-transfer interrupt (*Half-Complete*) and the transfer-complete interrupt (*Full-Complete*). Inside the interrupt service routines (ISRs), only flags waking up the `SigDelTask` task in the FreeRTOS system are set (Listing 5.2).

Listing 5.2: DMA interrupt service routines waking up the measurement thread.

```
1     void HAL_DFSDM_FilterRegConvHalfCpltCallback(
      DFSDM_Filter_HandleTypeDef *hdfsdm_filter)
2     {
3         osThreadFlagsSet(SigDelTaskHandle, FLAG_HALF_READY);
4     }
5
6     void HAL_DFSDM_FilterRegConvCpltCallback(
      DFSDM_Filter_HandleTypeDef *hdfsdm_filter)
7     {
8         osThreadFlagsSet(SigDelTaskHandle, FLAG_FULL_READY);
9     }
```

Such an architecture (referred to as *ping-pong* buffering) makes it possible to collision-free and synchronously process the first half of the buffer by the processor core, while the

hardware continuously writes subsequent samples to the second half, and vice versa.

Owing to the physically limited size of the buffer, the direct determination of the power quality parameters from the required 10 periods in a single iterative cycle is impossible. This problem was solved by using structural accumulating variables that retain their state between successive calls of the interrupt functions. The current phase angle θ, estimated by the SOGI-PLL algorithm, was used for the precise counting of the full wave periods. Since its value wraps around in the range from 0 to 2π, at the moment when the updated angle takes a value smaller than the previous one, the program precisely identifies the beginning of a new period (Listing 5.3). After ten such events are counted, the target parameters are calculated, and the accumulating registers are reset. This ensures the accuracy of the measurement window regardless of the DMA buffer capacity and the grid frequency.

Listing 5.3: Fragment of the data-processing loop with zero-crossing detection based on the angle θ from the SOGI-PLL.

```
1    static void Process_data(int32_t v_raw_shifted, int32_t
     i_raw_shifted, int i,
2    uint32_t *tx_raw_idx, uint32_t *tx_control_idx)
3    {
4       // 1. Conversion of raw values taking into account the factors
5       float V_f32 = (float)v_raw_shifted * voltage_gain_factor;
6       float I_f32 = (float)i_raw_shifted * current_gain_factor;
7
8       // 2. Processing of the signals by the SOGI algorithms
9       SOGI_return_t sogi_u_result = Run_SOGI_PLL(V_f32);
10      SOGI_return_t sogi_i_result = Run_SOGI_QSG(I_f32, sogi_u_result
     .omega);
11
12      float sogi_freq = sogi_u_result.omega / (2.0f * PI);
13      sogi_freq = SP_filter_pll_freq(sogi_freq);
14
15      // 3. Zero detection and frame synchronisation in the
     GRID_PARAM_DATA mode
16      if(meas_mode == GRID_PARAM_DATA && sogi_u_result.theta <
     prev_theta)
17      {
18         // Rest of the code
19
```

5.7 Transmission of measurement data via Ethernet

In order to ensure efficient and stable transmission of data to the host software (Serial Studio) via the TCP server, the transmission process was implemented using defined structures and dedicated memory areas. Similarly to the case of the input data from the DFSDM module, the transmit buffers for the Ethernet interface had to be located in a

specially allocated RAM sector, to which the MAC controller built into the microcontroller has direct access.

Three independent buffering arrays, corresponding to the three previously described measurement modes, were declared in the device memory. Their definitions are presented in Listing 5.4. The buffer size for the raw mode (`RAW`) corresponds to half of the main measurement buffer, which is closely related to the cyclic ping-pong operation mechanism. In the diagnostic mode (`CONTROL`), the buffer capacity is five times smaller owing to the decimation applied in the code (sending every tenth measurement sample divided into two halves of the cycle). For the aggregating mode (`GRID`), the smallest buffer, holding three complete result frames, was reserved.

Listing 5.4: Allocation of the transmit buffers in a dedicated RAM section.

```
1  __attribute__((section(".dma_buffers"))) RawFrame_t
   eth_tx_raw_buffer[HALF_BUF_SIZE];
2  __attribute__((section(".dma_buffers"))) ControlFrame_t
   eth_tx_control_buffer[HALF_BUF_SIZE / 5];
3  __attribute__((section(".dma_buffers"))) GridParamFrame_t
   eth_tx_grid_param_buffer[PARA_GRID_FRAME_CNT];
```

In order to ensure strict bit-level consistency of the transmitted data, each network frame was formed by means of C structures using the `__attribute__((packed))` compiler directive. The use of this attribute is absolutely crucial in machine communication, because it forces the compiler to pack the variables one after another, preventing the addition of empty alignment bytes typical of the 32-bit ARM processor architecture.

The network frame structures are presented in Listing 5.5. At the beginning of each packet, a two-byte synchronisation word with a constant value of `0xAAAA` was placed. In binary representation, this constitutes an alternating sequence of ones and zeros (`1010101010101010`). This solution is widely used in telemetry systems, because it facilitates the receiving software (the parser of the Serial Studio application) finding the beginning of the proper data packet in the continuous TCP protocol stream. The entire rest of the payload in each of the frames consists of 32-bit variables of the `float` type.

The use of the packed binary format drastically reduces the communication overhead compared with sending data in the form of plain text encoded in the ASCII standard (e.g. in the JSON format). This is particularly important in the `RAW` and `CONTROL` modes, in which the device sends thousands of frames per second. The Serial Studio software intercepts the network stream and then, on the basis of the provided configuration file, maps the individual bytes onto charts and indicators displayed in real time.

Listing 5.5: Structures of the transmit frames used in the individual measurement modes.

```
1    // Structure of the RAW frame (10 bytes: SYNC + V_RAW + I_RAW)
2    typedef struct __attribute__((packed))
3    {
4        uint16_t sync_word;
5        float v_raw;
6        float i_raw;
7    } RawFrame_t;
8
9    // Structure of the CONTROL frame (38 bytes: SYNC + 9 x float)
10   typedef struct __attribute__((packed))
11   {
12       uint16_t sync_word;
13       float v_raw;
14       float i_raw;
15       float pll_theta;
16       float pll_omega;
17       float v_alpha;
18       float v_beta;
19       float i_alpha;
20       float i_beta;
21   } ControlFrame_t;
22
23   // Structure of the GRID PARAM frame (78 bytes: SYNC + 19 x float)
24   typedef struct __attribute__((packed))
25   {
26       uint16_t sync_word;
27       float U_avg;
28       float I_avg;
29       float U_rms;
30       float I_rms;
31       float U1_rms;
32       float I1_rms;
33       float P;
34       float P1;
35       float Q1;
36       float S;
37       float S1;
38       float PF;
39       float DPF;
40       float DF;
41       float THD_u;
42       float THD_i;
43       float fundamental_freq;
44       float Q;
45       float U_peak;
46       float I_peak;
47   } GridParamFrame_t;
```

5.8 Data visualisation – the Serial Studio environment

In accordance with the design assumptions, the microcontroller software was integrated with a host application running on a PC. The *Serial Studio* [33] application was selected for the reception, decoding, and visualisation of the telemetry data stream. Its flexible architecture makes it possible to define a custom communication protocol on the basis of JSON configuration files and parsing scripts.

Data reception is carried out via a TCP socket. Owing to the fact that the TCP protocol transmits a continuous stream of bytes without determining the boundaries of the individual messages, it was necessary to use a frame-synchronisation mechanism. In the application configuration, a synchronisation word (`0xAAAA`), corresponding to the `sync_word` parameter sent by the microcontroller, was defined. The detection of this sequence by the host program initiates the process of extracting the proper data.

The transformation of the raw byte array into human-readable floating-point values is carried out by means of a dedicated script written in JavaScript. Owing to the ARM Cortex-M7 processor architecture, the data is transmitted in the *Little-Endian* format (the least significant byte at the lowest address). The implemented parser accounts for this convention, as presented in Listing 5.6.

The parameter array returned by the parsing script is then mapped onto specific elements of the virtual operator panel. To facilitate the analysis of data in the different operating modes of the device (RAW, CONTROL, GRID), independent parsers and dedicated graphical-interface views were prepared.

The visualisation structure was divided into two main screens. The first of them, a screenshot of which is presented in Fig. 5.2, constitutes the panel of numerical indicators. It is responsible for presenting the current, updated parameter values in the form of readable digital tiles. The second screen, a view of which is presented in Fig. 5.3, is used for the continuous recording and analysis of trends in the time domain using dynamically refreshed charts.

Listing 5.6: Fragment of the JavaScript script responsible for parsing the binary GRID-mode frame.

```
/**
* Function decoding the raw binary frame into an array of values.
* @param {array} frame - Raw binary data (byte array)
* @returns {array} Array of the decoded floating-point parameters
*/
function parse(frame) {
   // 1. Verification of the frame length (72 bytes required for 18 float variables)
   if (frame.length < 72) {
      return [];
   }

   // 2. Copying the data to the local memory buffer
   let buffer = new ArrayBuffer(72);
   let bytes = new Uint8Array(buffer);
   for (let i = 0; i < 72; i++) {
      bytes[i] = frame[i];
   }

   // 3. Extraction of the 32-bit values (Little-Endian = true)
   let view = new DataView(buffer);
   let V_avg  = view.getFloat32(0, true);
   let I_avg  = view.getFloat32(4, true);
   let U_rms  = view.getFloat32(8, true);
   let I_rms  = view.getFloat32(12, true);
   let V1_rms = view.getFloat32(16, true);
   let I1_rms = view.getFloat32(20, true);
   let P      = view.getFloat32(24, true);
   let P1     = view.getFloat32(28, true);
   let Q1     = view.getFloat32(32, true);
   let S      = view.getFloat32(36, true);
   let S1     = view.getFloat32(40, true);
   let PF     = view.getFloat32(44, true);
   let DPF    = view.getFloat32(48, true);
   let DF     = view.getFloat32(52, true);
   let THD_u  = view.getFloat32(56, true);
   let THD_i  = view.getFloat32(60, true);
   let f      = view.getFloat32(64, true);
   let Q      = view.getFloat32(68, true);

   // 4. Return of the decoded array to the graphical interface
   return [V_avg, I_avg, U_rms, I_rms, V1_rms, I1_rms, P, P1, Q1, S, S1, PF, DPF, DF, THD_u, THD_i, f, Q];
}
```

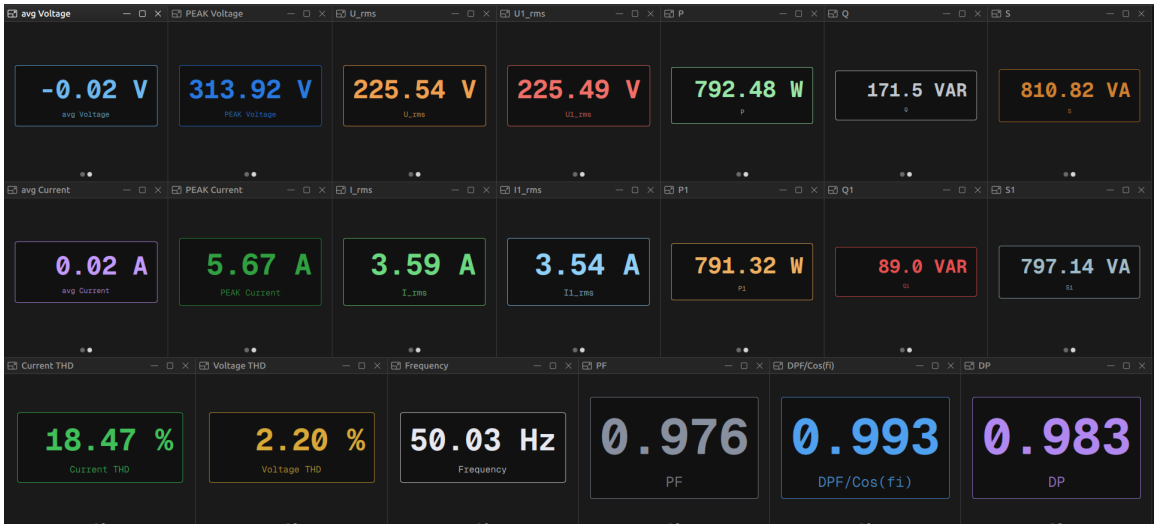


Figure 5.2: View of the operator panel in the Serial Studio application – the numerical-indicator section presenting the aggregated power quality parameters. Source: own elaboration.



Figure 5.3: View of the operator panel in the Serial Studio application – the charts section enabling the analysis of the trends of the grid parameters over time. Source: own elaboration.

Chapter 6

Tests and validation of the measurement system

6.1 Calibration of the measurement channels

The first stage in verifying the correct operation of the designed device was the calibration of the voltage channel and the current channel. This process was divided into two phases: compensation of the zero error and determination of the gain coefficient.

The zero-error compensation procedure did not require any specialised apparatus. For the voltage channel it consisted in physically short-circuiting the measurement terminals, whereas for the current channel an open circuit was ensured, guaranteeing the absence of current flow. As a result of the automatic calibration procedure implemented in the meter's software (averaging the results over a period of one second), offset values were written to the hardware registers of the DFSDM module. For the voltage channel, the raw value read from the converter was 420, which translates into a constant error of the order of 44.87 mV. For the current channel, the raw value was -31, which corresponds to an error of the order of -0.24 mA. From that point onwards, these values are automatically subtracted from every read sample.

The next step was to determine the gain coefficient for the voltage channel. This required preparing a measurement stand capable of generating highly stable reference values. For this purpose, the programmable laboratory power supply *EA-PS 9200-25* was used. It was used to generate a direct voltage of 200 V DC, which was applied to the meter's input terminals. This value was simultaneously and independently verified using a reference multimeter. After performing a one-second acquisition in auto-calibration mode, the microcontroller computed the final multiplier (scale coefficient) of $1.0683e-04$. This value was permanently stored in the device's external EEPROM memory.

6.2 Verification of the accuracy of the voltage channel

In order to validate the performed calibration and to examine the linearity of the voltage channel, a series of measurements was carried out over the full working range from -200 V to 200 V DC. Data from the microcontroller were acquired in raw mode (`RAW`), saved to `.csv` files using the Serial Studio application, and then post-processed in the GNU Octave environment.

An example time waveform of the recorded signal for the set voltage of -40 V is presented in Fig. 6.1. The grey colour marks the raw samples returned by the device, the red dashed line marks the reference value from the power supply, whereas the green curve represents the signal smoothed using a moving-average filter (with $n = 100$) applied to improve the readability of the visualisation.

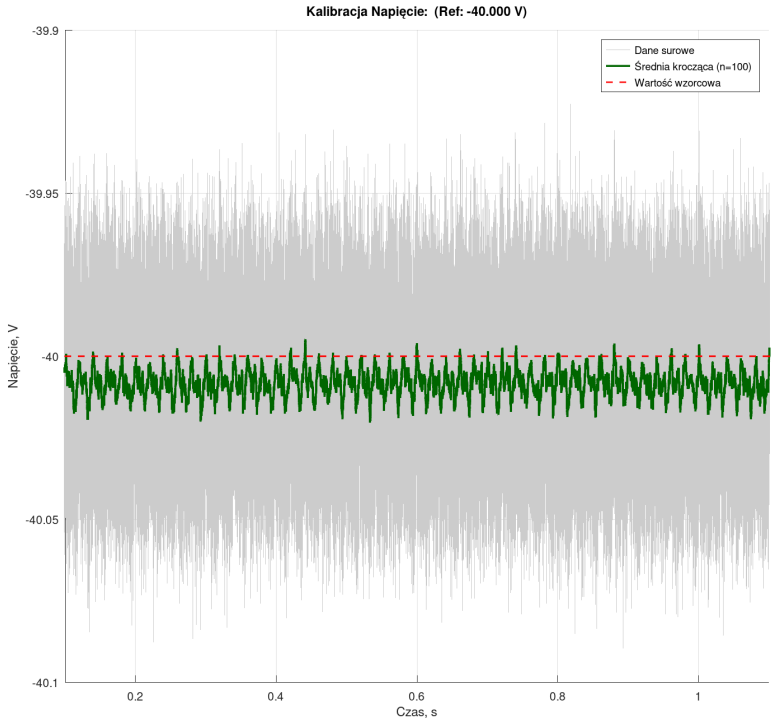


Figure 6.1: Time waveform of raw and smoothed voltage samples for the reference value of -40 V. Source: own elaboration in GNU Octave.

Analysing the waveform, one can observe a distinct oscillatory character of the recorded voltage signal. This phenomenon directly reflects the inherent ripple of the laboratory power supply used for the tests. It was observed that for extreme values, close to the limits of the power supply's operating range (e.g. at 200 V or -200 V), the character of the voltage changes to a sawtooth shape and the ripple amplitude increases noticeably. Such a case for the voltage of 200 V is illustrated in Fig. 6.2. Since, when verifying the calibration

process, the results are averaged from a 1-second measurement, these oscillations do not have a major influence. However, they could have slightly determined the stability of the reference point during the calibration procedure itself.

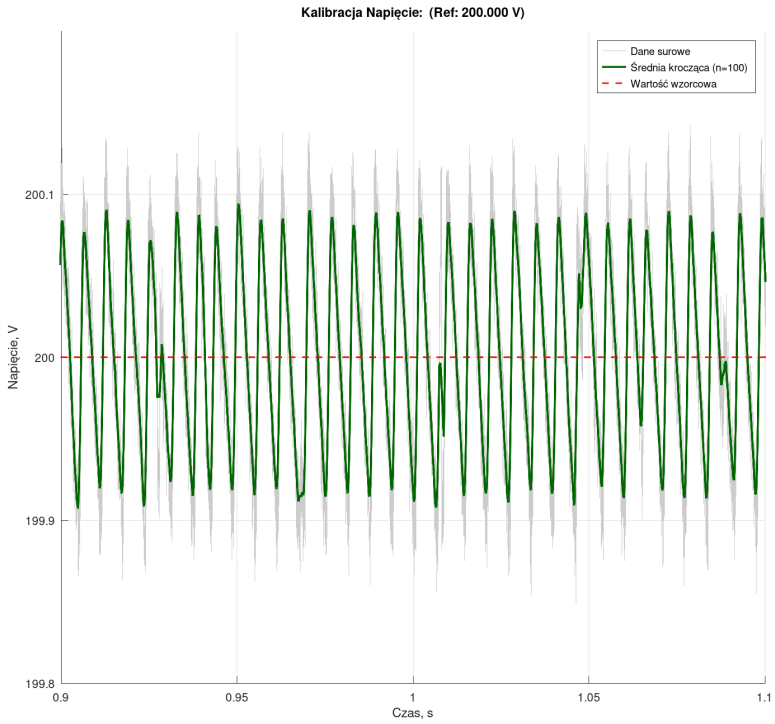


Figure 6.2: Time waveform of the voltage for the reference value of 200 V, illustrating the increased ripple amplitude of the power supply. Source: own elaboration.

From the data obtained for the individual measurement points, mean values and deviations were determined. A summary of the results, including the absolute error (the difference between the computed mean and the reference value), is presented in Table 6.1.

On the basis of Table 6.1, the characteristic of the absolute and relative error as a function of the reference voltage was generated and presented in Fig. 6.3.

Analysing the presented characteristic, the following conclusions were formulated:

1. **High overall accuracy:** Despite the non-linear and oscillatory character of the error plot, the absolute deviations of the readings over the entire range of ±200 V do not exceed the limit of ±0.011 V.

2. **Symmetry about the zero point:** It can be observed that the plot is symmetrical about the point (0,0), which implies that the ADC circuit used has a predictable character of the measurement error in the case of a change of polarity.

3. **Minimisation of the error at the calibration point:** As expected, the error takes the lowest values around 200 V, which is a direct consequence of performing a single-point calibration of the gain coefficient precisely for that supply value.

Table 6.1: Summary of measurement results verifying the accuracy of the DC voltage channel.

Reference value [V]	**Measured mean** [V]	**Absolute error** [V]	**Relative error** [%]
-200,00	-200,0038	-0,0038	-0,0019
-150,00	-149,9909	0,0091	0,0061
-100,00	-99,9994	0,0006	0,0006
-75,00	-75,0097	-0,0097	-0,0130
-50,00	-50,0031	-0,0031	-0,0062
-40,00	-40,0082	-0,0082	-0,0205
-30,00	-30,0101	-0,0101	-0,0337
-20,00	-20,0095	-0,0095	-0,0476
-10,00	-10,0070	-0,0070	-0,0703
0,00	0,0010	0,0010	—
10,00	10,0045	0,0045	0,0445
20,00	20,0072	0,0072	0,0361
30,00	30,0072	0,0072	0,0241
40,00	40,0064	0,0064	0,0160
50,00	50,0004	0,0004	0,0008
75,00	75,0069	0,0069	0,0092
100,00	99,9960	-0,0040	-0,0040
150,00	149,9919	-0,0081	-0,0054
200,00	199,9992	-0,0008	-0,0004

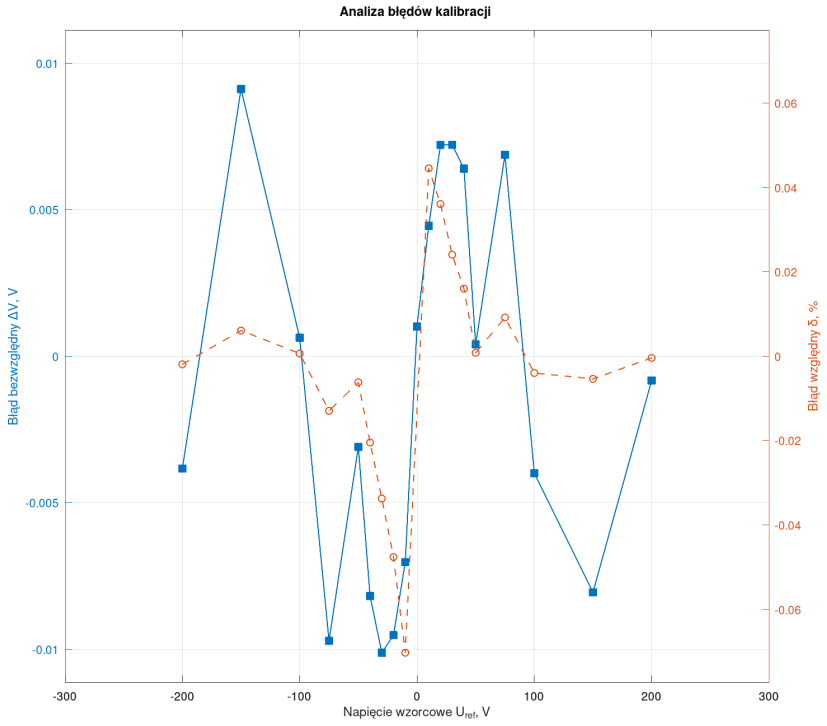


Figure 6.3: Characteristic of the absolute and relative error of the voltage channel as a function of the set DC reference voltage. Source: own elaboration.

4. **Inaccuracy of the zero compensation:** The entire absolute-error characteristic shows a slight shift towards positive values (asymmetry about the OX axis). This indicates that the offset determined at the beginning, of the order of 0 V, was not ideal and a small, uncompensated DC component remained in the circuit.

5. **Behaviour of the relative error:** The largest values of the relative (percentage) error were recorded in the immediate vicinity of zero volts. This is a completely natural phenomenon for A/D converters – at a constant level of noise and absolute error, the decreasing reference value in the denominator of the formula causes a mathematical, asymptotic increase in the relative error.

6.3 Verification and calibration of the current channel

6.3.1 Test setup and the phenomenon of thermal drift

The calibration of the current channel posed a considerably greater metrological challenge than in the case of the voltage channel. The use of the direct method, based on a series shunt resistor, is associated with the inevitable phenomenon of heat dissipation in the measuring element. An increase in the temperature of the resistor structure leads to a change in its resistance, which directly translates into a measurement error referred to as thermal drift. This phenomenon becomes more pronounced the greater the current flowing through the circuit.

A modified test setup was used to carry out the calibration process and the validation of the measurements. The *EA-PS 9200-25* power supply operated in voltage-stabilisation mode (set to a safe level of 10 V). As a precise source forcing the current flow, the programmable DC electronic load *Tenma 72-13210* [43] was used. The use of a relatively low supply voltage was dictated by the necessity not to exceed the maximum permissible power dissipated by the active load (amounting to 300 W). The process of calibrating the current gain coefficient itself was carried out for a reference value equal to 10 A.

Most of the measurement points in the basic accuracy test were collected using the pulse method – consisting in quickly switching on the set current, immediately acquiring the data and breaking the circuit. This methodology was intended to minimise the influence of the resistor's self-heating on the reading. The exception were the points for high loads (10 A, ± 16 A and -20 A), which were extracted from longer trials verifying the behaviour of the circuit in steady states, with the use of forced cooling.

6.3.2 Analysis of time waveforms

In order to illustrate the influence of temperature on the measurement channel, the waveforms for different current values were juxtaposed. Fig. 6.4 presents the measurement for a current of 1 A. Similarly to the voltage measurement, a slightly oscillatory character of the signal originating from the power supply is visible here, yet the measured value remains stable over time (no noticeable heating). The situation changes in Fig. 6.5, where for a current of 8 A a linear trend can already be observed – a slow increase in the read value resulting from thermal drift.

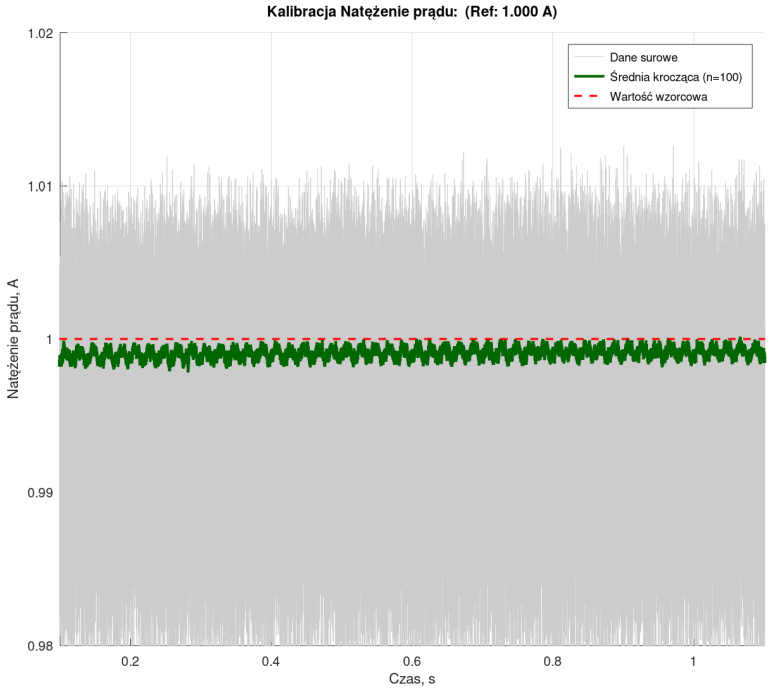


Figure 6.4: Time waveform of the measurement of a 1 A current. Source: own elaboration.

Key observations are provided by the analysis of the long-duration test for a current of 16 A (Fig. 6.6), recorded for nearly 3 minutes. At the beginning of the measurement, with the resistor not yet heated, the recorded value lies below the reference value. The reading then increases in an inertial manner, tending towards thermal stabilisation. The small, sudden spikes visible in the characteristic correlate with minor adjustments of the direction of the cooling air stream. The turning point on the plot is the 120th second of the experiment – at this point the fan was precisely directed at cooling the current shunts. This caused an immediate drop in the temperature of the element and, consequently, a decrease and stabilisation of the readings in the immediate vicinity of the actual value (around the 138th second – this point was adopted for the final error analysis). A phenomenon of identical inertial character but with the opposite mathematical sign was recorded for a forcing of -16 A (Fig. 6.7).

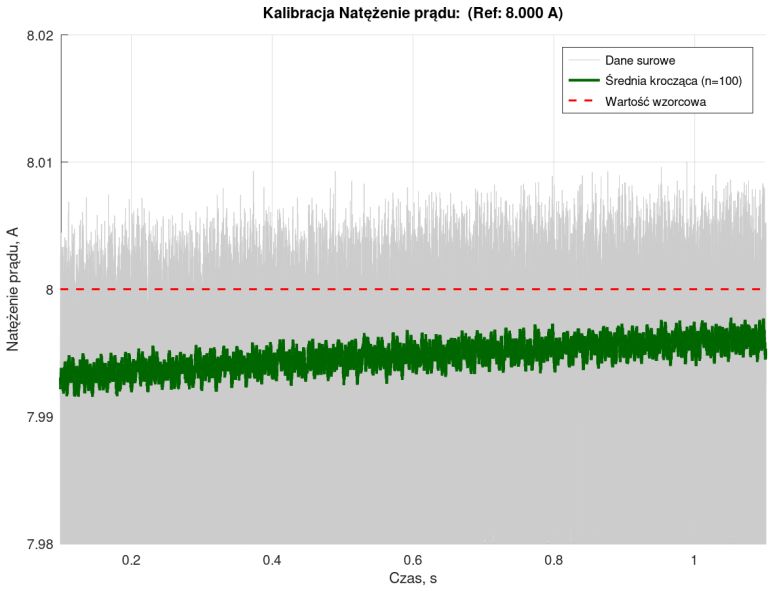


Figure 6.5: Time waveform of the measurement of an 8 A current. Source: own elaboration.

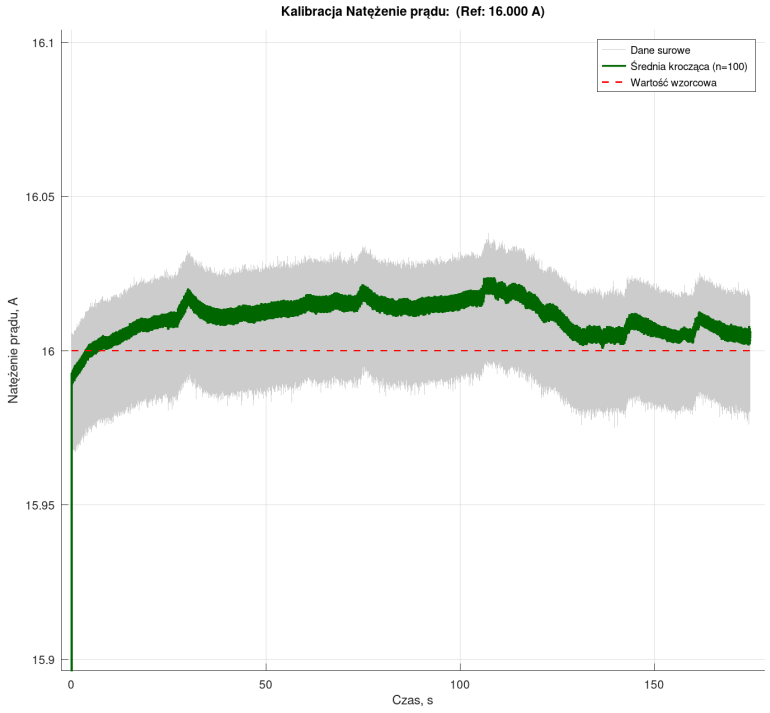


Figure 6.6: Influence of thermal drift on the measurement of a 16 A current over 180 seconds. Source: own elaboration.

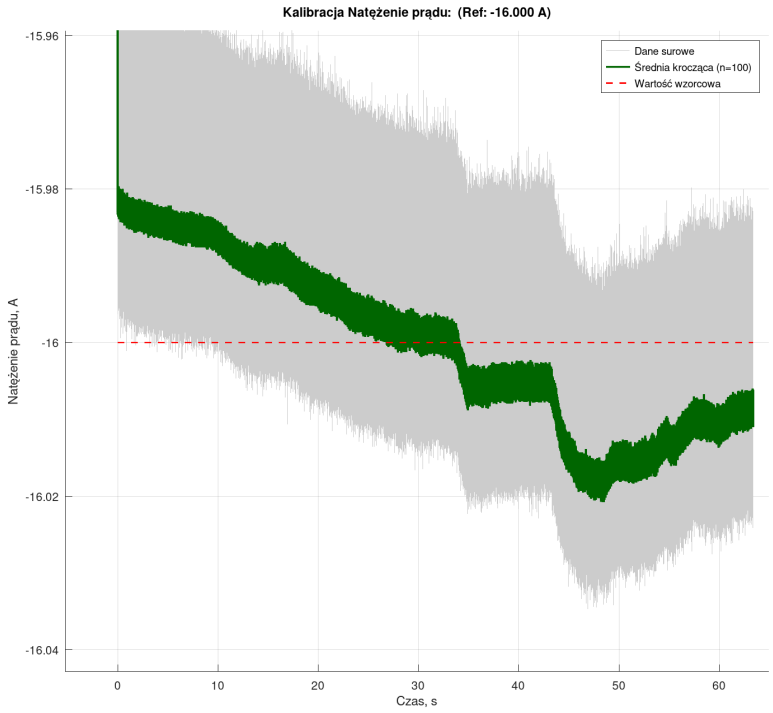


Figure 6.7: Time waveform of the measurement of a -16 A current. Source: own elaboration.

6.3.3 Assessment of measurement accuracy

The final values from the current measurements were juxtaposed with the reference values in Table 6.2.

The data from the table were transferred onto a plot of the error as a function of the set current value, presented in Fig. 6.8. This characteristic perfectly illustrates the influence of the applied measurement methodology (the hybridisation of fast and thermally stabilised measurements) on the final errors. The points collected using the pulse method (without temperature stabilisation) show a linear character of the error and symmetry about the centre of the coordinate system, and their sign is negative (the result did not manage to grow together with the temperature). In turn, the boundary points (adopted after stabilisation and with appropriate airflow) ended up on the opposite side of the error axis, retaining a higher value owing to the optimal thermal conditions.

The *post factum* analysis indicates that a more rigorous metrological approach would have been to perform each point in a fully established thermal state after waiting an appropriate time. Nevertheless, the total relative error values obtained over the entire analysed range do not exceed the limit of $\pm 0.13\%$. This should be regarded as a satisfactory result, which fully confirms the usefulness and calibration of the built current channel for the target power-quality studies under real conditions.

Table 6.2: Summary of measurement results verifying the accuracy of the DC current channel.

Reference value [A]	**Measured mean** [A]	**Absolute error** [A]	**Relative error** [%]
-20,00	-20,0247	-0,0247	-0,12
-16,00	-16,0170	-0,0170	-0,11
-13,00	-12,9874	0,0126	0,10
-10,00	-9,9933	0,0067	0,07
-8,00	-7,9950	0,0050	0,06
-5,00	-4,9973	0,0027	0,05
-3,00	-2,9982	0,0018	0,06
-1,00	-0,9996	0,0004	0,04
1,00	0,9991	-0,0009	-0,09
3,00	2,9977	-0,0023	-0,08
5,00	4,9982	-0,0018	-0,04
8,00	7,9948	-0,0052	-0,07
10,00	10,0092	0,0092	0,09
13,00	12,9924	-0,0076	-0,06
16,00	16,0208	0,0208	0,13

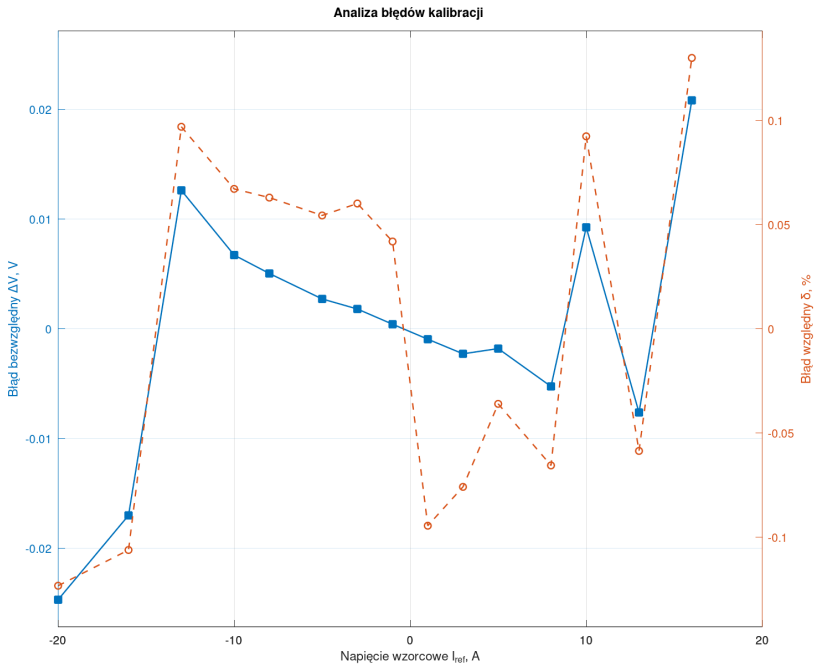


Figure 6.8: Characteristic of the absolute and relative error of the current channel. Source: own elaboration.

6.4 Object tests and validation with a Class A analyser

6.4.1 Test methodology and measurement stand

The final, key stage in the verification of the designed measurement system was to compare its readings with the results from a professional, industrial power quality analyser, the *Fluke 435-II* (a Class A device). For this purpose, a dedicated test setup was prepared, enabling simultaneous acquisition of data from the same circuit by both instruments.

Given that the industrial current transformers of the Fluke analyser are adapted to measure currents of considerable magnitudes, whereas the tested household appliances were characterised by relatively low power consumption, a technique of multiplying the magnetic flux was applied. The phase conductor supplying the load was wrapped eight times around the core of the current clamp of the reference analyser. The current and power readings from the Fluke device were then internally divided by eight. The prototype meter was connected into the circuit in the classical manner – in series.

Eight diverse electrical devices were selected for the tests, representing different load characteristics (resistive, inductive, non-linear switched-mode):

1. A resistive heater with a nominal power of 2 kW.
2. A compressor supplied by a single-phase induction motor (1.1 kW).
3. A three-phase motor (250 W) supplied via a single-phase frequency inverter (a Lenze inverter).
4. A classic fluorescent lamp.
5. A fan (from a heater).
6. A lamp equipped with an LED light source (6 W).
7. A switched-mode power supply of a Lenovo laptop in the no-load state (without the computer connected).
8. A switched-mode power supply of a Lenovo laptop under load (battery charging).

In parallel with the set of aggregated parameters, time waveforms from both devices were recorded in order to verify the shape of the read current and voltage waves. An example visual comparison of the time waveform for one of the loads is presented in Fig. 6.9.

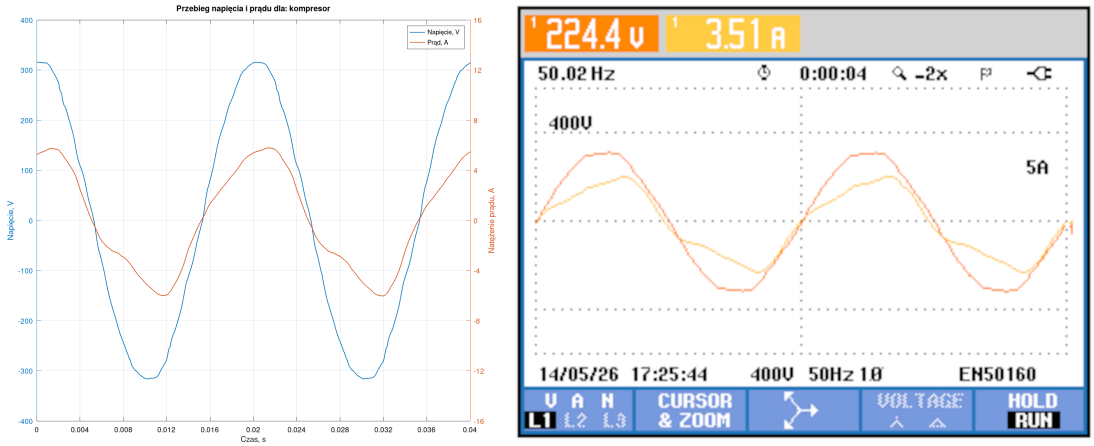


Figure 6.9: Comparison of the time waveforms of current and voltage: on the left, data exported from the prototype; on the right, a screenshot from the Fluke 438-II analyser. Source: own elaboration.

6.4.2 Summary of measurement results

The data acquired from both devices concerning the power components, RMS values and distortion coefficients were compiled in Tables 6.3 and 6.4. The prototype device is referred to in the tables as the *Meter*.

Table 6.3: Summary of the raw measurement results for devices with higher power consumption.

Parameter	Resistive heater		Compressor		Motor with inverter		Fluorescent lamp	
	Fluke	Meter	Fluke	Meter	Fluke	Meter	Fluke	Meter
U_{rms} [V]	221,80	223,41	224,18	225,18	225,95	226,26	226,12	226,10
U_{pk} [V]	309,10	311,34	312,30	313,56	314,20	314,79	314,70	314,69
I_{rms} [A]	7,87	7,92	3,44	3,39	0,91	0,92	0,47	0,47
I_{pk} [A]	10,94	11,04	5,47	5,45	3,15	3,26	0,83	0,86
P [W]	1740,00	1769,85	764,00	742,91	109,00	109,53	101,00	102,69
Q [VAR]	17,00	-8,80	170,00	175,65	-175,00	-176,17	-27,00	-27,36
S [VA]	1740,00	1769,88	783,00	763,39	206,00	207,45	105,00	106,27
PF	1,00	1,00	0,98	0,97	0,53	0,54	0,97	0,97
$\cos(\phi)$	1,00	1,00	0,99	0,99	1,00	0,97	0,97	0,98
THD_I [%]	2,40	2,08	19,50	19,80	154,20	151,22	13,50	14,14
f [Hz]	50,02	50,04	49,97	50,03	50,02	50,03	50,00	50,01

On the basis of the collected data, the absolute errors (Δ) and the relative errors in percent (δ) were determined, taking the values from the Fluke analyser as the reference values. The results of the error analysis were compiled in Tables 6.5 and 6.6, and their graphical representation in the form of bar charts was illustrated in Figs. 6.10, 6.11, 6.12, 6.13, 6.14 and 6.15.

Table 6.4: Summary of the raw measurement results for devices with lower power consumption.

Parameter	Fan		LED lamp		Power supply (no-load)		Power supply (loaded)	
	Fluke	**Meter**	**Fluke**	**Meter**	**Fluke**	**Meter**	**Fluke**	**Meter**
U_{rms} [V]	226,78	225,93	226,03	226,09	226,63	226,49	226,15	225,92
U_{pk} [V]	315,60	314,91	314,80	314,89	315,40	314,99	315,50	314,88
I_{rms} [A]	0,17	0,16	0,04	0,05	0,02	0,02	0,24	0,22
I_{pk} [A]	0,26	0,25	0,19	0,19	0,04	0,06	1,81	1,23
P [W]	24,00	22,92	6,00	6,20	0,00	0,18	22,00	21,23
Q [VAR]	28,00	27,94	-8,00	-8,71	-4,00	-4,09	-48,00	-44,39
S [VA]	37,00	36,13	10,00	10,69	4,00	4,09	53,00	49,20
PF	0,64	0,63	—	0,58	—	0,04	0,42	0,44
$\cos(\phi)$	0,65	0,64	—	0,94	—	0,05	0,98	0,96
THD_I [%]	15,00	13,98	131,20	126,51	25,50	32,71	199,90	192,44
f [Hz]	50,03	50,01	50,03	50,05	50,01	50,02	50,01	50,01

Table 6.5: Summary of the absolute error (Δ) and the relative error (δ) for higher-power devices.

Parameter	Resistive heater		Compressor		Motor with inverter		Fluorescent lamp	
	Δ	δ [%]	Δ	δ [%]	Δ	δ [%]	Δ	δ [%]
U_{rms} [V]	1,61	0,72	1,00	0,45	0,31	0,14	-0,02	-0,01
U_{pk} [V]	2,24	0,72	1,26	0,40	0,59	0,19	-0,01	0,00
I_{rms} [A]	0,05	0,66	-0,05	-1,45	0,01	0,76	0,00	0,00
I_{pk} [A]	0,10	0,90	-0,02	-0,44	0,11	3,61	0,03	4,17
P [W]	29,85	1,72	-21,09	-2,76	0,53	0,49	1,69	1,67
Q [VAR]	-25,80	-151,76	5,65	3,33	-1,17	0,67	-0,36	1,33
S [VA]	29,88	1,72	-19,61	-2,50	1,45	0,70	1,27	1,21
PF	0,00	-0,04	-0,01	-0,70	0,01	1,30	0,00	-0,32
$\cos(\phi)$	0,00	0,00	0,00	0,22	-0,03	-2,64	0,01	0,70
THD_I [%]	-0,32	-13,13	0,30	1,55	-2,98	-1,94	0,64	4,78
f [Hz]	0,02	0,03	0,06	0,13	0,01	0,02	0,01	0,01

Table 6.6: Summary of the absolute error (Δ) and the relative error (δ) for lower-power devices.

Parameter	**Fan**		**LED lamp**		**Power supply (no-load)**		**Power supply (loaded)**	
	Δ	δ [%]	Δ	δ [%]	Δ	δ [%]	Δ	δ [%]
U_{rms} [V]	-0,85	-0,38	0,06	0,03	-0,14	-0,06	-0,23	-0,10
U_{pk} [V]	-0,69	-0,22	0,09	0,03	-0,41	-0,13	-0,62	-0,20
I_{rms} [A]	-0,01	-5,94	0,01	18,25	0,00	-9,50	-0,02	-9,25
I_{pk} [A]	-0,01	-3,08	0,00	2,16	0,02	37,50	-0,58	-32,12
P [W]	-1,08	-4,52	0,20	3,39	0,18	—	-0,77	-3,50
Q [VAR]	-0,06	-0,22	-0,71	8,82	-0,09	2,13	3,61	-7,53
S [VA]	-0,87	-2,34	0,69	6,90	0,09	2,23	-3,80	-7,16
PF	-0,01	-1,05	—	—	—	—	0,02	4,88
$\cos(\phi)$	-0,01	-1,60	—	—	—	—	-0,02	-2,50
THD_I [%]	-1,02	-6,79	-4,69	-3,57	7,21	28,26	-7,46	-3,73
f [Hz]	-0,02	-0,03	0,02	0,05	0,02	0,03	0,00	0,00

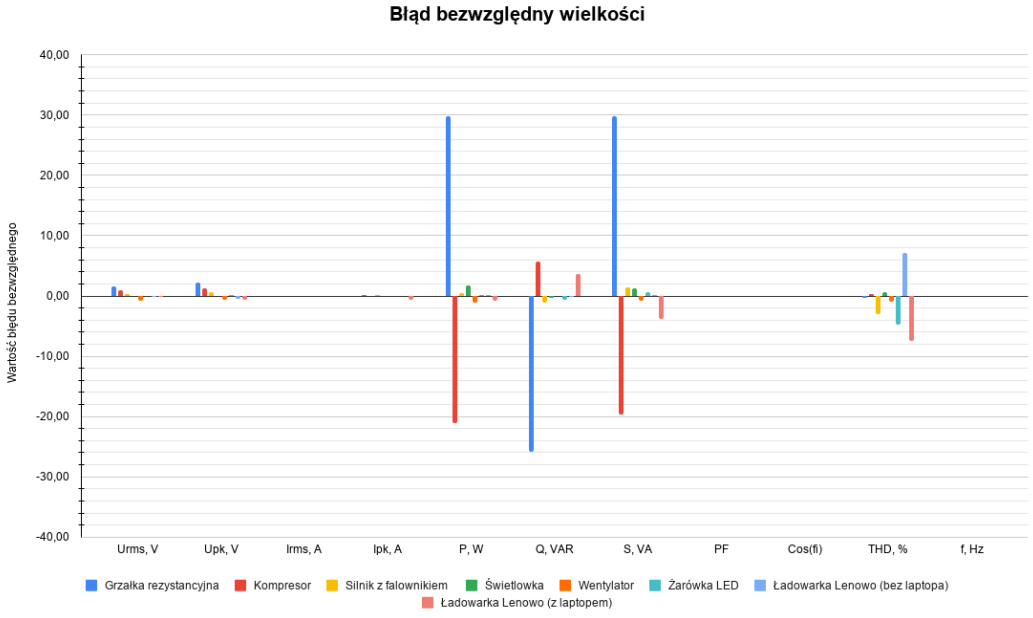


Figure 6.10: Absolute error of the measurements as a function of the tested parameter. Source: own elaboration.

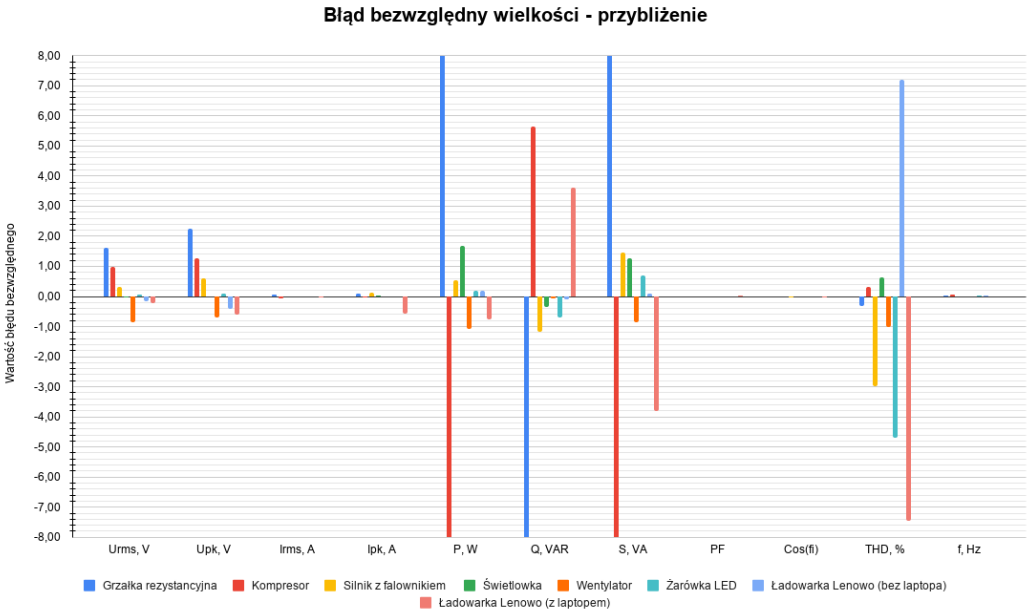


Figure 6.11: Absolute error of the measurements as a function of the tested quantity - close-up. Source: own elaboration.

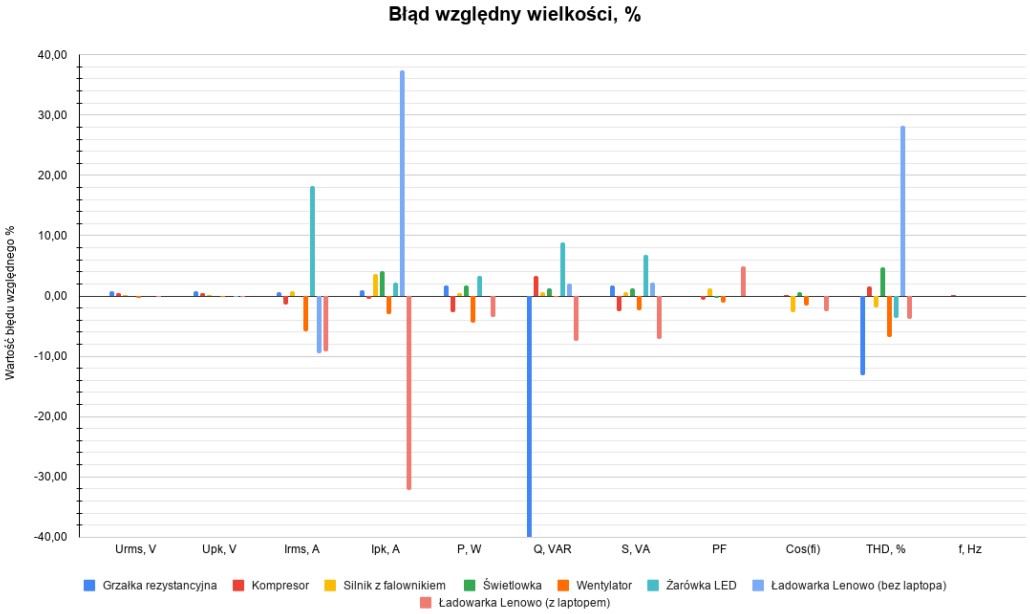


Figure 6.12: Relative error of the measurements as a function of the tested quantity. Source: own elaboration.

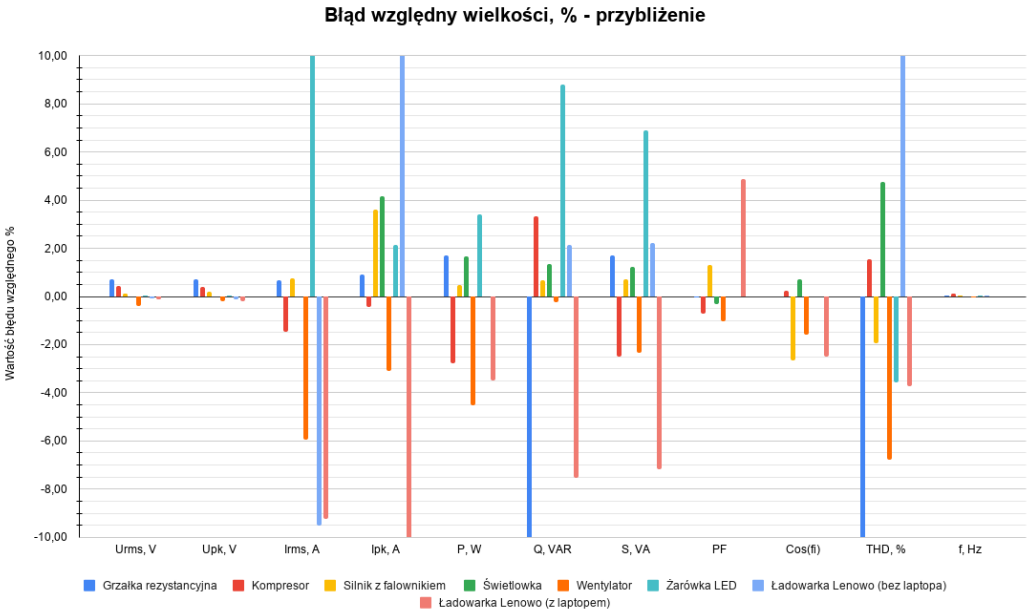


Figure 6.13: Relative error of the measurements as a function of the tested quantity - close-up. Source: own elaboration.

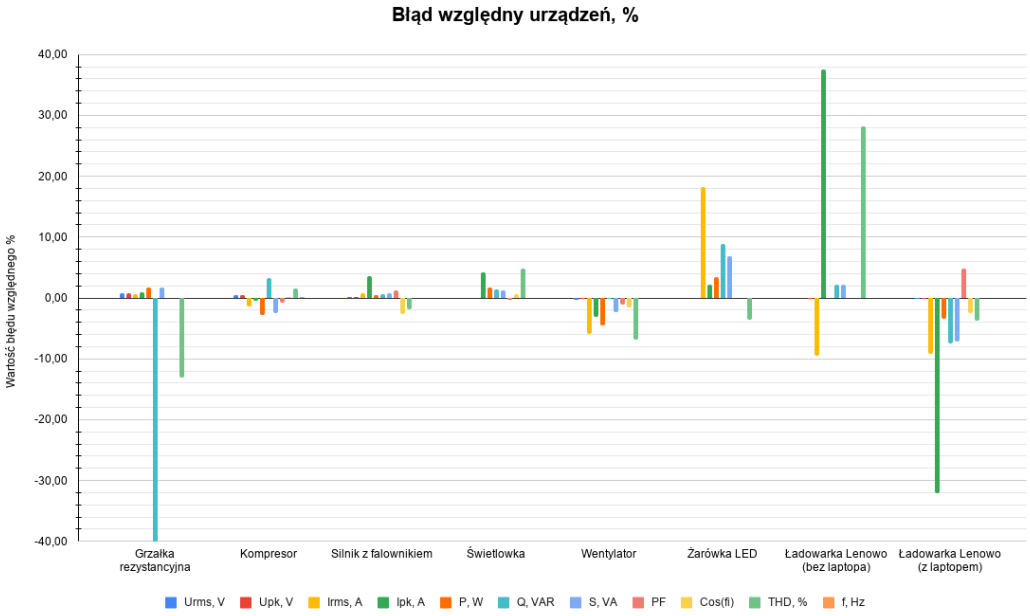


Figure 6.14: Relative error of the measurements as a function of the tested device. Source: own elaboration.

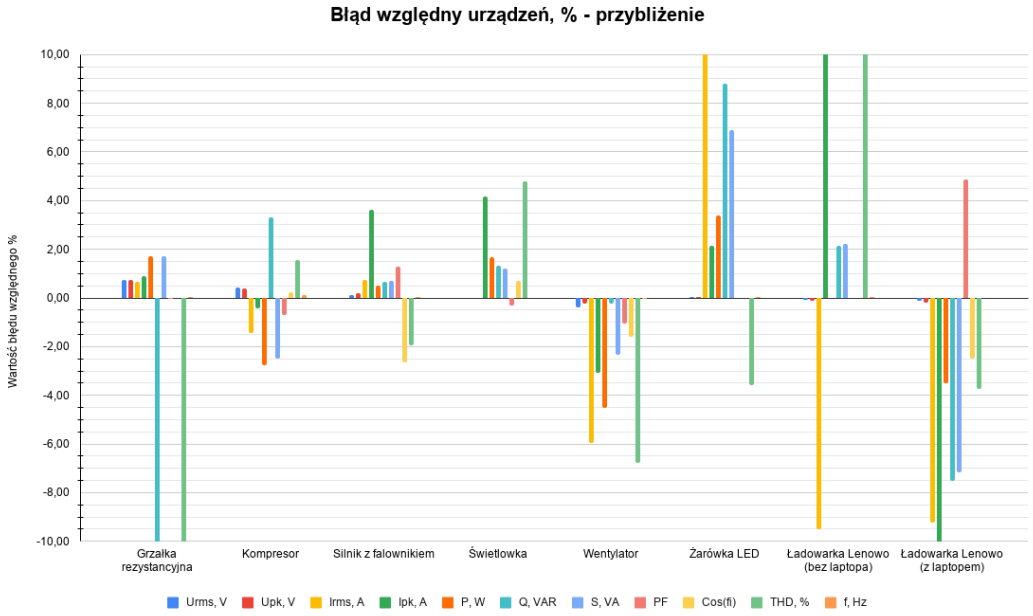


Figure 6.15: Relative error of the measurements as a function of the tested device - close-up. Source: own elaboration.

6.4.3 Error analysis and metrological conclusions

On the basis of the generated characteristics and the tabular data, an analysis was carried out, formulating the following conclusions:

- **High precision of the voltage channel and frequency:** The system exhibits high accuracy for the measurement of the RMS voltage (U_{rms}) and the peak voltage (U_{pk}), where the relative error in all trials remained below 1%. The frequency measurements were almost error-free (maximum error $\delta = 0.13\%$), which proves the high effectiveness of the applied SOGI-PLL algorithm.

- **Errors in the measurement of low-magnitude currents:** The largest absolute errors for the values I_{rms} and the peak currents I_{pk} were observed in the case of low-power devices. This follows directly from the fact that the measured values were very small and any relative error translates into a large absolute error. The differences in the measurements probably result from two reasons: the unsuitability of the Fluke meter for measuring such small values, and the faster sampling of the signal by the Fluke meter compared with the prototype (200 kS/s vs 78 kS/s). This translates into errors of other quantities for lower-power devices.

- **Anomaly in the measurement of reactive power for the resistive load:** A significant discrepancy was observed in the measurement of the reactive power of the heater, for which the Fluke indicated 17 VAR (inductive character), whereas the prototype circuit indicated -8.8 VAR (capacitive character). This probably results

from different computational algorithms. According to the specification [9], the Fluke analyser computes the fundamental reactive power directly from the difference of the phase angles: $Q_1 = U_1 I_1 \sin(\varphi_u - \varphi_i)$. In the prototype device this power is determined on the basis of the cross products of the orthogonal signals generated by the SOGI-PLL algorithm. The character of the heater is strongly resistive, which translates into a very small phase shift; the slight residual non-compensation of the prototype's measurement channel may cause large errors in this case.

- **Algorithmic differences in the THD calculations:** The fairly large percentage deviations in determining the total harmonic distortion of the current (THD_I) are a natural phenomenon when comparing devices of different mathematical architecture. The Fluke analyser computes THD on the basis of the classical Fast Fourier Transform (FFT), strictly realising a 200 ms window. The prototype meter estimates THD on an ongoing basis, using the ratio of the computed broadband (RMS) power to the fundamental component power from the SOGI circuit. Both of these algorithms behave differently for highly distorted currents, generating a natural margin of divergence.

- **Systematic character:** The errors for a given device maintain a constant sign (they lie on one side of the axis in the bar charts). This proves the absence of random disturbances and confirms that the discrepancies have a purely systematic basis, depending directly on the computational methodology and the non-linearity of the converters in the given operating bands. For high-power devices (compressor, motor, heater) these errors decrease to acceptable values.

6.5 Analysis of the load profiles of selected household appliances

The next stage of the research, after the metrological verification of the measurement circuit itself, was the analysis of the energy-consumption characteristics of selected household appliances in common use. The research focused on recording the time waveforms of the computed parameters as well as the direct current and voltage waveforms. This made it possible to illustrate the non-linear character of modern loads and its influence on the power quality in the supply network. Devices with different load characteristics were selected for the tests: an electric kettle, a treadmill, a table jigsaw and two models of vacuum cleaner (of an older and a newer generation).

6.5.1 Resistive load – electric kettle

The first tested device was a standard electric kettle, representing a load of a purely resistive character. In accordance with the theoretical expectations, the current waveform has the shape of an ideal sinusoid and is fully in phase with the supply voltage, which is confirmed by the oscillogram presented in Fig. 6.16.

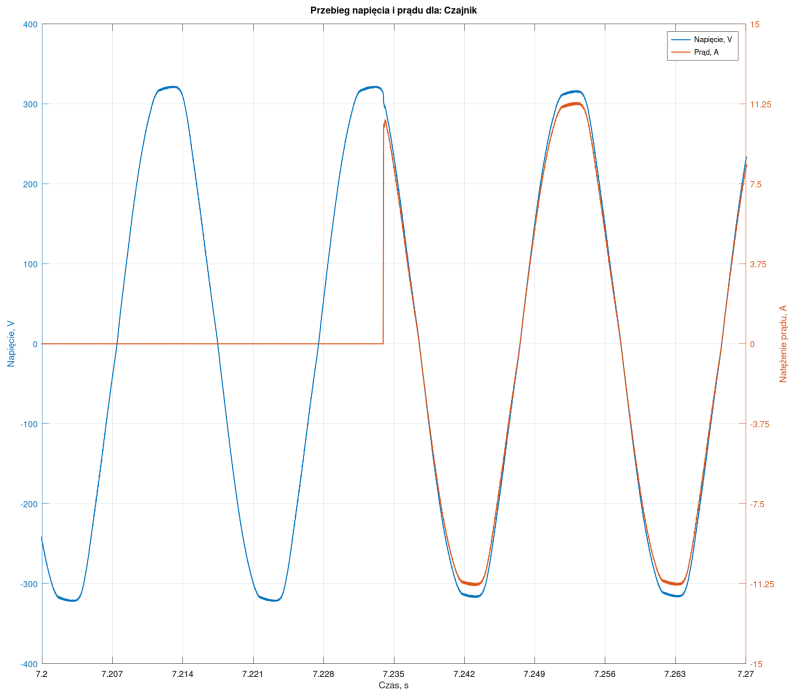


Figure 6.16: Time waveform of the voltage and current for the electric kettle. Source: own elaboration.

The purely active character of the energy consumption is also confirmed by the plot of the total powers (Fig. 6.17). The values of the active power (P) and the apparent power (S) coincide almost perfectly. An interesting phenomenon is the visible reactive-power spikes arising at the moments of switching the device on and off. Additionally, in the steady state the meter records a negligible value of reactive power of a capacitive character, which is an effect of the presence of parallel parasitic capacitances of the supply conductors, or it is an error resulting from the cause described in 6.4.3.

Owing to the very high current consumption of the device, on the plot of the aggregate analysis of the RMS values (Fig. 6.18) a clear drop in the value of the mains voltage (U_{rms}) was observed at the moment of the heater's operation. This is a direct consequence of the voltage drop across the impedance of the supply line of the domestic installation.

The course of the total harmonic distortion THD (Fig. 6.19) proves that the resistive load does not generate its own higher current harmonics – the value of THD_I precisely follows the distortions of the supply voltage itself (THD_U). The distinct peak in the

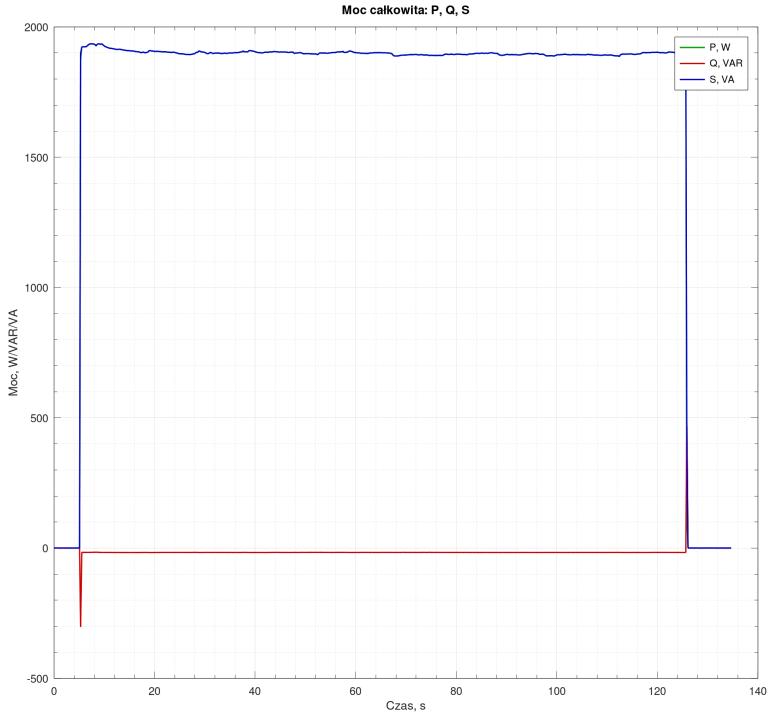


Figure 6.17: Time plot of the total active, reactive and apparent power for the electric kettle. Source: own elaboration.

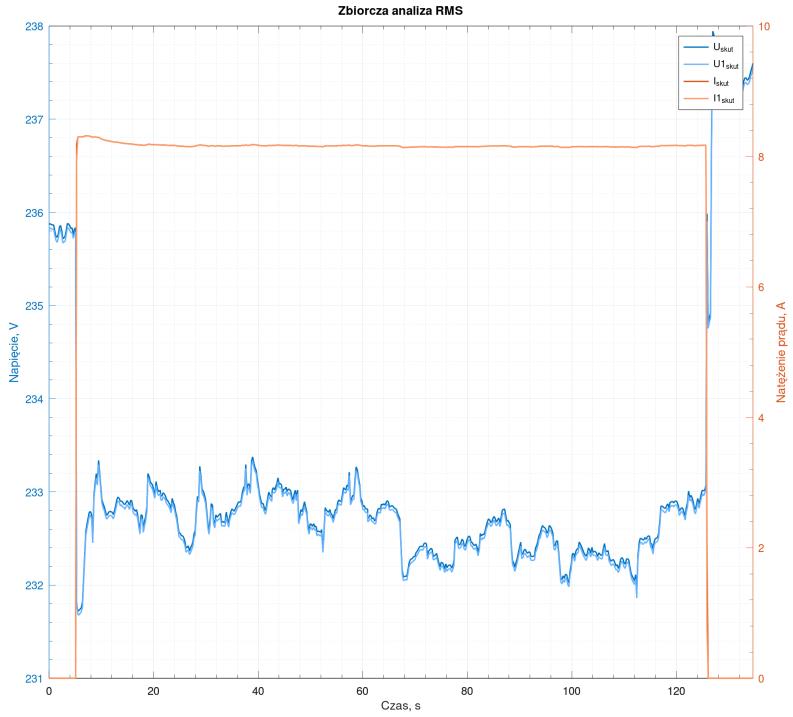


Figure 6.18: Aggregate analysis of the course of the RMS values of voltage and current for the electric kettle. Source: own elaboration.

voltage THD plot at the moment of switching on coincides in time with the aforementioned drop in the RMS value and the transient state.

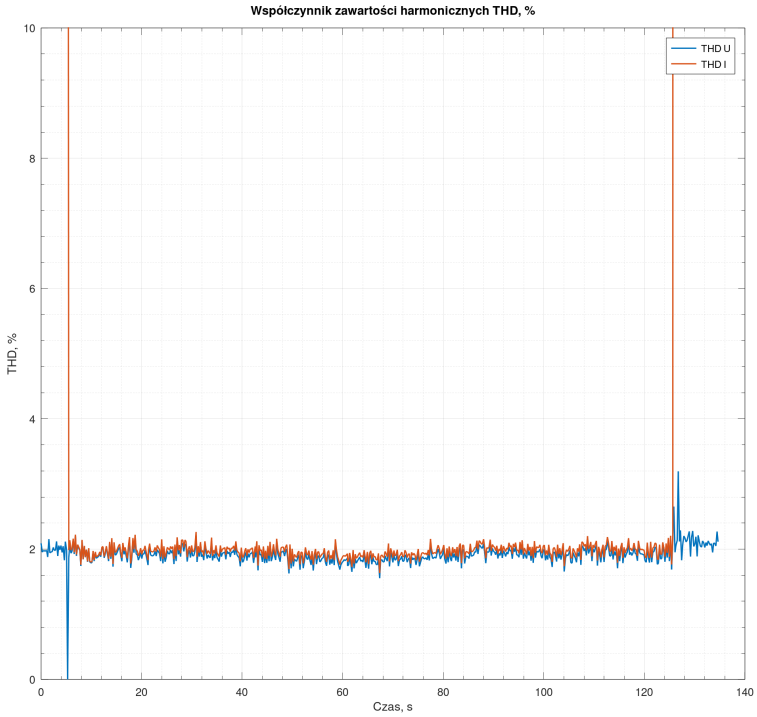


Figure 6.19: Course of the total harmonic distortion (THD) of the voltage and current for the electric kettle. Source: own elaboration.

6.5.2 Non-linear load – treadmill

The next tested object was a treadmill. The measurement methodology assumed starting the device, smoothly accelerating it to the maximum speed without load, and then physically loading the running belt by a running user.

As illustrated in Fig. 6.20, the current consumption has a strongly non-linear character. The current remains close to zero for most of the duration of the voltage half-wave, rising rapidly only around the peak of the voltage wave. On the falling edges of the current pulses, distinct oscillations of a higher frequency are visible. Such a current shape suggests the presence of a DC motor supplied through a simple rectifier circuit with filter capacitors of relatively small capacitance.

This character is directly reflected in the computed power factors (Fig. 6.21). The displacement power factor precisely describing the relation for the fundamental component ($DPF/\cos(\varphi_1)$) takes values close to unity. However, the true power factor of the device (PF), taking into account all harmonics, falls to a level of about 0.7. This proves that the circuit generates significant amounts of non-active power, which does not result from the

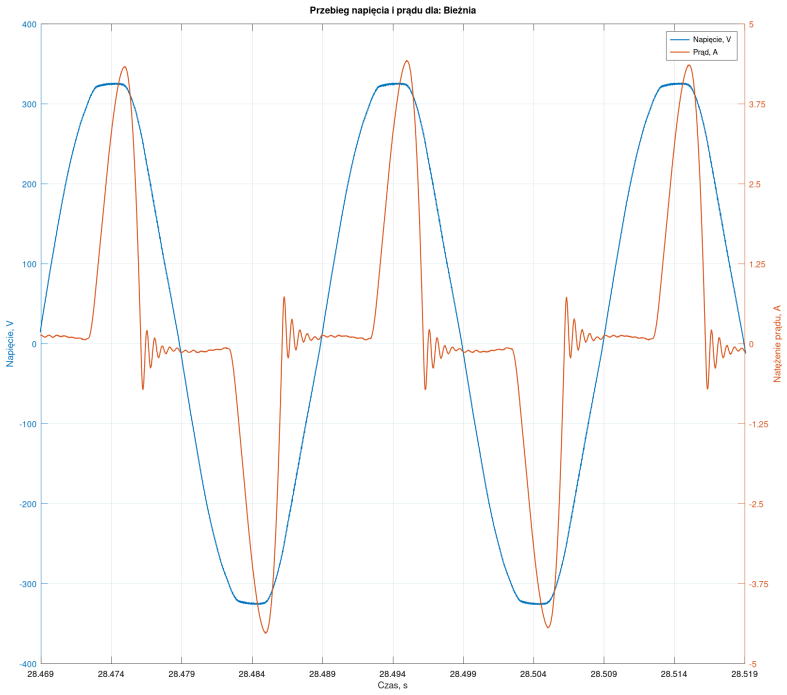


Figure 6.20: Time waveform of the voltage and current for the treadmill. Source: own elaboration.

classical phase shift but from the distortion power generated by the higher harmonics.

The physical loading of the device (the user's running) becomes clearly visible in the power plot (Fig. 6.22) as a sudden increase in the consumed power and strong fluctuations of the signal correlated with the cyclic strikes of the feet against the running belt. The variable load of the DC motor results in a dynamic change of the shape of the consumed current, which is most strongly reflected in the parameter PF, introducing distinct oscillations into it. At the same time, the displacement angle of the fundamental component (DPF) remains almost unchanged.

6.5.3 Asymmetric load – table jigsaw

The table jigsaw was tested by starting it at maximum speed with no load, and then simulating the working load by cutting a wooden element.

On the basis of the recorded oscillograms (Fig. 6.23), an extremely asymmetric current consumption was observed. The device draws energy from the network exclusively during one half-wave of the voltage (in this case the negative half-wave), remaining idle during the positive half-wave. The shape of the current is additionally strongly distorted and noisy. This indicates the use of a simple rotational-speed controller that cuts off the half-wave.

Such aggressive control of the energy consumption is reflected in the extremely low quality parameters. Both the true power factor (PF) and the displacement power factor

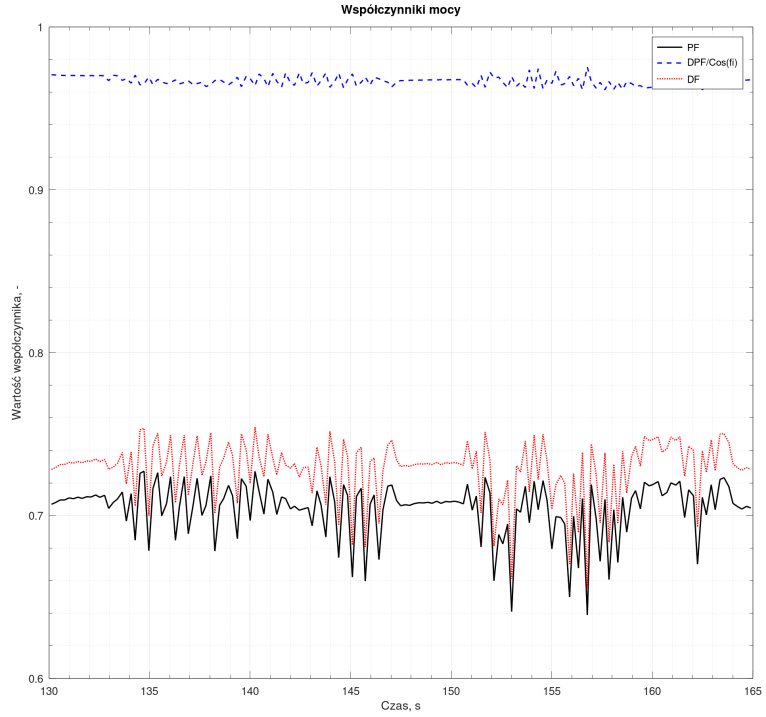


Figure 6.21: Course of the power factors (PF, DPF, DF) for the treadmill. Source: own elaboration.

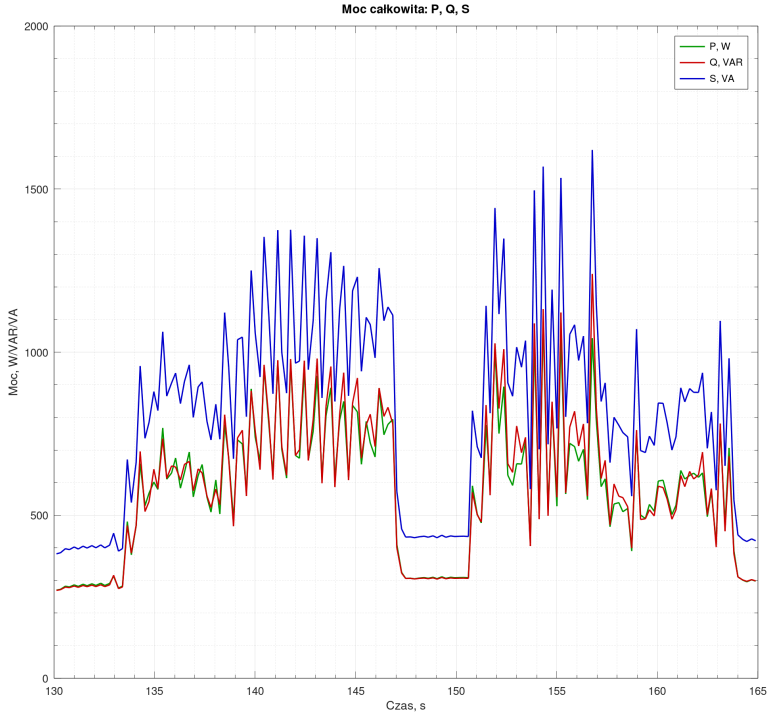


Figure 6.22: Time plot of the total active, reactive and apparent power for the treadmill. Source: own elaboration.

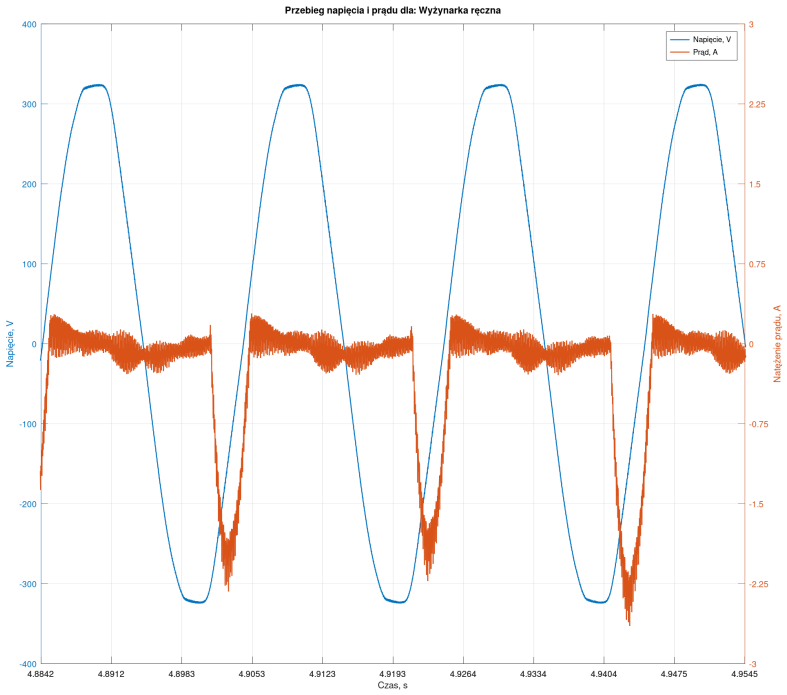


Figure 6.23: Time waveform of the voltage and current for the table jigsaw. Source: own elaboration.

(DPF) take very small values (Fig. 6.24).

Additionally, from the power and harmonic-content plots (Figs. 6.25, 6.26) it follows that idle running generates a coefficient THD_I reaching over 100%, and the main component drawn from the network in this state is reactive power. The transition to the loaded state (the cutting process) causes a noticeable improvement of the coefficients and a reduction of the THD parameter.

6.5.4 Comparison of power-supply technologies – vacuum cleaners

In order to illustrate the progress in the field of designing power supplies for household appliances, a comparative study of two vacuum cleaners was carried out: an older model of high power and a newer model of lower nominal power. The test procedure for both devices consisted in starting them with free airflow, and then momentarily completely blocking the suction pipe, simulating extreme loading or a clogged filter.

The juxtaposition of the current waveforms (Fig. 6.27) shows that the newer vacuum cleaner is characterised by a considerably gentler current consumption, closer to a sinusoid, compared with the older model, whose current is slightly jagged and bent in the vicinity of zero.

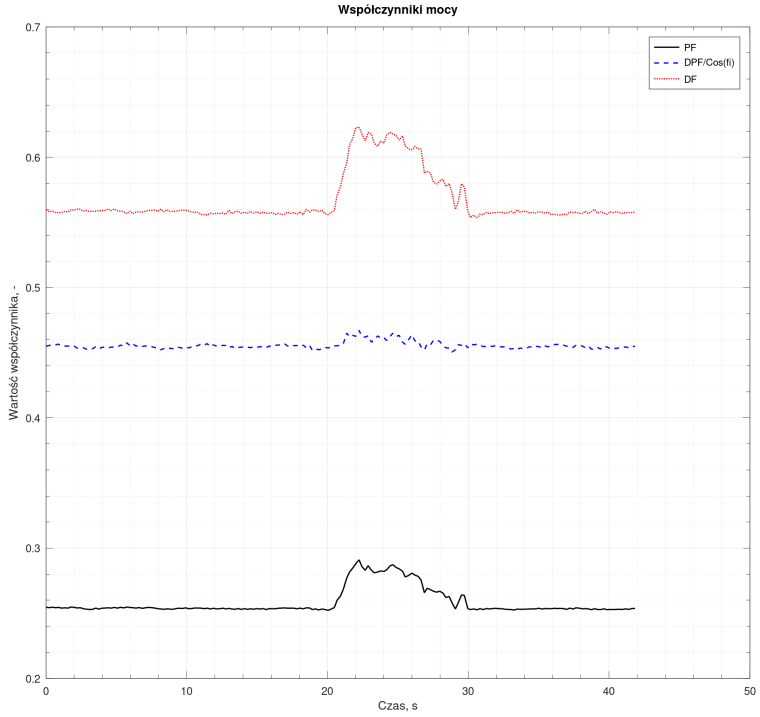


Figure 6.24: Course of the power factors (PF, DPF, DF) for the table jigsaw. Source: own elaboration.

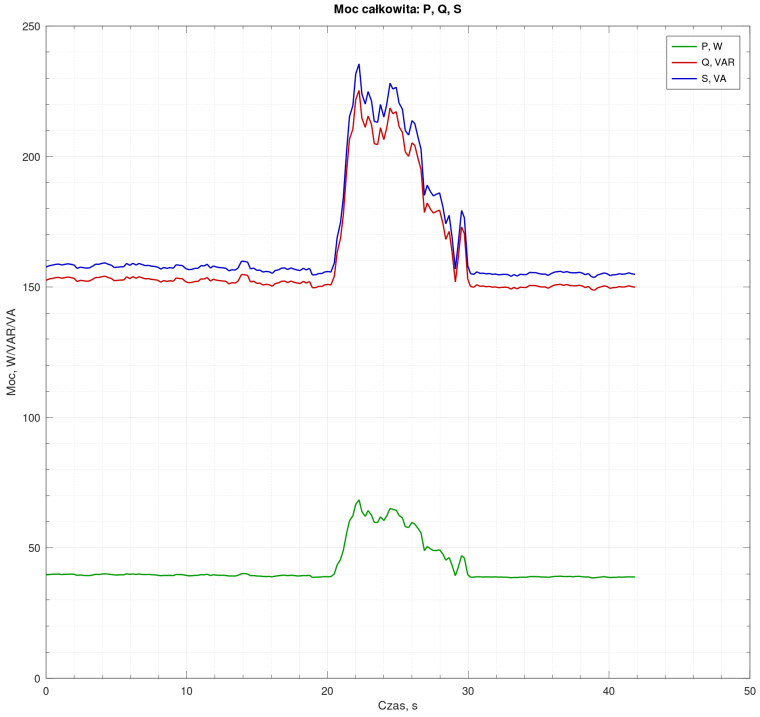


Figure 6.25: Time plot of the total active, reactive and apparent power for the table jigsaw. Source: own elaboration.

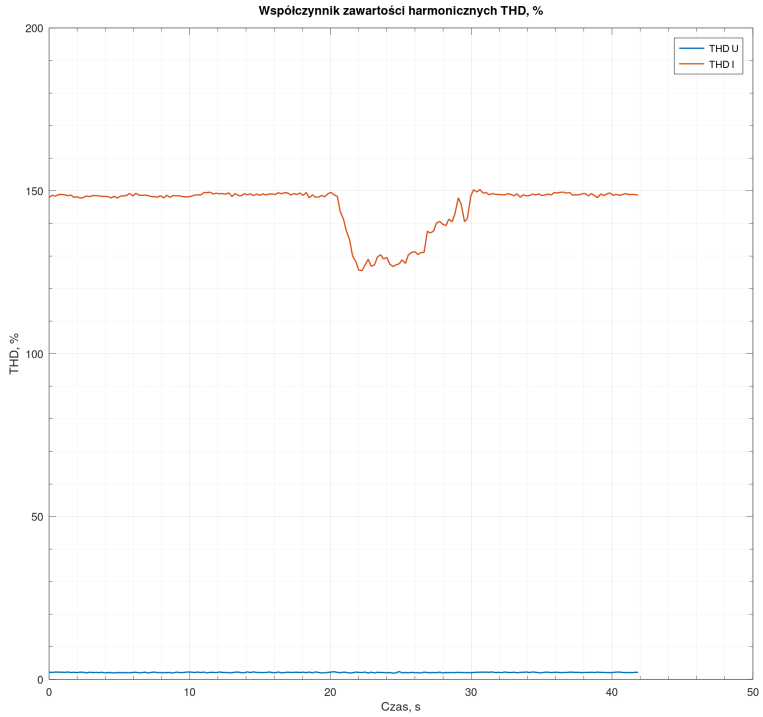
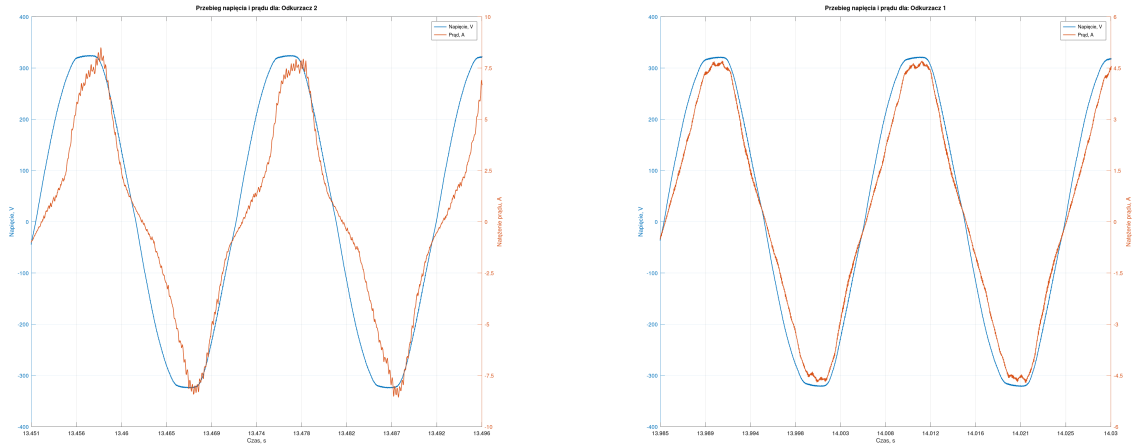


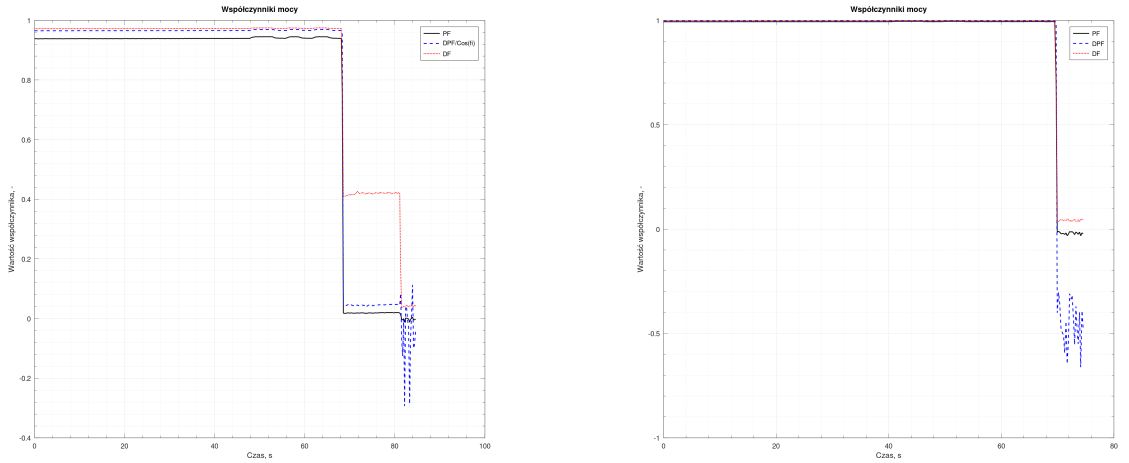
Figure 6.26: Course of the total harmonic distortion (THD) of the voltage and current for the table jigsaw. Source: own elaboration.



(a) Older model of the vacuum cleaner (b) Newer model of the vacuum cleaner

Figure 6.27: Comparison of the oscillograms for the vacuum cleaner of the older and newer generation. Source: own elaboration.

The differences are particularly visible in the distortion parameters. The newer vacuum cleaner generates harmonics at a level of merely $THD_I \approx 9\%$, whereas the older model reaches $THD_I \approx 24\%$. The improvement in the quality of the consumed energy in the new model also translates into the true power factor PF, which remains almost at the ideal level of 1.0, whereas for the old device slight drops are recorded (Fig. 6.28). This indicates the use of a better motor-supply circuit in the newer device.



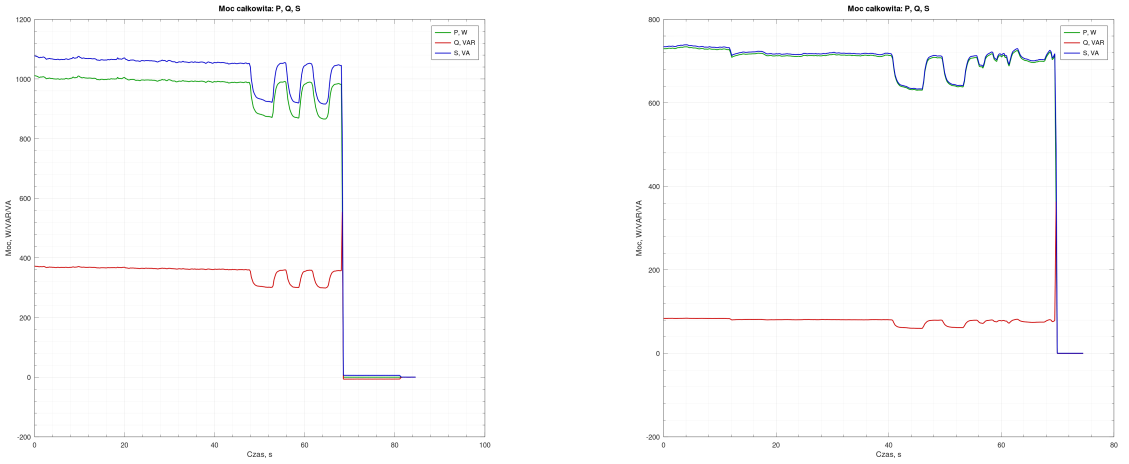
(a) Older model of the vacuum cleaner (b) Newer model of the vacuum cleaner

Figure 6.28: Comparison of the power factors (PF, DPF, DF) for both generations of vacuum cleaners. Source: own elaboration.

A phenomenon surprising to the average user, yet fully explicable physically, observed in both cases, is the reaction of the devices to the blocking of the suction pipe. As can be seen in the power plots (Fig. 6.29), blocking the airflow results in a sudden and drastic drop in the consumed electrical power. This phenomenon follows directly from the fact that in fan and turbine systems the torque loading the motor (and thus the aerodynamic work to be performed) is directly proportional to the mass of the air being moved. Blocking the pipe cuts off the mass flow of air; the rotor rotates in a rarefied environment (performing less useful work), owing to which the motor speeds up, but at the same time draws considerably less active power from the power network.

6.6 Analysis of the daily load profile of residential premises

In order to verify the operation of the designed device under long-term and variable conditions, an analysis of the load profile of a typical residential premises was carried out. Owing to the requirement of series measurement of the prototype circuit, a temporary modification inside the domestic electrical switchboard was necessary. The measurements



(a) Older model of the vacuum cleaner (b) Newer model of the vacuum cleaner

Figure 6.29: Comparison of the profiles of the total powers (P, Q, S) for the tested vacuum cleaners during the blocking of the suction pipe. Source: own elaboration.

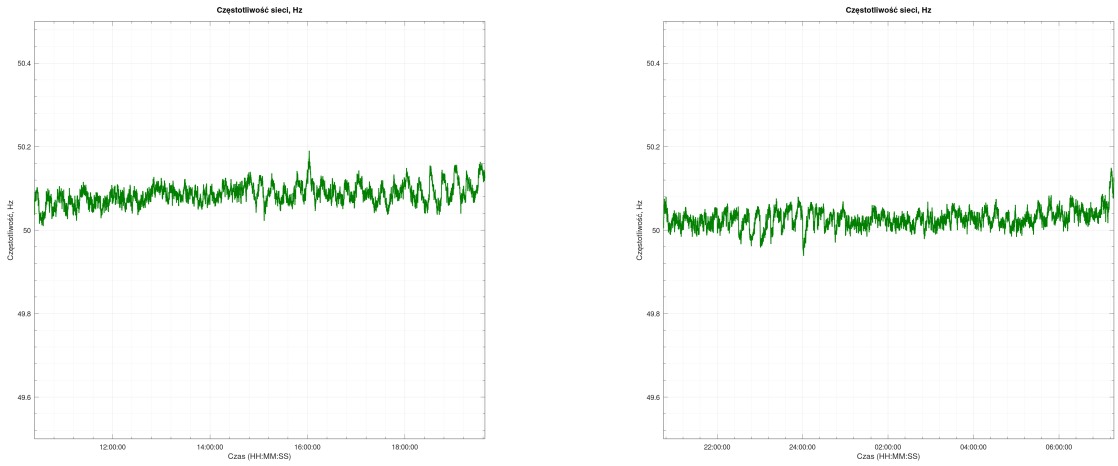
covered a separated group of single-phase circuits, which included the lighting, the general-purpose socket outlets, the bathroom circuits (including the washing machine) and the kitchen circuits (excluding the three-phase circuit of the hob and the oven).

The data acquisition was divided into two independent intervals in order to highlight the differences in the usage characteristics of the installation: a daytime interval (recording from about 10:30 to 20:00) and a night-time interval (recording from about 21:00 to 07:00).

6.6.1 Analysis of frequency and power

The analysis of the collected data began with the verification of the frequency of the mains voltage. The recorded waveforms for the daytime and night-time periods (Fig. 6.30) indicate that the frequency did not remain at the ideal level of 50.00 Hz, showing a slight shift towards higher values. These oscillations, however, remain with a large margin within the tolerance limits defined by the transmission systems and the PN-EN 50160 standard.

Key observations are provided by the analysis of the courses of the active power (P), reactive power (Q) and apparent power (S) compiled in Fig. 6.31. The overall character of the apartment's load in both time intervals is unambiguously capacitive (negative values of the power Q). The capacitive reactive power remains at a relatively constant baseline level, with a gradual, stepwise change noticeable depending on the time of day – from about -300 var during the day, through about -400 var in the evening hours, up to about -500 var at night. This results from the continuous operation of the capacitive interference-suppression filters in home appliances and consumer electronics, the network infrastructure (routers, home servers) and the switched-mode power supplies remaining in standby mode.



(a) Daily profile (10:30 – 20:00)

(b) Night-time profile (21:00 – 07:00)

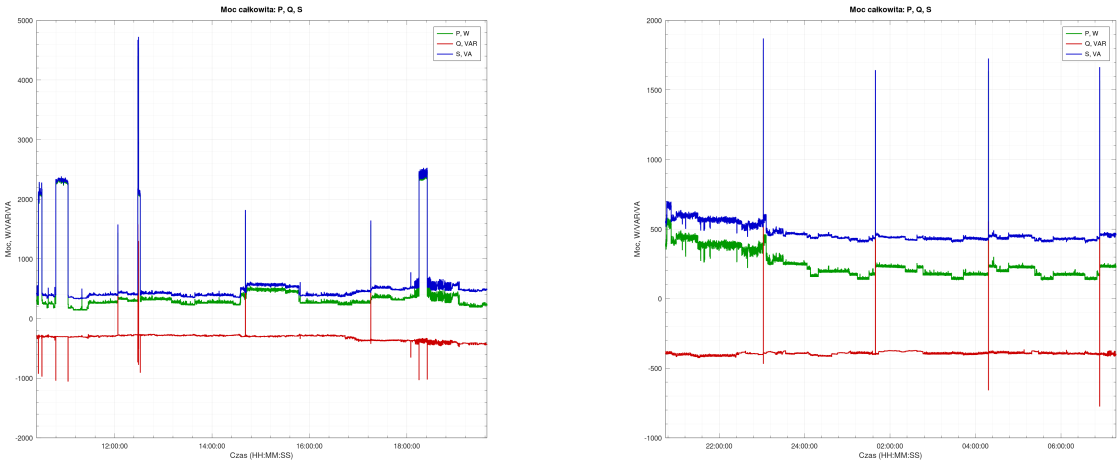
Figure 6.30: Course of the frequency of the mains voltage in the tested domestic installation. Source: own elaboration.

The baseline active-power consumption shows an increasing tendency around 18:00–19:00, which is a natural consequence of switching on the domestic lighting, and then noticeably falls after 23:00. In the night-time profile the cyclic switching on of the compressors of the refrigeration devices (fridge, freezer) is clearly visible. Particular attention is drawn by the very regular, short-lived peaks of active-power consumption reaching as much as 1300 W, occurring approximately every two hours. They correspond to the switching on of the highly resistive heater of the defrosting system (the so-called *No Frost*) in the refrigerator. In turn, in the daytime profile one can easily identify the manual switching on of loads of high active power, such as an electric kettle, a vacuum cleaner or a dishwasher.

6.6.2 Analysis of the power factors and harmonics

The characteristic, capacitive-non-linear profile of the domestic load discussed earlier strongly affects the power factors (Fig. 6.32). The displacement power factor of the fundamental component (DPF corresponding to the classical $\cos(\varphi_1)$) remains at a relatively high level throughout the whole day. A phenomenon requiring particular analysis is, however, the drastic drop in the total, true power factor (PF) during the night-time period, reaching at times a level of merely 0.35.

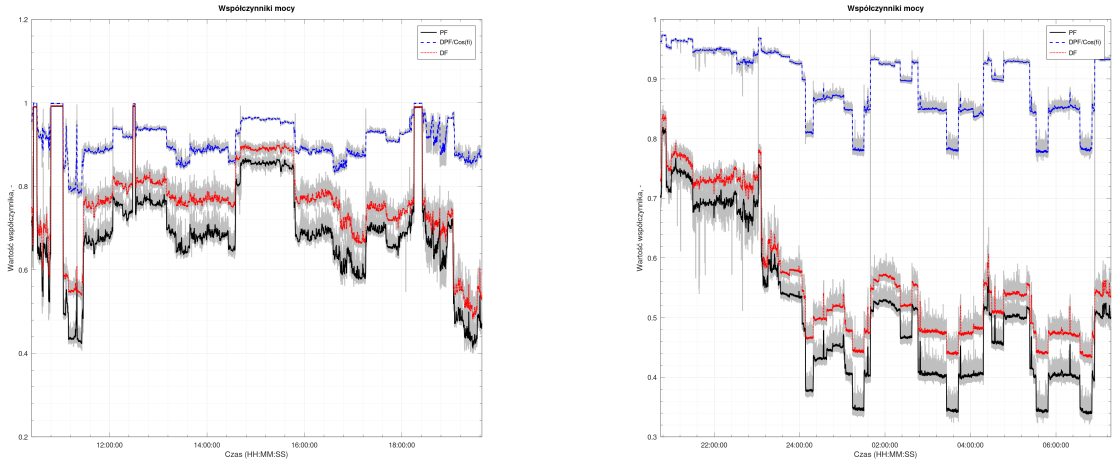
This state is fully justified from the point of view of the specific operation of electronic circuits. At night, most of the resistive and inductive loads of high power (improving the overall active balance of the network) remain switched off. The dominant load then becomes numerous, small switched-mode DC power supplies (chargers) operating in the no-load state or under negligible load. These circuits generate high distortions of the



(a) Daily profile (10:30 – 20:00)

(b) Night-time profile (21:00 – 07:00)

Figure 6.31: Compilation of the profiles of the total active power (P), reactive power (Q) and apparent power (S) in the domestic installation. Source: own elaboration.



(a) Daily profile (10:30 – 20:00)

(b) Night-time profile (21:00 – 07:00)

Figure 6.32: Course of the power factors (PF, DPF, DF) in the tested installation. Source: own elaboration.

consumed current, which, at a very small active power, causes a significant reduction of the distortion factor (DF), pulling down the true parameter PF along with it. These studies prove the effectiveness of the designed meter in detecting hidden power-quality problems in households, which often escape the attention of traditional energy-consumption meters.

Chapter 7

Summary and directions of development

7.1 Final conclusions

The main aim of this thesis was to design, build and program a compact, precise meter of power quality parameters, constituting a low-cost alternative to expensive commercial solutions. The metrological tests carried out, the calibration process, and the tests under real-world conditions (measurements of household appliances and the daily profile of a residential premises) clearly confirm that the main design assumption was successfully achieved.

The device correctly acquires data using isolated Sigma-Delta converters, performs complex real-time mathematical calculations based on the SOGI-PLL algorithm, and transmits the aggregated results via the Ethernet interface (TCP/IP stack) to the host application. The developed text interface based on the SCPI standard provides the device with flexibility in terms of configuration and calibration. It has been demonstrated that the applied method of direct series current measurement, combined with appropriately selected digital filters (DFSDM), makes it possible to achieve high measurement precision, enabling effective identification of phenomena degrading power quality in the grid.

7.2 Analysis of the design problems encountered and their mitigation

As with any prototype of an electronic device, a number of design errors and technological limitations became apparent at the stage of commissioning and testing the hardware. Their identification and physical mitigation constituted an important part of the research process.

7.2.1 Memory limitations and abandonment of the HMI and USB interfaces

The initial concept of the system assumed equipping the device with a physical human-machine interface (HMI) consisting of an OLED display, buttons, an encoder, and an additional USB communication connector. Ultimately, however, the implementation of these subsystems in the software was abandoned.

The main reason for this decision was the architectural limitations of the selected microcontroller. The STM32H750VBT6 device used has a powerful computing core and a huge amount of RAM, but its internal non-volatile Flash memory amounts to only 128 kB. It is dedicated to working with external QSPI memories, which were not provided for in the prototype schematic. In the current version of the software, the support for the FreeRTOS system, the LwIP stack, the CMSIS-DSP library, and the SCPI parser, with maximum code compression applied by the compiler, occupies about 95% of the available Flash space. Adding libraries supporting the USB port and graphics support for the display would physically exceed the microcontroller's resources.

7.2.2 Error in the fault indication circuit (eFuse)

At the schematic design stage, a logic error was made in the visual indication circuit of the TPS2596 electronic fuse. The LED indicating the occurrence of a fault (the `FLT` signal) was connected to the 5 V power line located after the eFuse. As a consequence, at the moment of a fault (e.g. a short circuit), the fuse immediately cut off the output voltage, which deprived the LED of power and made it impossible to light up. This problem was solved in the prototype by mechanically cutting the PCB trace and soldering a jumper connection supplying the LED with the reference voltage from the input of the device (before the fuse).

7.2.3 Underestimation of the thermal losses of the measurement shunt

The original design assumed the use of a standard, single measurement resistor in an SMD 2512 package with a manufacturer-declared power of 3 W. At the early testing stage, the fact was not taken into account that the stated rated power is guaranteed only under rigorously controlled laboratory conditions (with a die temperature that may reach 125–175°C).

During the first load tests with a current of about 10 A, the resistor heated up to over 100°C within a short time. This generated an unacceptably high thermal drift error for the measurement circuit and posed a risk of thermal damage to the laminate and the adjacent components. This forced the manual scraping of the solder mask from the

current pours and the soldering in this place of two specialised resistors with an elevated structure, working in OAR technology, which significantly reduced the thermal resistance and stabilised the measurement.

7.2.4 Hardware conflict of the pins and the transition to the RMII standard

The most serious hardware error was the incorrect assignment of signals for the MII network interface. During the configuration process in STM32CubeMX, the `PC2_C` and `PC3_C` pins were automatically assigned to network communication and routed on the printed circuit board. During software trials, it turned out that the pins with the `"_C"` suffix in the LQFP100 package used are dedicated analog pins and have no hardware connection to the GPIO switching matrix, which completely prevented their use in the MII standard.

Due to the impossibility of remapping the MII bus pins, a migration to the reduced RMII interface was necessary. This required significant modifications to the prototype:

1. Use of the series resistors as solder points for making jumper connections.

2. Replacement of the external 25 MHz crystal oscillator with a 50 MHz version (required by the RMII specification).

3. Physical routing of the new 50 MHz clock signal to the dedicated pin clocking the MAC controller in the STM32.

7.3 Directions of further development (Version 2.0)

Despite the difficulties encountered, the final version of the device is characterised by high application potential. It is planned to make the full project documentation (schematics, Gerber production files, and source code) available on the GitHub platform under an open (*open-source*) licence. At the same time, it is planned to begin work on a thoroughly redesigned version 2.0.

The key modifications planned in the new hardware iteration include:

- **Modularisation of the design (two PCBs):** Separation of the digital part (processor, power supply, communication interfaces) from the measurement module. Such an architecture will allow flexible use of different measurement boards, e.g. a three-phase variant or a version cooperating with non-invasive current transformers and Rogowski coils, depending on the needs of the end user.

- **Improvement of the memory subsystem:** Addition of an external Flash memory (e.g. on the QSPI bus) in order to circumvent the architectural limitations of the processor, as well as the implementation of a dedicated EEPROM chip.
- **Redesign of the power supply:** Adaptation of the power input to the industrial 24 V DC standard and the modern USB-C connector, with an optional implementation of power supply directly via the PoE (*Power over Ethernet*) network cable.
- **Mechanical design:** Creation of an enclosure and a standardised holder enabling the device to be mounted directly on a DIN rail in domestic and industrial electrical switchboards.

Removing the Flash memory limitations will also open the way to a significant expansion of the firmware. It is planned to fully restore support for the local HMI interface (OLED display and buttons) and to implement the industrial Modbus protocol (both in the Modbus TCP version and the hardware RTU/RS485 version) for configuration and readout purposes. A very promising research direction is also the implementation in the device of a data recording structure compatible with the PQDIF standard (in accordance with IEEE Std 1159.3). This would allow the direct analysis of the files generated by the prototype in popular, professional computer programs used for power quality analysis, bringing the functionality of the *open-source* project on a par with advanced commercial solutions.

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Appendices

Technical documentation

Table D1: List of electronic components of the measurement module

Designation (Ref)	**Quantity**	**Value / Type**	**Manufacturer part number (MPN)**
C1–C5, C7, C9–C12, C47, C55–C61	18	100 nF	—
C24, C27, C36, C38	4	100 nF	C0603C104K5RACAUTO
C6	1	4.7 μF	—
C13, C14	2	1 nF, 2 kV	885342208024
C16	1	22 nF	AC0402KRX7R9BB223
C17	1	4.7 nF, 2 kV	1210B472K202CT
C18, C42, C43, C44	4	10 pF	885392005106
C19	1	10 nF	06035C103JAT2A
C20, C25	2	100 nF	C3216NP01H104J160AA
C21, C35, C37, C40, C51	5	1 μF	12063C105KAT2A
C22	1	22 μF	CL31A226MOCLNNC
C23, C26	2	1 μF	CGA3E1X7R1E105K080AC
C28, C30, C33, C34	4	1 nF	12065C102KAT2A
C29, C31, C32	3	1 nF	C0603C102K5RAC7411
C39, C41, C49, C50	4	2.2 μF	—
C45, C46, C48	3	100 nF	GRM155R61A104KA01J
C52	1	470 pF	KGM05ACG1H471JN
C53, C54	2	5 nF	08051C502KAT2A
D1	1	LED 5V	—
D2	1	Red LED	—
D3	1	TVS diode	ESD203B102ELE6327XTMA1
D4	1	TVS diode	GG040205100N2P
D5	1	Green LED	—
D6	1	LED USB PWR	—

Continued on next page...

Table D1 – continued from previous page

Designation (Ref)	Quantity	Value / Type	Manufacturer part number (MPN)
D7	1	LED 3.3V	—
D8	1	LED FAULT	—
D9, D10	2	ESDA5V3L	771-PESD36VS2UT215
FB1, FB2	2	1000 Ω	—
IC1	1	LAN8740AI-EN	579-LAN8740AI-EN-TR
IC2	1	SN74LVC14A	595-SN74LVC14APWRG3
IC3, IC4	2	PS2802-1	—
J1	1	BOOT connector	—
J2	1	USB OTG	538-47590-0001
J3	1	GRID connector	710-691502710003
J4	1	5V connector	—
J5	1	OUT connector	710-691502710003
J6	1	PROG DEBUG	—
J7	1	RJHSE-5380	—
J8	1	UART DEBUG	—
J9	1	OLED connector	—
J10	1	ADC EXT	TBC05-03-1-G-G
J11	1	SPI EXT	—
J12	1	OUT 1	—
J13	1	OUT 2	—
J15	1	Spare connectors	—
J17	1	I/O connector	—
Q2	1	SI3493DDV	SI3493DDV-T1-GE3
R3–R6	4	75 Ω	ERA-2AEB750X
R7	1	1 MΩ	RC2512JK-071ML
R8–R10, R37, R64, R65	6	49.9 Ω	—
R11, R12, R14, R15, R21	5	715 kΩ	667-ERA-6AEB7153V
R13	1	10 kΩ, 0.1%	667-ERA-3AEB103V
R16, R17	2	94.2 kΩ	RN73H1JTTD94R2F50
R18, R24, R25, R35	4	47 kΩ	—

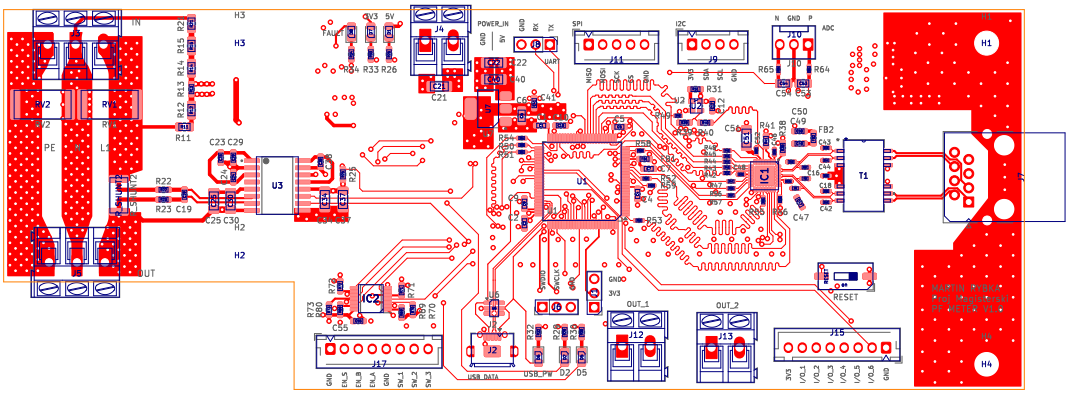
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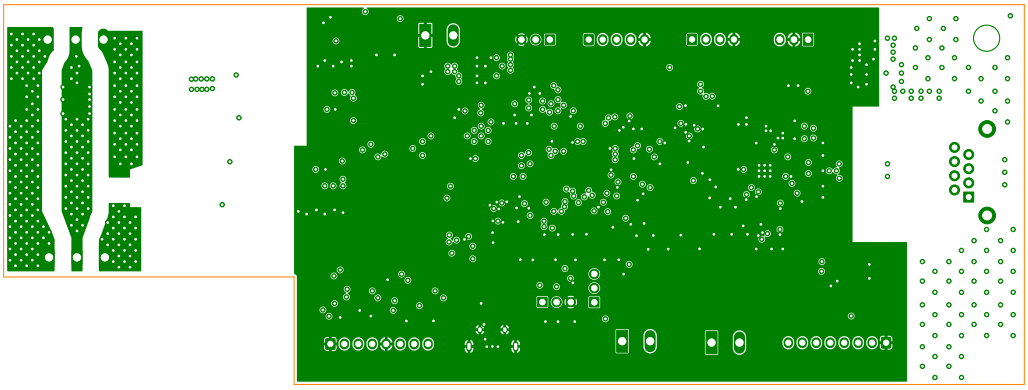
Designation (Ref)	Quantity	Value / Type	Manufacturer part number (MPN)
R19, R36	2	1 kΩ	—
R20, R27, R28, R31, R41, R69–R73, R80	11	10 kΩ	—
R22, R23	2	10 Ω	RCA060310R0FKEAHP
R26, R29, R30, R32–R34	6	680 Ω	—
R38	1	12.1 kΩ	RT0603FRE0712K1L
R39, R40	2	33 Ω	—
R42–R48, R56–R59	11	56 Ω	RT0402DRE0756RL
R49, R61	2	1.5 kΩ	MCS04020C1501FE000
R50–R55	6	100 Ω	RR0510P-101-D
R60	1	4.7 kΩ	MCS04020C4701FE000
R67, R74, R75, R76	4	150 Ω	—
R_SHUNT	2	20 mΩ	OAR5R020FLF
RESET1	1	Tact switch	—
RV1, RV2, RV3	3	Varistor	MOVS4032V230
T1	1	Transformer	815-ALANL100X1DE12DT
U1	1	STM32H750VBT6	STM32H750VBT6
U2	1	DSC1001DI1	998-1001DI1-050.0000
U3	1	AMC3306M25	AMC3306M25DWER
U4	1	TPS259620	595-TPS259620DDAR
U5	1	AMC3336	AMC3336DWER
U6	1	USBLC6-4SC6	511-USBLC6-4SC6Y
U7	1	TLV1117-33	595-TLV1117LV33DCYR

Visualisation of the printed circuit board layers

This section presents the views of the individual layers of the designed printed circuit board.

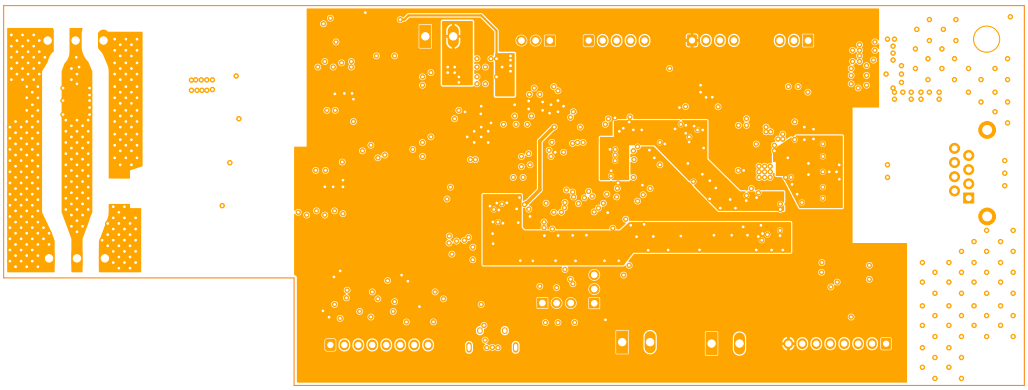


Layer L1 (Top)

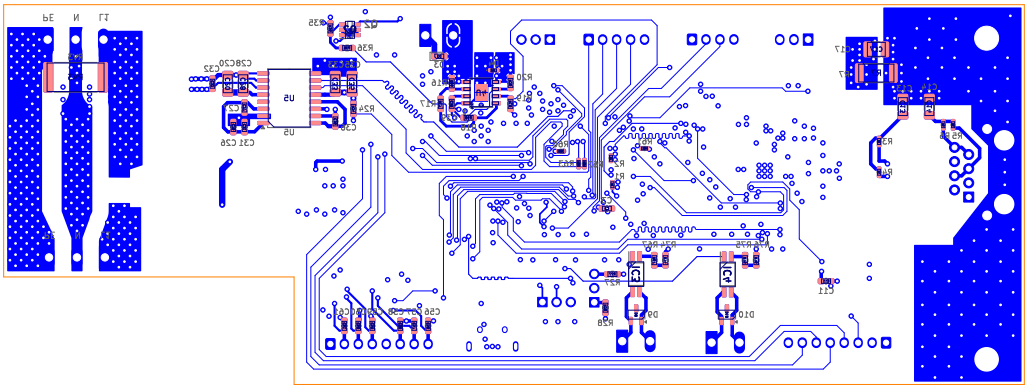


Layer L2 (In1 – GND)

Figure D1: View of the top layers of the printed circuit board (L1 and L2). Source: own elaboration.



Layer L3 (In2 – VCC)



Layer L4 (Bottom)

Figure D2: View of the bottom layers of the printed circuit board (L3 and L4). Source: own elaboration.

List of abbreviations and symbols

List of abbreviations

A/C analog-to-digital

ADC analog-to-digital converter (Analog-to-Digital Converter)

AGD/RTV household appliances and radio/television equipment

AMC family of isolated Sigma-Delta modulators by Texas Instruments (incl. the AMC3306 and AMC3336 devices)

ARM processor core architecture (Advanced RISC Machine)

ASCII character encoding system (American Standard Code for Information Interchange)

BOM list of components (Bill of Materials)

BPF band-pass filter (Band-Pass Filter)

C/A digital-to-analog converter

CMTI immunity to common-mode transient voltages (Common-Mode Transient Immunity)

CT clamp-on current transformer (Current Transformer clamp)

CTR current transfer ratio (Current Transfer Ratio)

DC direct component / direct current (Direct Current)

DF distortion factor (Distortion Factor)

DFSDM digital filter for Sigma-Delta modulators (Digital Filter for Sigma-Delta Modulators)

DMA direct memory access (Direct Memory Access)

DPF displacement power factor of the fundamental component (Displacement Power Factor)

DSP digital signal processing (Digital Signal Processing)

EDA software for designing electronic circuits (Electronic Design Automation)

EEPROM non-volatile memory (Electrically Erasable Programmable Read-Only Memory)

EMC electromagnetic compatibility (ElectroMagnetic Compatibility)

EMI electromagnetic interference (Electromagnetic Interference)

ENOB effective number of bits (Effective Number of Bits)

ENTSO-E European Network of Transmission System Operators for Electricity

ESD electrostatic discharge (Electrostatic Discharge)

ESR equivalent series resistance (Equivalent Series Resistance)

FCS frame check sequence (Frame Check Sequence)

FFT fast Fourier transform (Fast Fourier Transform)

FLL frequency-locked loop (Frequency-Locked Loop)

FPGA field-programmable gate array (Field-Programmable Gate Array)

FPU hardware floating-point unit (Floating-Point Unit)

FreeRTOS real-time operating system (Free Real-Time Operating System)

GPIO general-purpose input/output (General-Purpose Input/Output)

HAL hardware abstraction layer (Hardware Abstraction Layer)

HMI human-machine interface (Human-Machine Interface)

I2C serial, bidirectional communication bus (Inter-Integrated Circuit)

IEC International Electrotechnical Commission

IEEE Institute of Electrical and Electronics Engineers

IP Internet protocol (Internet Protocol)

ISR interrupt service routine (Interrupt Service Routine)

PQ power quality

JSON data interchange format (JavaScript Object Notation)

LDO low-dropout linear regulator (Low-Dropout Regulator)

LED light-emitting diode (Light-Emitting Diode)

LPF low-pass filter (Low-Pass Filter)

LwIP lightweight network protocol stack (Lightweight IP)

MAC media access controller (Media Access Control)

MII media-independent interface (Media Independent Interface)

Modbus industrial communication protocol (variants: Modbus TCP and RTU/RS485)

MOV metal-oxide varistor (Metal-Oxide Varistor)

MSOGI-PLL multiple SOGI-based phase-locked loop (Multiple SOGI-PLL)

OAR open-construction resistor (Open Air Resistor)

OLED organic light-emitting diode (Organic Light-Emitting Diode)

OOK on-off keying (On-Off Keying)

OSR oversampling ratio (Oversampling Ratio)

OTG USB connection standard (On-The-Go)

PC personal computer (Personal Computer)

PCB printed circuit board (Printed Circuit Board)

PF true power factor (True Power Factor)

PHY physical layer device (Physical Layer)

PI proportional-integral controller

PLL phase-locked loop (Phase-Locked Loop)

PoE power over Ethernet (Power over Ethernet)

PQDIF power quality data interchange format (Power Quality Data Interchange Format)

QSPI quad serial peripheral interface (Quad Serial Peripheral Interface)

RAM random access memory (Random Access Memory)

RJ45 standardised Ethernet network connector (8P8C)

RMII reduced media-independent interface (Reduced Media Independent Interface)

RMS root mean square value of a signal (Root Mean Square)

RS485 serial data transmission standard (EIA-485)

RTU binary transmission mode in the Modbus protocol (Remote Terminal Unit)

RX data receive line (Receive)

SAR successive approximation register converter (Successive Approximation Register)

SCPI standard commands for programmable instruments (Standard Commands for Programmable Instruments)

SMD surface-mount device (Surface Mount Device)

SMPS switched-mode power supply (Switched-Mode Power Supply)

SOGI second-order generalized integrator (Second Order Generalized Integrator)

SOGI-PLL phase-locked loop with a second-order generalized integrator (Second Order Generalized Integrator – Phase-Locked Loop)

SPI serial peripheral interface (Serial Peripheral Interface)

SR slew rate (Slew Rate)

SWD serial wire debug interface (Serial Wire Debug)

TCP transmission control protocol (Transmission Control Protocol)

TCP/IP layered communication protocol stack model (Transmission Control Protocol / Internet Protocol)

TCR temperature coefficient of resistance (Temperature Coefficient of Resistance)

THD total harmonic distortion (Total Harmonic Distortion)

TPF true power factor (True Power Factor)

TVS transient voltage suppressor (protection diode) (Transient Voltage Suppressor)

TX data transmit line (Transmit)

UART universal asynchronous receiver-transmitter (Universal Asynchronous Receiver-Transmitter)

UDP connectionless transport-layer protocol (User Datagram Protocol)

USB universal serial bus (Universal Serial Bus)

UVLO under-voltage lockout (Under-Voltage Lockout)

ZCD zero-crossing detection (Zero-Crossing Detection)

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